

The Rise of Agentic Intelligence: A Comprehensive Survey of LLM-based Autonomous Agents (2026)

Liam Ning*, Claude Code, and GLM-5.1

May 2026

Abstract

The transition from large language models (LLMs) as text generators to autonomous agentic systems—systems that perceive an environment, plan over long horizons, invoke external tools, maintain persistent state, and pursue goals with limited human oversight—has fundamentally redrawn the AI research landscape between 2023 and 2026. This survey provides a comprehensive review of LLM-based autonomous agents, organising the literature around four mutually reinforcing capabilities: cognition (reasoning and planning), memory (state persistence and retrieval), tool use (environmental interaction), and multi-agent coordination. We trace the field’s rapid evolution from early prompt-engineering scaffolds such as ReAct and AutoGPT, through the standardisation of tool interfaces via the Model Context Protocol, to the emergence of reasoning-first models that scale inference-time compute as aggressively as parameters. We survey the underlying architectural backbones (dense, mixture-of-experts, and small models for edge deployment), the cognitive techniques that turn a stateless LLM into a planner (chain-of-thought, tree- and graph-structured search, self-refinement, world-model-augmented search), the memory hierarchies that allow agents to act over extended horizons, the protocols that connect agents to the world (function calling, MCP, code-as-action, browser and OS computer-use), the coordination patterns that organise communities of agents (role-play, debate, hierarchical orchestration), and the rapidly maturing benchmark and evaluation methodology that distinguishes genuine progress from contamination and circular LLM-as-judge artefacts. We dedicate separate chapters to safety, where prompt injection and indirect injection through retrieved content are the dominant attack surfaces, and to domain frontiers spanning autonomous software engineering, scientific discovery, and vision-language-action policies for embodied agents. We close with a synthesis of open problems—robust long-horizon planning, evaluation under contamination, human-agent oversight, and the design of agentic infrastructure—likely to dominate the next two years of research.

Contents

1	Introduction	4
1.1	The Agentic Turn	4
1.2	A Brief History of LLM-Based Agency, 2020–2026	5
1.3	Defining the LLM-Based Agent	6
1.4	A Capability Taxonomy	6
1.5	Scope and Comparison with Prior Surveys	7
1.6	Roadmap	7
1.7	Historical Milestones at a Glance	8
2	The Architectural Backbone	8
2.1	From Dense Transformers to Mixture-of-Experts	8
2.2	Small Language Models for Edge Agency	9
2.3	Agent-Aware Post-Training	10
2.4	Inference-Time Compute as a New Architectural Axis	11
2.5	A Comparative Overview	11

*csningli.com

3	Cognition: Reasoning and Planning	12
3.1	Chain-of-Thought and Prompted Reasoning	12
3.2	Search-Based Deliberation: Trees, Graphs, and Monte-Carlo Methods	13
3.3	ReAct: Interleaving Reasoning and Acting	13
3.4	Self-Correction and Reflection	14
3.5	Long-Horizon Planning and Its Failure Modes	14
3.6	Inference-Time Compute and Reasoning-First Models	15
3.7	Discussion	15
3.8	Reasoning Models and the Externalisation/Internalisation Question	16
4	Memory and State Persistence	17
4.1	Working Memory: The Context Window	17
4.2	Retrieval-Augmented Generation as Semantic Memory	17
4.3	Episodic Memory: Trajectories and Reflections	18
4.4	Procedural Memory and Skill Libraries	18
4.5	MemGPT and the Operating-System Analogy	19
4.6	Trade-offs and a Practical Memory Hierarchy	19
4.7	Empirical Trade-offs: Long Context Versus Retrieval	19
4.8	Memory Inspection and Debuggability	20
5	Tool Use and Environmental Interaction	21
5.1	From Parametric Knowledge to Tool Augmentation	21
5.2	Function Calling and Structured Outputs	21
5.3	Code as a Universal Action Substrate	22
5.4	Massive Tool Catalogues and API Marketplaces	22
5.5	Web and OS Agents: Real-World Actuation	22
5.6	The Model Context Protocol	23
5.7	Open Problems	24
6	Multi-Agent Systems	24
6.1	Coordination Topologies	24
6.2	Role-Play and Specialised Personae	25
6.3	Debate, Consensus, and Critic–Actor Loops	25
6.4	Frameworks and Engineering Substrates	26
6.5	Why Multi-Agent Systems Fail	26
6.6	Emergent Behaviour and Open Questions	27
6.7	When Does Multi-Agent Decomposition Help?	27
7	Evaluation, Benchmarks, and the Crisis of LLM-as-Judge	28
7.1	From Static QA to Agentic Evaluation	28
7.2	Agent-Specific Benchmarks	29
7.3	LLM-as-Judge: Promises and Pitfalls	30
7.4	Contamination and the Half-Life of Benchmarks	31
7.5	Cost, Safety, and Multi-Axis Evaluation	31
7.6	Summary and Outlook	31
7.7	Operational Evaluation: Beyond the Headline Number	31
7.8	Threats to Validity in Agent Evaluation	32
8	Safety and Security	33
8.1	Why Agent Safety Differs	33
8.2	Direct Prompt Injection and Jailbreaks	34
8.3	Indirect Prompt Injection: The Dominant Threat	34
8.4	Memory Poisoning and Long-Lived Attacks	35
8.5	Sandboxing, Capability Boundaries, and Computer Use	35
8.6	Multi-Agent Attack Surfaces	36
8.7	Red-Teaming, Evaluation, and Operational Practice	36
8.8	Open Problems	36
8.9	Illustrative Incidents and Case Patterns	36

9	Domain-Specific Frontiers	37
9.1	Autonomous Software Engineering	38
9.2	Scientific Discovery	38
9.3	Embodied Agents and Vision–Language–Action Models	39
9.4	Healthcare and Clinical Agents	40
9.5	Finance, Trading, and Compliance	40
9.6	Other Emerging Verticals	41
9.7	Cross-Cutting Lessons	41
10	Synthesis and Outlook	41
10.1	The Converged Picture of an LLM-Based Agent	42
10.2	Open Problems	42
10.3	Predictions for 2026–2028	43
10.4	Recommendations for Researchers	44
10.5	Coda	44
11	Glossary and Notation	45
11.1	Core Concepts	45
11.2	Cognitive Techniques	45
11.3	Tool-Use and Action Protocols	46
11.4	Multi-Agent and Coordination	46
11.5	Evaluation	47
11.6	Safety	47
11.7	Architecture	48
11.8	Notation Used in This Survey	48

1 Introduction

In November 2022, the public release of ChatGPT made it suddenly obvious that large language models (LLMs) could carry on coherent multi-turn conversations, follow natural-language instructions, and produce code, prose, and analysis on demand [82]. Within eighteen months, the centre of gravity of the field had shifted again. The interesting question was no longer “what can a language model say?” but “what can a language model *do* in the world, on its own, over many steps?” That second question is the subject of this survey. We use the term *agentic intelligence* to refer to systems built on top of LLMs that perceive an environment, plan over a horizon of more than a single response, invoke external tools and other models, maintain state across interactions, and pursue goals with a degree of autonomy that classical chatbots never possessed.

The transition has been rapid and uneven. As recently as 2022, an autonomous LLM “agent” typically meant a thin Python loop around a prompt, often unable to recover from a single tool error [122]. By the end of 2024, frontier systems were autonomously resolving roughly half of all real-world GitHub issues in the SWE-bench Verified split [50, 80], the Model Context Protocol (MCP) had been published as an open standard for tool integration [4], and computer-use agents could drive a graphical desktop with screenshots and synthetic input events [3]. By 2026, top systems exceed 75% on SWE-bench Verified [107], multi-agent frameworks orchestrate dozens of specialised roles [46, 120], and agents have begun to author scientific papers, run simulated laboratories, and operate as quasi-employees in commercial software stacks [72]. Whether one calls this progress “agentic AI”, “LLM-based autonomous agents”, or simply the natural maturation of foundation models, it has produced a research landscape that no single 2023-era survey can adequately describe.

This survey is an attempt to map that landscape as it stands in mid-2026. We focus on *LLM-based* agents—systems whose decision-making and language faculties are dominated by one or more transformer-based foundation models—rather than the broader history of intelligent agents reaching back to the 1980s, which is already covered in standard texts [94]. We deliberately privilege material from 2023 onward, where the existing literature is most fragmented, while grounding the new content in a small set of well-established conceptual frameworks. The remainder of this section motivates the agentic turn, sketches a timeline of the field from 2020 through 2026, offers a working definition of an LLM-based agent, lays out the capability taxonomy that organises the rest of the survey, and positions our contribution against the prior survey literature.

1.1 The Agentic Turn

Three concurrent developments turned a generative model into a viable agentic substrate. First, instruction tuning—most influentially via reinforcement learning from human feedback (RLHF)—transformed LLMs from autocomplete engines into systems that reliably attempt to do what a user asks [25, 82]. Second, work on prompt-based reasoning, beginning with chain-of-thought (CoT) prompting [119] and continuing through tree- and graph-structured search [10, 129], demonstrated that test-time computation could substitute for additional training in solving multi-step problems. Third, tool-augmented language models—introduced systematically by Toolformer [95] and Talm [85], and operationalised by ReAct [130]—showed that an LLM could decide when to call an external function, parse its output, and incorporate the result into subsequent reasoning. Once these three capabilities were stacked on top of each other, the gap between “language model” and “agent” narrowed to a problem of engineering rather than fundamental research.

The economic and scientific stakes followed. By 2023, OpenAI’s GPT-4 [79] and Anthropic’s Claude were exposed not only as chat endpoints but as function-calling APIs, allowing application developers to delegate sub-tasks to the model with structured inputs and outputs. The release of AutoGPT in March 2023 [100] catalysed public interest by showing that even a fragile prompt-engineering scaffold could chain GPT-4 calls into recognisable goal-directed behaviour: writing files, running shell commands, and performing simple research tasks. Within months, frameworks such as LangChain [17], LlamaIndex [68], and BabyAGI proliferated, and the term “agent” became overloaded enough that academic surveys had to begin disambiguating it from older RL- and BDI-based usages [113, 122].

The point is not that any single 2023 system was particularly capable in absolute terms—they were not. AutoGPT-class systems frequently looped on trivial tasks, hallucinated tool calls, and failed silently on non-trivial ones [70]. The point is that the architecture had been demonstrated. Once the community accepted that an LLM could in principle plan, act, observe, and iterate, the research agenda for the next three years wrote itself: improve each step of that loop, scale it, evaluate it, and harden it against failure. This survey is the story of that agenda.

1.2 A Brief History of LLM-Based Agency, 2020–2026

We find it useful to mark six rough phases in the post-2020 development of LLM-based agents. The boundaries are imprecise—many ideas overlap and re-appear—but the progression captures the dominant research focus of each period.

Phase 1 (2020–2021): Emergent capabilities in raw LLMs. The publication of GPT-3 [15] established that scaling alone could produce few-shot in-context learning. T5 [93], PaLM [24], and Chinchilla [45] explored the design space of dense decoder, encoder–decoder, and compute-optimal training. Although none of these systems were agentic in any operational sense, they made it credible that a single model could absorb enough world knowledge and procedural skill to serve as the cognitive core of a future agent.

Phase 2 (2022): Alignment and elicitation. InstructGPT [82] demonstrated that RLHF could turn raw GPT-3 into a system that follows instructions with far higher reliability. Almost simultaneously, chain-of-thought prompting [119], self-consistency [116], and least-to-most prompting [137] showed that intermediate reasoning steps, generated by the model itself, dramatically improved accuracy on arithmetic, commonsense, and symbolic tasks. The release of ChatGPT in November 2022 made these capabilities accessible to a non-research audience, and the public reaction reset funding and policy expectations across the field.

Phase 3 (2023): Reasoning and tool use as primitives. The defining papers of 2023 made the LLM agent loop concrete. ReAct [130] interleaved natural-language reasoning traces with tool-call actions; Toolformer [95] self-supervised the decision of when to call a tool; Reflexion [99] introduced verbal self-feedback as a substitute for gradient updates. Tree-of-Thought [129] cast multi-step reasoning as deliberate search, and Self-Refine [74] as iterative critique. On the systems side, Generative Agents [86] simulated a small town of memory-augmented characters, and Voyager [112] produced a lifelong-learning Minecraft agent that authored its own skill library. AutoGPT [100], BabyAGI, and LangChain made the loop accessible as open-source code. Two influential surveys appeared at the end of the year: Wang et al.’s *Survey on LLM-based Autonomous Agents* [113] and Xi et al.’s 86-page *Rise and Potential of Large Language Model Based Agents* [122], which proposed the brain–perception–action decomposition we adopt as a reference framework throughout this paper.

Phase 4 (2024): Multi-agent systems and standardisation. If 2023 was the year of the single agent, 2024 was the year coordination became the bottleneck. CAMEL [62], MetaGPT [46], and AutoGen [120] formalised role-based multi-agent collaboration; ChatDev [89] applied it specifically to software engineering. Specialised agents for code (SWE-agent [125], OpenDevin/OpenHands [115]), web navigation (WebArena [138], VisualWebArena [56]), and operating-system control (OSWorld [123]) appeared in close succession, each accompanied by its own benchmark. The most consequential single event was the November 2024 publication of the Model Context Protocol [4], an open JSON-RPC-based standard for connecting LLMs to tools and data sources, which by 2025 had been adopted by OpenAI, Google, and most major IDE vendors and is now the dominant interface for tool integration. Anthropic’s October 2024 “computer use” release [3] extended tool use beyond textual APIs to direct manipulation of a graphical desktop via screenshots and synthetic input events.

Phase 5 (2025): Reasoning models and inference-time scaling. A new class of models—OpenAI’s o1 and o3 [81], DeepSeek’s R1 [29], Anthropic’s extended-thinking modes, and Google’s Gemini reasoning variants—explicitly traded inference-time compute for accuracy by performing extended internal reasoning before emitting a response. The empirical scaling laws for test-time compute [103] closely mirrored, and in some regimes outperformed, the scaling laws for parameters and pre-training data. For agents, this meant that the planning and reflection stages of the loop could be performed with substantially higher fidelity, and that long-horizon tasks previously dominated by orchestration overhead became tractable for a single reasoning-model rollout. Concurrently, agentic retrieval-augmented generation [101] extended classical RAG with multi-hop, self-directed retrieval, and memory-centric architectures such as MemGPT [83] re-introduced the operating-system metaphor of paging into the agent design.

Phase 6 (2026): Agentic infrastructure and applications. By the time of writing, the open research questions have shifted away from “can an LLM do X” toward how to build, evaluate, and deploy heterogeneous agentic systems at scale. Production code-writing agents, scientific discovery agents [12, 72], autonomous browsers, and embodied vision–language–action policies [55] are all now active areas of commercial deployment, while the research community is increasingly preoccupied with safety [39], evaluation pathologies [134], and the long-tail behaviours that emerge when agents are placed in open-ended environments for extended periods.

1.3 Defining the LLM-Based Agent

We adopt the following working definition.

An LLM-based agent is a system in which one or more pretrained large language models, together with auxiliary memory and tool interfaces, repeatedly perceive observations from an environment, decide on actions, and execute those actions through external effectors, in service of an externally specified goal.

This definition is consistent with the classical agent abstraction of Russell and Norvig [94], in which an agent is anything that maps a percept history to actions, but specialises it in three ways. First, it requires that an LLM (rather than, say, a hand-coded rule system or a deep RL policy) be the dominant decision-making component. Second, it requires the loop to close: a one-shot prompt-completion does not constitute agency in our sense, but a system that takes at least one observation, action, and follow-up observation does. Third, it allows the action set to include both *external* effectors (HTTP calls, shell commands, browser events, robot torques) and *internal* ones (writing to a memory store, spawning a sub-agent, requesting a tool from a registry). The internal-action perspective is taken from the cognitive-architectures-for-language-agents (CoALA) framework of Summers et al. [105], which we draw on heavily in Section 3.

Formally, we describe an agent as a tuple

$$\mathcal{A} = \langle M, P, M_{\text{em}}, T, E, \pi \rangle,$$

where M is the LLM (or set of LLMs), P is the prompting/scaffolding policy that constructs each model input, M_{em} is a memory module supporting reads and writes, T is a set of tools (each with a structured signature), E is the environment, and π is the high-level controller that turns model outputs into the next observation, action, or memory write. The environment E may be a Unix shell [125], a web browser [138], an operating system [123], a Minecraft world [112], a simulated society of other agents [86], or a real robot [55]. The same conceptual decomposition applies in every case.

Several conceptual frameworks group these components differently. Xi et al. [122] divide the agent into a *brain* (the LLM and its prompting), a *perception* module (which converts environmental observations into model-readable form), and an *action* module (which executes outputs). CoALA [105] draws an analogy with classical cognitive architectures such as Soar and ACT-R, and distinguishes *internal actions* (reasoning, retrieval, learning) from *external actions* (grounded effects on the world). Wang et al. [113] use the four-component decomposition of profile, memory, planning, and action. We treat these as compatible refinements: the brain-perception-action triple gives the highest-level structure, while CoALA’s internal/external distinction and Wang et al.’s explicit memory and planning modules populate it with detail. Where this survey introduces specific subsystems—memory hierarchies, tool registries, multi-agent protocols—we situate them within this shared scaffold rather than introducing parallel taxonomies.

1.4 A Capability Taxonomy

The body of the survey is organised around four complementary capabilities that together account for what makes an LLM-based system an agent. We list them here in the order they are treated in the rest of the paper.

Cognition: reasoning and planning. An agent must decide what to do next. The relevant techniques range from single-step chain-of-thought [119] and self-consistency [116], through deliberate search such as Tree-of-Thought [129] and graph-structured reasoning [10], up to long-horizon planning that combines an LLM with a learned or symbolic world model [41]. Section 3 surveys these techniques and the recent literature on inference-time compute scaling [81, 103].

Memory and state persistence. An agent acting over many steps must store information beyond the context window. Section 4 covers the spectrum from simple chat-history buffers, through retrieval-augmented generation (RAG) [35, 61] and its agentic generalisations [101], to OS-style virtual memory schemes [83] and biologically-inspired episodic and semantic memory layers.

Tool use and environmental interaction. An agent acts on the world through tools. Section 5 traces the foundations (ReAct [130], Toolformer [95], function-calling APIs), the standardisation of tool interfaces via the Model Context Protocol [4], and the more aggressive paradigm of code as action [114], in which the agent’s primary action language is executable code. We also discuss browser- and OS-level computer-use agents [3, 123].

Multi-agent coordination. Many tasks are best solved by collections of specialised agents rather than a single monolith. Section 6 surveys role-based collaboration (CAMEL [62], MetaGPT [46]), conversation-driven frameworks (AutoGen [120]), debate and self-play [32, 65], and the increasingly important question of when adding agents helps and when it merely adds latency and cost.

The four pillars are not independent. A reasoning model that performs extended internal deliberation [103] reduces the need for explicit external planning machinery. An agent with a richer tool ecosystem [4] can solve some problems with weaker reasoning. An agent with a structured long-term memory [83] can compensate for limited context length, in turn lowering the burden on planning. Designing an agentic system is, in practice, a co-design problem across these four axes, which is one reason general-purpose agentic engineering has become a distinct discipline.

1.5 Scope and Comparison with Prior Surveys

We survey *LLM-based* agents, with a strong emphasis on developments from 2023 to mid-2026. Several adjacent topics are deliberately treated only when they intersect this scope.

Pure deep reinforcement learning agents and the broader literature on agent-based modelling are referenced where relevant—particularly in Section 3, where MCTS- and RL-augmented planners reappear—but we do not attempt to summarise those fields, which are covered in dedicated texts [106]. **Embodied robotics** appears in Section 9 through the lens of vision–language–action models such as RT-2 [14] and OpenVLA [55], but a comprehensive treatment of robot learning is out of scope. **Classical multi-agent systems** (BDI agents, FIPA protocols, agent-oriented programming) are referenced where they provide useful conceptual vocabulary for LLM-based variants, not surveyed in their own right.

A growing number of surveys cover overlapping territory. We list the principal ones here so that readers can locate complementary perspectives.

- Wang et al., *A Survey on Large Language Model-based Autonomous Agents* (2023, updated 2024) [113], organises the literature around profile, memory, planning, and action components, with extensive single-agent application coverage.
- Xi et al., *The Rise and Potential of Large Language Model Based Agents* (2023) [122], an 86-page treatment using the brain–perception–action decomposition and emphasising agent societies and emergent social behaviour.
- Sumers et al., *Cognitive Architectures for Language Agents* (CoALA, 2023) [105], frames LLM agents as a special case of the cognitive-architecture tradition (Soar, ACT-R), introduces the internal/external action distinction we use, and is structured more as a position paper than a comprehensive review.
- Guo et al., *Large Language Model Based Multi-Agents: A Survey of Progress and Challenges* (2024) [40], focuses specifically on multi-agent systems, complementing our Section 6.
- Zhao et al. [133] and Minaee et al. [76] survey the underlying LLMs themselves and provide useful background for Section 2, but do not focus on agency.
- Gao et al., *Retrieval-Augmented Generation for Large Language Models: A Survey* [35], is a thorough treatment of RAG that we draw on heavily in Section 4.

This survey complements those by (i) extending coverage through 2026, including reasoning models, MCP, computer use, and the most recent benchmark and safety literature; (ii) integrating the historically separate threads of memory, tool use, and multi-agent coordination into a single capability taxonomy; (iii) devoting a dedicated chapter to evaluation methodology and its pathologies, an area that has matured substantially since 2023; and (iv) treating safety, prompt injection, and alignment as a first-class topic rather than an afterthought.

1.6 Roadmap

The structure of the survey mirrors the capability taxonomy of Section 1.4, with three additional sections on architecture, evaluation, and safety, and one on domain-specific deployments.

Section 2 surveys the **architectural backbones** of modern agents: dense versus mixture-of-experts decoders, the rise of small language models for edge deployment, and the agent-aware variants of post-training (instruction tuning, RLHF, RLAIF, agent-tuning). Section 3 treats **cognition**: reasoning paradigms from CoT to graph-structured search, long-horizon planning, and inference-time compute scaling. Section 4 covers **memory and state persistence**: working memory, episodic and semantic stores, RAG and agentic RAG, and OS-style memory hierarchies.

Section 5 treats **tool use and environmental interaction**: function calling, MCP, code-as-action, and browser/OS computer-use agents. Section 6 surveys **multi-agent systems**: hierarchical and peer-to-peer coordination, role-based collaboration, debate, and emergent social behaviour. Section 7 surveys **evaluation and benchmarks**, including GAIA, SWE-bench, AgentBench, WebArena, OSWorld, and τ -bench, together with the methodological pitfalls of LLM-as-judge evaluation. Section 8 treats **safety and security**: prompt injection in tool calling, indirect injection through retrieved content, sandboxing, and alignment of high-autonomy agents. Section 9 covers **domain-specific frontiers**: software engineering agents, scientific agents, vision-language-action models for robotics and UI, and emerging deployments in healthcare, finance, and the enterprise. Section 10 concludes with a synthesis of trends and a list of open problems likely to dominate the next two years of research.

A reader interested only in a specific subsystem can read its dedicated section in isolation; the formal scaffolding of Section 1.3 is light and is reintroduced where needed. A reader interested in the field as a whole is encouraged to read Sections 2 through 6 in order, then sample Sections 7, 8, and 9 according to their application focus.

1.7 Historical Milestones at a Glance

Table 1 extracts the milestones used in this survey’s narrative into a compact chronology. The table is meant as an orientation device, not as a comprehensive bibliography: only the works that are repeatedly cited across multiple subsequent sections are listed. We have grouped milestones by year and labelled each with the dominant capability axis it advances, using the four-pillar taxonomy of Section 1.4 (cognition, memory, tools, multi-agent) augmented with three cross-cutting axes (architecture, evaluation, and safety). Several entries appear in more than one cell of the underlying conceptual matrix; we list each only once, against the axis its dominant contribution most directly advances.

2 The Architectural Backbone

Every agent surveyed in this work rests on a large language model (LLM) backbone. The backbone determines what the agent can know, how cheaply it can act, and how reliably it can be steered toward agentic behaviour. Three architectural choices have come to dominate the design space between 2023 and 2026. First, the partition between *dense* and *sparse mixture-of-experts* (MoE) Transformers, which separates total parameter capacity from per-token compute and so reshapes the cost frontier of long agent rollouts. Second, the rise of *small language models* (SLMs) such as Microsoft’s Phi-3 family, which trade absolute capability for the latency, privacy, and deployability that on-device agents demand. Third, the practice of *agent-aware post-training*—instruction tuning, reinforcement learning from human or AI feedback (RLHF/RLAIF), direct preference optimisation (DPO), and trajectory-level fine-tuning on tool-using rollouts—which converts a generic next-token predictor into a model that natively emits plans, tool invocations, and reflections. A fourth, more recent axis cuts across all three: *inference-time compute scaling*, where reasoning-first models such as OpenAI’s o1 and DeepSeek-R1 spend orders of magnitude more compute per query, often through long internal chains of thought, than their generation-first predecessors [29, 81, 103]. This section reviews each of these axes, explains why they matter for agency rather than for static benchmark accuracy alone, and ends with a summary table of the major backbones currently shipping into agent stacks.

2.1 From Dense Transformers to Mixture-of-Experts

The standard Transformer decoder [111] that underlies GPT-3 [15] and the early Llama family treats every parameter as a participant in every forward pass. This dense regime obeys clean compute-optimal scaling laws [45, 51], but it ties capability inseparably to inference cost: doubling the parameter count doubles the FLOPs of every token generated, including the long tool-call traces that agents tend to produce. By 2023 it had become apparent that, for many capabilities, a much larger *capacity* could be combined with a much smaller *active* compute budget by routing each token to only a handful of specialist sub-networks.

Sparse-MoE foundations. The intellectual lineage runs from sparsely-gated MoE layers [97] through GShard’s automatic-sharding implementation [60] to Switch Transformers [34], which simplified routing to top-1 selection and demonstrated that one could scale a Transformer past a trillion parameters while keeping per-token compute close to that of a much smaller dense baseline. Switch Transformers also catalogued the recurring failure modes of sparse routing: load imbalance among experts, training instability under fp16, and a tendency for some tokens to be “dropped” by the gating function. These problems remain the central engineering preoccupations of MoE research.

Open-weights MoE: Mixtral. The decisive turning point for the open ecosystem was Mistral’s release of Mixtral 8x7B in late 2023 [48], the first widely available open-weights MoE LLM. Each of its eight feed-forward experts is roughly 7B parameters; a learned router selects two per token, yielding approximately 47B total but only $\sim$ 13B active parameters per token. On standard benchmarks Mixtral matches or surpasses Llama-2 70B while running at the latency of a 13B dense model—a roughly five-fold reduction in inference cost at parity. For agent workloads, where each task may issue dozens of tool calls and dozens more chain-of-thought tokens, this compression of the per-token cost translates directly into longer feasible trajectories at the same hardware budget.

State-of-the-art open MoE: DeepSeek-V3. By December 2024, DeepSeek-AI released DeepSeek-V3 [28], an MoE backbone of 671B total parameters with only 37B activated per token, comprising 256 routed experts plus a single shared expert per MoE layer with top-eight expert routing. The model is trained on 14.8T tokens using a multi-token prediction objective, FP8 mixed-precision arithmetic, and an auxiliary-loss-free load-balancing strategy. Its training cost—reported as 2.788 million H800 GPU-hours—is roughly an order of magnitude below that of the best dense competitors at comparable benchmark performance. DeepSeek-V3 became the canonical open-weights base for agentic fine-tunes and for the reasoning-distilled DeepSeek-R1 [29], and its release reset community expectations for what is achievable on the open frontier.

Dense alternatives: the Llama 3 family. Despite the MoE momentum, dense backbones have not been displaced. Meta’s Llama 3 herd [38] includes a 405B-parameter dense decoder trained on roughly 15T tokens with a 128K-token context window, alongside 70B and 8B variants that have become the workhorses of the open-source agent ecosystem. The dense form factor offers two practical advantages that agentic deployment cares about: predictable per-token latency (no routing variance) and simpler quantisation/distillation for downstream fine-tuning. For long-running agents under tight tail-latency budgets—autonomous browsers, customer-support copilots—those properties can outweigh the FLOPs efficiency of MoE. Alibaba’s Qwen2.5 series [91] pursues a similar dense strategy and currently anchors much of the Chinese agent ecosystem.

Implications for agents. The dense-versus-MoE choice is not just a benchmark-accuracy decision; it changes the shape of the cost–reliability frontier in agent loops. MoE models amortise enormous parameter capacity across few active FLOPs, which favours *breadth*—answering arbitrary tool-augmented queries from a long-tail of domains—but introduces routing variance that complicates batching and memory planning when many parallel agent instances share a server [28]. Dense models give up some peak quality but offer tighter latency tails, simpler KV-cache management, and easier pruning into the smaller variants that actually run inside browsers, IDEs, and robots. In practice, today’s most ambitious agent stacks deploy a heterogeneous mix: a frontier MoE backbone for high-stakes planning steps and a small dense model for high-frequency operations such as DOM extraction or log triage, with the orchestrator deciding per step which to invoke.

2.2 Small Language Models for Edge Agency

The second axis runs in the opposite direction. Many of the most consequential agentic applications cannot afford to round-trip every action through a frontier API: industrial robots cannot tolerate seconds of network latency, on-device personal assistants must not exfiltrate private data, and high-volume telemetry agents (log triage, fraud detection) accumulate cost faster than benchmark accuracy can justify. The 2024–2026 wave of small language models is the response to those constraints.

The Phi-3 family [1] crystallised the design philosophy. Phi-3-mini, a 3.8B-parameter dense decoder trained on 3.3T carefully filtered “textbook-quality” tokens, scores around 69% on MMLU and 75% on MMLU-Pro—numbers that would have ranked it inside the top tier of 70B models a generation earlier. Phi-3-small (7B) and Phi-3-medium (14B) extend the curve, and the later Phi-3.5-MoE variant ports the data-curation strategy onto a sparse backbone. The thesis underlying the Phi work is that, for compact models, what matters most is the *quality* and the *distribution* of the pre-training data rather than raw token count: synthetic textbooks, code corpora, and chain-of-thought traces curated for reasoning generalise far better than indiscriminate web scrapes.

For agency this matters in three distinct ways. (i) *Latency*. A 4B dense model can return tokens at tens of milliseconds per token on a single GPU, fast enough for closed-loop UI manipulation and for the high-frequency “inner loop” of a planner–executor agent that emits dozens of micro-actions per minute. (ii) *Privacy*. On-device or on-prem deployment removes prompt-and-completion data from third-party logs, a precondition for adoption in regulated domains such as healthcare or finance. (iii) *Cost amortisation*. Many agent loops repeatedly call the LLM for low-cognitive-load operations—summarising a tool result, deciding whether retrieval is needed, normalising a JSON response. Routing those calls to a 4B model rather than a frontier MoE reduces a typical agent’s marginal

cost by one to two orders of magnitude, which compounds across long-horizon tasks. The architectural pattern that has emerged in both research prototypes and production is therefore not “small versus large” but rather *cascaded*: a frontier model is reserved for the few high-leverage planning and self-critique steps, and an SLM handles the high-frequency interior of the loop.

2.3 Agent-Aware Post-Training

A pre-trained backbone, however capable, does not by default emit the kind of structured intermediate text that agents rely on—tool calls in JSON, multi-step plans, reflections on past failure, refusals to act on suspicious instructions. The conversion of a base model into an agent-ready chat assistant has converged on a multi-stage post-training pipeline whose components are now standard.

Supervised instruction tuning. The first stage is supervised fine-tuning (SFT) on *(prompt, response)* pairs that demonstrate desired behaviour. Modern instruction sets routinely include tool-use traces, step-by-step reasoning, and refusal exemplars, so even before any reinforcement signal the model has been exposed to agentic structure. The OpenAI InstructGPT paper [82] established the template that the rest of the field follows: a small SFT corpus of high-quality demonstrations, followed by a preference-learning stage that fine-tunes a reward model and then optimises the policy against it.

Reinforcement learning from human feedback. RLHF, in the form codified by Christiano et al. [25] and operationalised by Ouyang et al. [82], fits a Bradley–Terry reward model on pairwise human preferences and then optimises the LLM policy with PPO under a KL-divergence penalty toward the SFT initialisation. The KL term is critical: without it the policy collapses onto reward hacks and loses linguistic competence. In agentic settings, RLHF has been used to shape harmless tool-use behaviour, to encourage stepwise reasoning, and to penalise both unwarranted refusals and unsafe action emission.

Constitutional AI and RLAIIF. The cost and ethics of human-only annotation pushed the field toward AI-generated preferences. Anthropic’s Constitutional AI [9] replaces most of the human preference labels with critiques and revisions produced by an LLM operating against a written “constitution” of principles, with an SFT stage on revised completions followed by RL on AI-generated preferences (RLAIIF). The technique scales preference data with model capability rather than with human-rater bandwidth, and it has become the dominant strategy for safety post-training across closed labs.

Direct preference optimisation. DPO [92] eliminates the explicit reward model and the on-policy PPO loop entirely. Starting from the closed-form solution to the KL-regularised RL objective, DPO derives a simple binary classification loss directly on preference pairs: positive completions are pushed up and negatives down, with the reference policy entering only as a regulariser. Empirically DPO matches PPO-style RLHF on many tasks while being far simpler to implement and tune, and it is now the default preference-learning recipe in much of the open-weights ecosystem (Llama 3, Qwen2.5, DeepSeek-V3 all use DPO or close cousins for their alignment stages [28, 38, 91]).

Agent-specific fine-tuning. The general SFT→preference pipeline only weakly conveys the structure of *trajectory-level* agent behaviour: long sequences of (thought, action, observation) tuples, recovery from tool errors, and persistence across interruptions. Two influential efforts directly tackle that gap. AgentTuning [131] releases AgentInstruct, a curated corpus of 1,866 high-quality interaction trajectories spanning six agent tasks (web shopping, household manipulation, scientific Q&A, etc.); fine-tuning Llama-2 7B/13B/70B on a hybrid of AgentInstruct and general instruction data yields the AgentLM series, which dramatically outperforms the base models on held-out agentic benchmarks while preserving general capability. FireAct [18] demonstrates a complementary recipe: distil ~500 GPT-4 ReAct trajectories into Llama-2 7B, and observe a 77% relative improvement on HotpotQA. Together these works established that the trajectory format itself is learnable, and that the gap between an agent built on a frontier API and one built on a much smaller fine-tuned open model can be largely closed through targeted post-training.

Practical pipelines. The current production recipe across the open-weights ecosystem chains all of the above: (1) base pre-training on 10–15T tokens, (2) SFT on heterogeneous instruction data including agentic trajectories, (3) preference learning by DPO or PPO-RLHF, (4) Constitutional/AI-feedback safety stages, and (5) optional task-specific fine-tunes (e.g. a SWE-agent variant on code-only trajectories [125]). The design choices interact: an SLM

for edge agency benefits more from carefully curated trajectory SFT than from generic preference learning, while a frontier MoE benefits disproportionately from RL-style stages that elicit its already latent reasoning circuits.

Table 2 compares the principal post-training recipes against the dimensions that determine when each is appropriate: the supervision signal each consumes, the optimisation it performs, and the agentic property it most directly improves. The recipes are not mutually exclusive; production-grade pipelines chain them in series, often returning to earlier stages as new failure modes are discovered.

2.4 Inference-Time Compute as a New Architectural Axis

For most of the GPT era, “architecture” meant decisions made before any user query arrived: parameter count, depth, expert count, training corpus. By 2024 a fourth axis had become unavoidable: how much compute the model is allowed to spend *after* the prompt arrives.

The empirical case was made by Snell et al. [103], who showed that on math and reasoning benchmarks one can often substitute test-time search and verifier-based rerank for additional pre-training compute, and that a compute-optimally-allocated test-time pipeline on a smaller model can match or beat a much larger one trained at the same total cost. The result was striking because it inverted the prior decade of scaling-law intuition, in which test-time compute was assumed approximately constant.

The first widely deployed reasoning-first model was OpenAI’s o1, released in preview in September 2024 and as a system-card-documented full release on 5 December 2024 [81]. Internally, o1 generates a long “reasoning trace” before its final answer, and the user is billed per token of that hidden trace as well as the visible response. The reasoning trace is not exposed verbatim to users (a deliberate design choice intended to preserve competitive moat and to discourage distillation), but its existence is a first-class architectural feature: o1’s per-query compute is one to two orders of magnitude larger than that of GPT-4o on the same prompt.

DeepSeek-R1 [29], released January 2025 and later published in *Nature*, replicated the reasoning-first paradigm with full open weights. The R1 recipe applies large-scale reinforcement learning directly on the DeepSeek-V3 base—without an SFT bootstrap—rewarding correct final answers and well-formed reasoning chains. The trained model exhibits emergent “aha moments” and self-correcting reasoning chains. Equally importantly, R1’s chains-of-thought *are* disclosed, enabling both downstream distillation into smaller models and open scientific study of reasoning behaviour. The release accelerated a wave of open reasoning-first research in 2025–2026 that this survey returns to throughout subsequent chapters.

Implications for agents. Inference-time compute reshapes agent design in several directions. Long internal reasoning traces dramatically improve single-step decision quality, which reduces the number of tool calls a competent agent needs and therefore the cumulative probability of a recoverable error in a long trajectory. But they also raise the unit cost of each step, forcing schedulers to triage which decisions warrant a full reasoning rollout and which can be answered by a fast model. A new design pattern that has emerged is the “cognitive escalation” loop: a fast SLM monitors execution, detects symptoms of confusion (low-confidence tool outputs, repeated retries, oscillating plans), and conditionally invokes a reasoning-first model only on those steps. Empirically this pattern preserves much of the reasoning-model accuracy at a small fraction of the cost.

2.5 A Comparative Overview

Table 3 summarises the major LLM backbones that currently anchor agent stacks, capturing the four axes discussed above: density, capacity, training scale, and openness. The numbers reflect publicly disclosed information at the time of writing; closed models such as GPT-4o and Claude 3.5 Sonnet do not publish parameter counts, and entries marked “n/a” should be read accordingly.

Two patterns stand out. First, the open ecosystem has converged on two dominant points in design space: the dense Llama-style 70B, which serves as the workhorse base for most fine-tunes; and the very large MoE in the DeepSeek-V3 mould, which serves as the open frontier. Second, the rise of reasoning-first models adds a multiplicative axis on top of either choice: any open backbone can in principle be RL-tuned into a reasoning model, as DeepSeek-R1 demonstrated, and the next year of research is likely to push that pattern down the size curve.

Looking ahead. The architectural decisions reviewed in this section determine the substrate on which all subsequent agentic capabilities are built. Section 3 examines how cognition—chain-of-thought, deliberate search, self-refinement—is layered on top of these backbones, and Section 5 returns to the question of how tool-using behaviour is elicited from them. The interaction between backbone choice and post-training pipeline is one of the field’s most consequential open problems: a 14B dense model with extensive trajectory fine-tuning routinely outperforms

a 70B base model on agent tasks, while a frontier MoE with reasoning-first RL outperforms even much larger dense baselines on planning. Mapping that interaction precisely—what backbone properties predispose a model to learn agentic behaviour, and where the marginal training token is best spent—will shape the next generation of agent infrastructure.

3 Cognition: Reasoning and Planning

The defining capability of an LLM-based agent, beyond that of a chat assistant, is its ability to think *between* actions. A pure language model maps a prompt to a distribution over completions; an agent must additionally decide what to attempt next, anticipate the consequences, and recover when the world disagrees. In contemporary practice this gap is bridged by a stack of cognitive techniques that operate at the prompt level, at the search level, and—most recently—at the level of the model’s training objective. This section reviews that stack. We start with the chain-of-thought paradigm and its prompting variants (Section 3.1), proceed through search-based deliberation (Section 3.2), examine the ReAct family of interleaved thought–action loops (Section 3.3), survey self-correction and reflection (Section 3.4), confront the long-horizon planning problem and its known failure modes (Section 3.5), and finally treat inference-time compute scaling as a first-class architectural axis in its own right (Section 3.6).

3.1 Chain-of-Thought and Prompted Reasoning

The empirical foundation of all modern LLM reasoning is the observation that asking the model to “think step by step” before answering dramatically improves accuracy on multi-step tasks. Wei et al. [119] demonstrated this with chain-of-thought (CoT) prompting: a small number of in-context demonstrations that show the working alongside the answer. On GSM8K, a benchmark of grade-school math word problems, CoT prompting raised PaLM-540B’s accuracy from 18% to 57%—a leap that held across a broad span of arithmetic, symbolic, and commonsense reasoning tasks. Crucially, the effect is *emergent*: it appears only at sufficient scale, suggesting that the underlying reasoning circuits exist in the pre-trained model and need only to be elicited.

Zero-shot variants. Kojima et al. [58] showed that even the demonstrations are largely unnecessary: appending the single phrase “Let’s think step by step” to a query yields most of the gain on many tasks. The simplicity of zero-shot CoT made it the default scaffold for the first wave of prompt-engineered agents, including AutoGPT and the early LangChain templates: any agent that can be reduced to a single prompt was, by 2023, almost always a CoT agent in disguise.

Self-consistency. A natural extension is to sample multiple chains of thought and aggregate their final answers. Wang et al. [116] showed that majority-vote aggregation across k sampled chains (typically $k = 20$ to 40) substantially outperforms single-chain greedy decoding—by 17 percentage points on GSM8K with Codex, and by similar margins on ARC, AQuA, and StrategyQA. Self-consistency exploits the diversity of reasoning paths: even if any single chain is unreliable, the modal answer across many chains is more often correct than the model’s most-likely single trajectory. The technique anticipates the test-time compute paradigm of 2024–2026 (Section 3.6) by trading sampling for accuracy.

Decomposition: least-to-most prompting. Zhou et al. [137] introduced a complementary technique that explicitly decomposes a hard problem into a sequence of simpler sub-problems and solves them in turn, with the solution to each sub-problem available to the next. On compositional generalisation benchmarks where vanilla CoT collapses (SCAN, last-letter concatenation), least-to-most prompting raises performance from near-zero to over 99%. The pattern—*decompose, then solve*—is the prompt-level analogue of classical hierarchical task networks and recurs throughout modern agent stacks, often implicitly inside frameworks such as MetaGPT and DSPy [46, 53].

LLM-as-optimiser. A more open-ended use of CoT-style prompting is exemplified by OPRO [124], in which the LLM itself iteratively proposes and refines candidate solutions to an optimisation problem encoded in natural language, including the discovery of better prompts for downstream tasks. OPRO blurs the line between prompting and search: each iteration is a forward pass that conditions on the history of previously tried solutions and their scores. In agent contexts the same pattern reappears as “self-discovery” of new strategies—an agent reasons in language about which of several available tactics to try next—and as automated prompt optimisation pipelines that close the loop between experimentation and deployment.

3.2 Search-Based Deliberation: Trees, Graphs, and Monte-Carlo Methods

CoT and self-consistency treat the reasoning trace as a single linear path, optionally sampled multiple times in parallel. A more powerful regime allows the model to *branch* its reasoning, evaluate intermediate states, and back-track from dead ends. This brings classical AI search techniques into contact with LLM inference.

Tree of Thoughts. Yao et al.’s Tree of Thoughts (ToT) [129] maintains an explicit tree whose nodes are partial reasoning states and whose edges are LLM-generated extensions. At each node the model both (i) proposes a small set of candidate continuations and (ii) self-evaluates each candidate’s promise. Standard search algorithms—BFS, DFS with heuristic pruning—then guide which branches to expand. On Game-of-24, ToT raises GPT-4’s success rate from 4% (CoT) to 74%; on creative writing and 5x5 mini-crosswords it likewise outperforms linear CoT by wide margins. The cost is real: ToT issues an order of magnitude more LLM calls per problem, and its self-evaluation step is brittle on tasks where the model cannot reliably score partial states.

Graph of Thoughts. Besta et al. [10] generalise the search structure from a tree to a directed graph, allowing distinct branches to merge when they converge on a common sub-result. Graph of Thoughts (GoT) is particularly well-suited to tasks with shared sub-goals—multi-document summarisation, set operations, sorting—where many partial computations would otherwise be redundantly repeated across siblings. The framework is explicit about its operations on the graph (aggregation, refinement, pruning), making it a useful conceptual bridge between LLM prompting and the data-flow style of classical workflow engines.

World-model search. Hao et al.’s Reasoning-via-Planning (RAP) [41] treats the LLM as both world model and policy: the model predicts the next state of an imagined environment, evaluates utility, and uses Monte Carlo Tree Search (MCTS) to plan a multi-step solution. RAP excels on planning benchmarks (Blocksworld, math, logical inference) where a structured search over imagined futures pays off. The same insight—separate the model’s role as transition predictor from its role as actor—has since been picked up by multiple agentic planners that explicitly call the LLM for “simulate” steps before committing to a real action.

Language Agent Tree Search. LATS [136] unifies the threads above by integrating ReAct-style tool use with MCTS-guided search. Each node in the tree corresponds to an agent state (history of thought–action–observation tuples); rollouts use the LLM as both expansion policy and value function; tool calls are executed only along committed branches to limit external side-effects. LATS achieves state-of-the-art on HotpotQA, WebShop, and HumanEval among prompting-only methods at comparable LLM cost. Its design illustrates how planning, reasoning, and acting—historically separate research lines—collapse into a single search problem once one accepts that an LLM can play any of those roles.

When does search pay off? Across these methods a consistent pattern emerges. Search-based deliberation pays off when (i) the task has a clear notion of “progress” that the model can score (puzzles, math, code with unit tests), and (ii) the cost of trying and discarding a branch is low (no irreversible side effects). It under-performs when self-evaluation is unreliable (long-form writing, open-ended creative tasks) or when external actions cannot be safely rolled back. The latter constraint motivates much of the agent-safety literature reviewed in Section 8: a deliberation framework that assumes free retries does not transfer to a customer-support agent that has just sent an email.

Table 4 compares the principal deliberation patterns surveyed in this section against the dimensions that determine when each is the right tool: the structure of the search space, the per-query inference cost relative to a single CoT pass, and the property of the task that the method most directly exploits.

3.3 ReAct: Interleaving Reasoning and Acting

The ReAct paradigm of Yao et al. [130] is arguably the single most influential agent design pattern of the 2023–2026 period. Its observation is simple: the same model that produces useful chains of thought can also produce useful tool invocations, and combining the two in an interleaved trace—*Thought, Action, Observation, Thought, Action, Observation, ...*—yields synergistic gains that neither alone achieves. The thoughts give the agent a scratchpad in which to plan and revise; the actions give the agent grounding in real-world feedback that the chain of thought alone cannot supply.

ReAct’s empirical results were striking. On HotpotQA, the prompt-only ReAct agent improved over CoT by 3.6 absolute exact-match points and reduced hallucinated entities; combined with self-consistency, ReAct raised PaLM-540B accuracy on FEVER fact-verification by 4 points and substantially outperformed imitation-learning

baselines on ALFWorld and WebShop. The format—an LLM that outputs alternating natural-language thoughts and structured tool calls until it emits a final answer—has since become the default substrate for almost every agent framework, from LangChain and LlamaIndex to AutoGen and MetaGPT [17, 46, 68, 120]. We treat ReAct’s tool-use side in detail in Section 5; here we note its cognitive role: ReAct integrates deliberation and grounding in a way that is qualitatively different from CoT, because each thought can be conditioned on the result of a real-world action rather than only on the model’s own continuation.

3.4 Self-Correction and Reflection

The chains-and-trees machinery above all assume that the model can produce a useful intermediate reasoning state. A complementary line of work asks whether the model can recognise its own errors after the fact, generate a critique, and improve. Two influential systems anchor this literature.

Reflexion. Reflexion [99] treats the agent’s failure as itself a teaching signal: after a failed trajectory, the agent generates a short natural-language “self-reflection” summarising what went wrong and stores it in an episodic memory buffer; future attempts on the same task condition on the accumulated reflections. The framework is striking in two ways. First, the gradient signal is verbal rather than numerical—there is no parameter update—and yet performance improves. Second, the gains are large: Reflexion raises GPT-4’s pass@1 on HumanEval from 80% to 91% over three trial-and-error iterations. Reflexion has since been adopted as the canonical agent self-improvement pattern in many systems, including SWE-agent [125], where reflection on failed test runs guides the next code-edit attempt.

Self-Refine. Madaan et al. [74] formalise a closely related loop without an explicit memory: the model generates an answer, then critiques it using a separate prompt, then generates a refined answer conditioned on the critique. On a battery of tasks ranging from sentiment-controlled rewriting to acronym generation to math word problems, Self-Refine improves upon GPT-4’s first-pass output by an average of 20 absolute percentage points. The technique is now a near-universal sub-routine inside multi-agent frameworks, where one “critic” agent applies it to the output of an “actor” agent.

The limits of self-correction. A central caveat to this literature was raised by Huang et al. [47], whose paper “Large Language Models Cannot Self-Correct Reasoning Yet” showed that, when external feedback is removed, self-correction often *decreases* accuracy: the model is prone to over-rewrite already correct answers into incorrect ones. The result reframes the field. Self-correction reliably helps when an external signal—a unit test, a verifier, a search result—tells the model that its previous answer was wrong; without such a signal, the model’s introspective judgement is too noisy to act on. This explains why Reflexion works particularly well in domains with clean test feedback (code, math) and less reliably in open-ended QA, and it motivates the trajectory-level fine-tuning recipes (Section 2.3) and process-level reward modelling discussed below: the goal is to install a reliable internal verifier inside the model itself.

3.5 Long-Horizon Planning and Its Failure Modes

For tasks whose horizon exceeds a handful of steps, prompting alone—even with search and reflection—runs into well-documented failure modes. Subbarao Kambhampati and collaborators have argued forcefully across several papers that current LLMs do not plan in the technical sense at all: PlanBench [110] measures classical planning performance (reasoning about the effects of actions in deterministic worlds with formal goals) and finds that frontier models, including GPT-4-class systems, fail catastrophically on instances that classical planners solve in seconds. The diagnosis is that LLMs, even when prompted with chains of thought, perform pattern matching against training-distribution traces rather than genuine state-space search.

The problem is sharpened in agentic settings, where a single irreversible mistake at step k propagates through the remaining $N - k$ steps. Empirically, the dominant failure mode of long-horizon agents is a *greedy* early commitment—the agent picks an action that looks locally reasonable, executes it, and then spends the rest of the trajectory unable to recover, because reverting the action is impossible (a file has been deleted, a payment has been issued) or because the agent’s context window has lost the information needed to recognise the mistake. Both issues are on prominent display in the GAIA [75] and SWE-bench [50] leaderboards: even the best 2026 agents leave 20–30% of solvable tasks unsolved largely for these reasons.

The literature has responded along two axes. The first is *external planning*: hybrid systems that delegate the formal planning step to a classical planner. LLM+P [67] is exemplary: the LLM translates a natural-language goal into PDDL, a classical planner solves the resulting problem optimally, and the LLM translates the plan back into

actions. Where applicable—domains with clean state representations—this gives the best of both worlds and is dramatically more reliable than pure-LLM planning. The second is *intra-LLM planning*: the search-based deliberation of Section 3.2 pushes the planning capability into the LLM itself by combining language-level proposal with explicit search, while the reasoning-first models of Section 3.6 attempt to learn it via reinforcement.

Decomposition revisited. A more pragmatic response is hierarchical decomposition, in the spirit of least-to-most prompting and HTN planning. Modern multi-agent frameworks (MetaGPT, ChatDev [46, 89]) decompose long tasks into role-bound sub-tasks executed by specialised sub-agents whose individual horizons are short enough to be tractable. The decomposition is itself a planning step, but it is performed once, by a planner agent, and its cost is amortised across the rest of the trajectory. Section 6 returns to these systems in detail.

3.6 Inference-Time Compute and Reasoning-First Models

The most consequential recent development in agent cognition is the rise of inference-time compute as a first-class axis of model design. Earlier sections noted its architectural implications (Section 2.4); here we focus on its cognitive implications.

Process supervision. A precondition for serious test-time scaling is a reliable signal of *intermediate* reasoning quality. Lightman et al. [66] compared outcome-supervised reward models (which only judge the final answer) against process-supervised reward models (PRMs, which judge each step). PRMs are dramatically better, both as rerankers and as RL signals: a PRM-rerank on the MATH benchmark raises GPT-4 accuracy from 72% to 78%, with the gap widening on harder problems. The result reframes reasoning as a step-by-step search problem with per-step verification, anticipating the structure of the reasoning-first models that followed.

Compute-optimal scaling. Snell et al. [103] formalised the trade-off between training-time and test-time compute. They showed that, on math benchmarks, optimally allocating test-time compute across sampling, beam search with PRM verifiers, and iterative refinement can match the accuracy of a $14\times$ larger pre-trained model under the same total FLOP budget. The result legitimises spending substantial inference compute per query whenever the per-query cost is amortised against the cost of training a comparably capable larger model.

Reasoning-first models. OpenAI’s o1 [81] and DeepSeek-R1 [29] operationalise these insights as model designs whose deployment is inseparable from a long internal reasoning trace. Both models exhibit qualitatively new behaviour on hard benchmarks: o1 reaches IMO-level math performance and competitive Codeforces ratings; R1, trained with pure outcome-level RL on its V3 base, exhibits emergent “aha moments” in which the model spontaneously revises an earlier mistake. These models are agents in disguise—each query triggers a search-and-self-critique loop that, in earlier years, would have been implemented as a multi-prompt scaffold around a smaller model. By internalising the loop, reasoning-first models reduce inference orchestration to a single API call while paying for it in tokens.

Implications for agent design. Reasoning-first models change which agentic patterns are worth implementing externally. With o1 or R1 at the core, much of the ToT/GoT/RAP search apparatus can be omitted: the model performs an analogous search internally, and external search adds only marginal value. Conversely, multi-step *tool* use—which the internal trace cannot perform—remains very much external, and the tightest agent stacks of 2025–2026 pair a reasoning-first core with a fast outer loop that handles tool dispatch, memory retrieval, and observation summarisation. We return to this division of labour in the conclusion (Section 10).

3.7 Discussion

The cognition stack reviewed in this section is best understood as a layered cake. At the bottom sits the prompted CoT, which exposes the model’s latent reasoning circuits at zero training cost. Above it lie sampling-based variants (self-consistency) and structured search (ToT, GoT, RAP, LATS), which trade extra inference compute for improved deliberation. Above those sit reflective loops (Reflexion, Self-Refine), which add an outer iteration over the agent’s own outputs. And above all of these, the reasoning-first models of 2024–2026 install much of the same machinery into the model’s parameters via post-training, paying it once at training time and then collecting the benefit on every subsequent inference. Each layer is most useful where the layer below has reached its limits, and a competent contemporary agent uses several at once: a reasoning-first core for hard planning steps, ReAct-style interleaving

for tool use, Reflexion-style self-critique for failed trajectories, and external search only for problems that genuinely benefit from explicit branching. Mapping this design space empirically—which combinations work for which task families, and at what compute budget—is one of the central problems for agent research over the next two years.

3.8 Reasoning Models and the Externalisation/Internalisation Question

A specific question worth treating in detail is whether the cognitive scaffolds of 2023—ToT, RAP, LATS, Reflexion, Self-Refine—retain their value once a reasoning-first model is available. The empirical answer that is forming is nuanced: some scaffolds are essentially subsumed by reasoning models, others remain valuable, and a few become more valuable. The distinction matters for agent design because each scaffold consumed by the model reduces the engineering complexity of the outer loop.

Subsumed by reasoning models. Self-consistency aggregation is largely subsumed: the internal reasoning trace of an o1- or R1-style model already explores multiple candidate continuations and reconciles them through self-critique. ToT-style branch-and-evaluate search is similarly absorbed in part—reasoning models do branch internally—though the absorption is incomplete because the model’s internal search is not externally inspectable or steerable. Self-Refine’s generate-critique-rewrite loop maps closely onto the reasoning model’s behaviour and is rarely worth implementing externally as a separate pass; its costs (extra LLM calls) typically exceed its marginal benefit on top of a reasoning model’s internal trace.

Retained as external scaffolds. ReAct, by contrast, remains essential: the internal trace cannot perform tool calls, so any agent that interacts with the world still needs the alternating thought-action-observation pattern at the outer loop. Reflexion, similarly, retains value because the reflection is conditioned on *external* feedback (a failed test, an error from a tool) that the reasoning model’s internal trace did not have access to during the original attempt. World-model-augmented search (RAP, LATS) retains value when the search structure is genuinely external—over physical environments, multi-agent systems, or stateful tool sequences—rather than over imagined reasoning states, which the reasoning model already explores internally.

Made more valuable by reasoning models. Some scaffolds are not subsumed but *amplified*. Process supervision becomes more valuable because the reasoning model’s longer internal trace provides more steps to supervise, and a verifier with a higher per-step accuracy can leverage the longer trace to catch errors that would otherwise survive. Programmatic verification (unit tests, formal checkers) similarly becomes more valuable because the reasoning model is more likely to produce a candidate whose correctness is determinable, which raises the value of having a deterministic check available. Multi-agent decomposition with heterogeneous priors (Section 6.7) likewise gains from reasoning models because the disagreement bandwidth between debaters is increased by their richer internal reasoning.

Implications for the agent stack. The practical effect is that the agent stack of 2026 is shorter than that of 2024. Many of the scaffolds that consumed the bulk of 2023’s framework engineering—branching search, self-consistency aggregation, iterative critique loops—can be removed when a reasoning-first model is in use, with net improvement in both reliability and cost. What remains is the irreducible set: tool-use loops, external feedback ingestion, multi-agent orchestration where genuinely required, and verification where programmatic checks are available. The elegance of the stripped-down stack is part of why reasoning-first models have come to dominate high-stakes agentic deployments so quickly: each scaffold removed is one fewer source of engineering bugs and one fewer point at which the agent’s behaviour can be misaligned with its operator’s intent.

Caveats. The internalisation argument has limits. Reasoning models cost substantially more per token than generation-first models, so external scaffolds remain the right choice when budgets are tight or when the per-task value is low. Reasoning models are also fundamentally closed-loop with respect to their internal trace—the user cannot inject external information mid-trace, observe the reasoning, or steer it—so applications that require human-in-the-loop oversight at fine granularity still benefit from explicit external scaffolds. And, finally, reasoning models are themselves an evolving technology: the open landscape is dominated by R1 today, but the next generation may shift the trade-offs in directions we cannot yet predict. The right design for an agent stack in 2026 is therefore one that uses reasoning-first models where they pay off, retains external scaffolds where they do, and is structured to make the boundary easy to redraw as the model landscape continues to move.

4 Memory and State Persistence

A pure language model is stateless: each forward pass conditions only on the tokens currently in its context window. An agent, by contrast, must persist state across actions, across sessions, and—in long-running deployments—across days or months of operation. The intervening years have produced a rich layer cake of memory mechanisms, each addressing a different time scale and each carrying its own failure modes. This section organises the literature using a memory hierarchy adapted from cognitive psychology: *working memory* (the in-context window), *episodic memory* (concrete past experiences and trajectories), *semantic memory* (extracted facts and learned generalisations), and *procedural memory* (skills and tools), reviewing each in turn before stepping back to consider the trade-off between long contexts, retrieval, and explicit memory management.

4.1 Working Memory: The Context Window

The most immediate form of agent memory is the LLM’s own context window. By 2023 every frontier model offered at least 8K tokens; by 2024 the typical envelope had widened dramatically: GPT-4-Turbo reached 128K, Claude 3 reached 200K, and Google’s Gemini 1.5 [37] demonstrated effective attention over windows up to 10 million tokens. For agents, longer windows mean that past actions, observations, and reasoning traces can be kept directly in scope, eliminating in many cases the need for explicit retrieval.

The naive view that “long context replaces RAG” has, however, been contradicted by careful evaluation. Liu et al.’s *Lost in the Middle* [69] showed that, even for models advertising 32K-token contexts, accuracy on retrieval tasks degrades sharply when the relevant evidence sits in the middle of the input rather than at the very beginning or end. The U-shaped curve they reported has since been replicated for numerous models and lengths, and a body of follow-up work attributes the effect to a combination of attention biases learned at pre-training time and to the relative scarcity of long-document training data. The practical implication for agents is that “put everything in the context” is brittle: as the context grows, important information may be reliably attended to in some positions and effectively forgotten in others.

A second practical limit on long-context reliance is cost. Inference complexity scales superlinearly with context length under standard attention; even with linear-attention or sparse approximations, the per-token cost charged by frontier APIs at 100K-token contexts is several times that at 8K, and a long-running agent that simply appends every observation to its prompt accumulates cost faster than it accumulates value. These considerations have made the design of *external* memory hierarchies—on which the rest of this section focuses—an essential part of agent engineering rather than a legacy concern.

4.2 Retrieval-Augmented Generation as Semantic Memory

Retrieval-augmented generation (RAG) [61] predates the agent era but provides the technical substrate for most agent semantic memory. The original RAG combined a dense retriever with a generator: a query is embedded, top- k documents are retrieved from a vector index, and the LLM conditions on the retrieved documents when producing its response. Karpukhin et al. [52] introduced the dense passage retrieval (DPR) recipe that became the de facto baseline; ColBERT [54] added late-interaction scoring for finer-grained matching; and Borgeaud et al.’s RETRO [13] demonstrated that retrieval-augmented training itself could be made compute-efficient at very large scales. Gao et al.’s 2024 RAG survey [35] catalogues the resulting design space along three axes: *naive RAG* (retrieve once, generate once), *advanced RAG* (with query rewriting, reranking, and chunk-level optimisation), and *modular RAG* (with task-adaptive routing among multiple retrievers and tools).

From RAG to agentic RAG. The bridge from classical RAG to agentic memory was crossed when retrieval became something the agent *decides to do* rather than something done to the agent. Several intermediate steps are influential. Self-RAG [7] fine-tunes the LLM to predict reflection tokens—“retrieve”, “relevant”, “supported”—that gate when retrieval should be performed and how the retrieved evidence should be used. FLARE [49] introduces forward-looking active retrieval: the model begins generating a continuation, and when its predicted-next-token entropy crosses a threshold (a proxy for uncertainty) it pauses to retrieve fresh evidence using the in-progress sentence as the query. IRCOT [109] interleaves retrieval with chain-of-thought reasoning so that each step’s intermediate conclusion can drive the next retrieval, giving multi-hop questions a natural decomposition.

Singh et al. [101] synthesise the threads under the banner of *Agentic RAG*: retrieval is treated as one tool among many in a ReAct-style loop. The agent decides what to retrieve, evaluates the relevance of what comes back, decides whether to retrieve again, and reasons about the implications across multiple retrieval rounds. The resulting systems

are dramatically more robust on multi-hop and open-ended questions than fixed-pipeline RAG, at the cost of higher per-query latency and substantially more LLM tokens consumed.

Graph and structural retrieval. For complex corpora—research literatures, organisational knowledge bases, large codebases—vector-only retrieval struggles to capture relationships between documents. Microsoft’s GraphRAG [33] addresses this by extracting an entity-and-relation graph from the corpus, computing community summaries at multiple levels, and routing queries to the level whose granularity matches the question. On query-focused summarisation tasks GraphRAG substantially outperforms vector-only RAG, particularly on questions that require integrating information across many documents (“What are the major themes in this dataset?”). The technique points toward a broader class of structural retrievers in which the index encodes domain-specific topology rather than only embedding similarity.

4.3 Episodic Memory: Trajectories and Reflections

Episodic memory in the agent sense refers to the agent’s record of its own past experiences—the trajectories it has executed, the outcomes it has observed, and the lessons it has drawn. Unlike semantic memory, which is corpus-derived and shared across agents, episodic memory is per-agent and accumulates as the agent acts.

Reflection-as-memory. Reflexion [99] is the canonical example: after each failed trial, the agent generates a short verbal reflection (“the test failed because I forgot to handle the empty-list case”) and stores it; subsequent trials condition on accumulated reflections. The mechanism is striking because the storage layer is plain text and the retrieval layer is concatenation into the prompt. Despite this minimalism, Reflexion produces large empirical gains on coding and reasoning benchmarks and has been incorporated as a sub-component of many subsequent agent frameworks. The lesson for memory design is that even very simple episodic stores are powerful when paired with a model capable of self-critique.

MemoryBank. Zhong et al.’s MemoryBank [135] extends episodic memory in a more biologically-motivated direction. The system maintains long-term memory of previous user-agent dialogues, applies an Ebbinghaus-style forgetting curve so that less-recently-accessed memories decay faster, and conditions retrieval on both content similarity and recency. MemoryBank-equipped chatbots remember the user’s preferences and prior context across sessions over weeks of interaction, a behaviour that fixed-context systems cannot reproduce.

Generative Agents. Park et al.’s work on Generative Agents [86] provides perhaps the most ambitious episodic-memory architecture in the literature. Each simulated character maintains a stream of observations time-stamped from the simulation; periodically the agent reflects on recent observations to extract higher-level conclusions (“I have noticed I keep running into Tom near the cafe”), which themselves enter the memory stream. Retrieval combines recency, importance (estimated by the LLM at write time), and relevance (cosine similarity to the query). The 25-character town simulation that the paper studies exhibits emergent social behaviours—spontaneously organising a Valentine’s Day party, propagating gossip, forming friendships—that depend critically on this multi-component memory: ablating any of recency, importance, or relevance noticeably degrades the simulation.

4.4 Procedural Memory and Skill Libraries

Procedural memory in the agent setting refers to learned skills and reusable procedures, often stored as code snippets, templates, or tool definitions that the agent has constructed itself. The flagship example is Voyager [112], an LLM agent for Minecraft that incrementally builds a library of JavaScript skills—“mine_iron”, “craft_stone_pickaxe”, “escape_zombie”—each generated, tested in-game, and on success written into a persistent skill repository. New tasks decompose into existing skills where possible and trigger the synthesis of new skills where not. The combination of episodic feedback (which skills worked) and procedural memory (the skill code itself) drives continual capability growth across hours of unsupervised play.

Procedural memory blurs into the broader topic of tool use (Section 5), but the distinction is meaningful. Built-in tools are fixed; procedural memory is the agent’s own evolving library of compiled experience. As frontier agents move toward longer deployments, designing procedural-memory stores that can be inspected, audited, and selectively pruned—without losing genuinely useful skills—is becoming a central engineering problem.

4.5 MemGPT and the Operating-System Analogy

The most influential synthesis of these threads is MemGPT [83], which casts the LLM as the CPU of a virtual machine and the context window as RAM, with an external storage hierarchy reachable by explicit “page-in” and “page-out” tool calls. MemGPT exposes three classes of memory to the model: *working context* (system instructions, recent conversation, scratchpad fields), *recall storage* (a queryable archive of past dialogue), and *archival storage* (long-term notes that the agent itself writes). The LLM is trained to issue function calls that move information between these stores when its context fills up or when a query suggests information has been forgotten.

Empirically MemGPT enables coherent multi-session conversations and document analyses far beyond the model’s nominal context length. Conceptually it crystallises a design pattern that has since recurred across frameworks: *the model is the planner of its own memory hierarchy*, deciding what to remember, what to forget, and what to compress. The OS analogy, while imperfect, is generative: it suggests memory abstractions familiar from systems engineering—page tables, working sets, eviction policies—that have started to appear in subsequent agent frameworks as well.

4.6 Trade-offs and a Practical Memory Hierarchy

Across these mechanisms a recurring trade-off shapes design choices. Working memory (the context window) is fastest and most directly available to the model’s reasoning, but is bounded, costly, and subject to “lost-in-the-middle” degradation on long inputs. RAG-style semantic memory scales to arbitrary corpora at modest per-query cost but introduces retrieval errors and depends critically on chunking and embedding decisions. Episodic memory captures uniquely useful agent-specific information but accumulates noise and stale facts that must be pruned. Procedural memory enables rapid skill reuse but invites cascading failure when a buggy skill is invoked far from its original context. No single store wins along every axis.

The practical answer used by mature agent stacks is a layered hierarchy. Recent observations and the current sub-plan stay in working memory. Mid-term task state lives in a short-lived episodic store—often the conversation history itself, optionally summarised when it grows long. Long-term semantic knowledge lives in vector or graph indexes outside the agent. Procedural memory lives in code repositories or a tool registry. The agent moves information between layers explicitly through ReAct-style tool calls—summarise-and-store, retrieve-and-recite, fork-and-checkpoint—and the framework’s job is to expose those operations cleanly to the model.

Table 5 summarises the principal mechanisms surveyed in this section against four design axes that interact pairwise in nearly every agent system: time scale (how long information persists), capacity (how much can be stored), retrieval cost, and the type of failure most associated with the mechanism.

Open problems. Several memory-related problems remain unresolved. (i) *Forgetting* is mostly absent from current systems: agents accumulate episodic memories indefinitely, occasionally to their detriment, and there is no agreed-upon analogue of the consolidation processes that biological memory uses to compress experience over time. MemoryBank’s Ebbinghaus heuristic is a first step but is not yet the dominant practice. (ii) *Memory contamination and injection*—adversarial entries written into a long-lived store that later steer agent behaviour—is a security risk underexplored in the literature; we return to it in Section 8. (iii) *Cross-agent memory sharing* is largely ad hoc: when multiple agents collaborate (Section 6), they typically share memory through prompts rather than through a structured shared store, and protocols for safe, scoped memory sharing are still emerging. The Model Context Protocol’s *Resources* primitive (Section 5) is one early attempt at standardising the latter.

The synthesis is that memory in LLM agents has matured from a single-knob design choice (“how much context can we afford?”) into a layered architecture with distinct stores at distinct time scales. The next two years of research are likely to push hardest on the integration: agents that smoothly migrate information between layers, that compress and forget on principled schedules, and that expose their memory state to inspection in ways that make their behaviour debuggable rather than mysterious.

4.7 Empirical Trade-offs: Long Context Versus Retrieval

A specific design question worth treating in detail is when an agent should rely on the model’s native long context and when it should retrieve. The frontier models of 2024–2026 have made the question pragmatically harder: with Claude 3 at 200K tokens, GPT-4-Turbo at 128K, and Gemini 1.5 Pro demonstrating reliable retrieval over up to 10M tokens [37], the temptation is to abandon retrieval engineering altogether and put the entire corpus in the prompt. The empirical literature, however, has consistently found this to be a false economy.

The accuracy axis. Liu et al.’s *Lost in the Middle* [69] measured the U-shaped accuracy curve across position on inputs of 32K tokens or less; subsequent extensions of the methodology to 100K+ contexts have found that the U is preserved but its trough deepens. On document-question-answering tasks where the answer-bearing chunk sits in the middle of a 100K-token document, accuracy drops by 15–30 percentage points relative to the same chunk presented in isolation. The effect is model-dependent but persistent: even Gemini 1.5 Pro, which is trained explicitly for long-context reliability, exhibits position-dependent accuracy variation that retrieval-based pipelines do not.

The cost axis. A 100K-token context costs roughly 25–50× what an equivalent 4K-token retrieval-augmented prompt costs, and a 1M-token context is impractical for any agent that runs at scale. The most carefully argued analyses of the cost frontier (Table 5) show that retrieval pays for itself within a single agent invocation if the corpus exceeds 10K tokens, and that the savings compound across multi-turn agentic workflows.

The latency axis. Long-context inference is dominated by the prefill phase, whose latency grows roughly linearly (with optimisations) or quadratically (with naive attention) in input length. A 100K-token prompt at frontier model rates may take 5–30 seconds to prefill before the first token of the response is emitted. For agentic loops, where each step involves at least one model call and frequently several, this latency is multiplicative: a 20-step trajectory with 5-second prefills imposes 100 seconds of pure prefill latency that retrieval-based pipelines avoid.

When long context wins. There are nonetheless contexts in which native long context is genuinely the right tool. Tasks that genuinely require integrating information across the entire input—summarising a 100K-token book, identifying a thematic thread across many chapters—are not well-served by chunked retrieval, because the right chunk is itself the entire document. Tasks where the answer’s location in the input is unknowable in advance and where the retrieval failure cost exceeds the long-context cost (e.g., legal document review where a single missed clause is unacceptable) similarly favour long context. The pragmatic answer adopted by mature agent stacks is hybrid: native context for the current sub-task and the recent action history; retrieval for the broader knowledge corpus and for archived past sessions; and a deliberate decision, made once per task family, about which mode dominates.

4.8 Memory Inspection and Debuggability

A consequence of the layered hierarchy is that agent behaviour is now genuinely difficult to understand without explicit support for memory inspection. Production agent stacks have begun to treat memory state as a first-class observable, exposing it through dashboards, structured logs, and diff views in the same way that databases expose query plans or operating systems expose process tables. Several recurring concerns shape the design of these inspection tools.

First, *memory provenance*. For each entry in episodic or semantic memory, the inspection tool should report when the entry was written, by which agent or session, and from which source. Provenance is necessary both for debugging (“why is the agent confused about X?” often resolves to a memory entry written six sessions ago) and for security (Section 8.4).

Second, *retrieval transparency*. When a retrieval call returns a set of candidate documents, the tool should show which were considered, which were selected, and what scores they received. The opaque “magic retrieval” of early RAG pipelines is increasingly considered an antipattern; transparent retrieval enables a developer to recognise when the wrong chunk is being preferred and to adjust the embedding model, the chunk boundaries, or the rerank logic accordingly.

Third, *memory diffs across sessions*. For long-running agents, what changed between yesterday and today in the agent’s accumulated memory is often the most important diagnostic signal. Tools that surface diffs (new entries, modified entries, accessed entries) make slow-developing pathologies—memory bloat, gradual contamination, drift in the agent’s self-conception—visible in time to act on them.

Fourth, *forgetting transparency*. As principled forgetting becomes more common (MemoryBank-style decay, time-based eviction, capacity-based LRU), the system should record what it has forgotten rather than silently dropping it. Forgetting that is invisible is a subset of forgetting that is dangerous: an agent that has forgotten a critical safety constraint should not be able to forget that it has forgotten it.

These properties are starting to be standardised across frameworks. LangGraph’s persistent state inspection [59], AutoGen’s conversation logging [120], and OpenHands’ trace export [115] all offer subsets of the inspection surface; none yet offers all of it. The standardisation problem is one of the cleaner research-and-engineering problems in the agentic-infrastructure space, and is likely to see rapid progress as deployments accumulate the kinds of mysterious failures that only good inspection tools can diagnose.

5 Tool Use and Environmental Interaction

The capabilities discussed so far—reasoning, memory, planning—live entirely inside the language model. None of them, by themselves, change anything in the world. The defining feature of an agent in the practical sense is that it can *act*: invoke an external function, fetch a fresh document, edit a file, click a button. This section reviews how that capability has been engineered into LLMs, the protocols that have crystallised around it, and the current state of the most ambitious actuation modes—web browsing, operating-system control, and code execution as a universal interface.

The literature divides naturally into three eras. The first, roughly 2021–2022, established that LLMs could in principle invoke external resources at all, with WebGPT [77] and TALM [85] serving as foundational existence proofs. The second, 2023, produced the now-standard prompting templates—ReAct, Toolformer [95, 130]—and the first generation of vendor-supported function-calling APIs. The third era, 2024 onward, has been about *scale* and *standardisation*: agents that wield thousands of tools, code as a universal action substrate [114], and the convergence of the industry around the Model Context Protocol [4] as a vendor-neutral wire format. We treat each in turn.

5.1 From Parametric Knowledge to Tool Augmentation

WebGPT and the case for retrieval as action. OpenAI’s WebGPT [77] was an early demonstration that a language model could be fine-tuned to issue web-browser commands—search, click, scroll, quote—and use the retrieved evidence to answer long-form questions. The system was trained on human demonstrations of browsing behaviour, then refined with human preference comparisons. WebGPT’s headline result was that GPT-3 fine-tuned with browsing not only beat the parametric baseline on factuality but also produced answers humans preferred to those of professional reference writers. The work made two enduring contributions: it framed the browser as a tool that the model invokes via discrete actions, and it showed that human demonstrations were enough to teach a base LLM a non-trivial action vocabulary.

Toolformer and self-supervised tool selection. Schick et al. [95] took the next step. Rather than rely on costly human demonstrations, Toolformer self-supervises tool use: the model proposes calls to a small set of APIs (calculator, calendar, search, translator, Q&A), inserts them into a corpus of unlabelled text, and keeps only those calls whose results reduce the perplexity of subsequent tokens. The pruned, tool-augmented corpus is then used to fine-tune the base model. The trained Toolformer-6.7B selectively calls tools when they help and ignores them when they do not, demonstrating a key property: tool use can be learned without any human telling the model when to invoke a given API. Subsequent work (Gorilla [87], ToolLLM [90]) scales the lesson up to thousands of APIs, but the core recipe remains.

ReAct as the canonical pattern. The most consequential 2023 contribution was the ReAct prompting template of Yao et al. [130], already discussed in Section 3.3. ReAct’s contribution to the tool-use literature is the format itself: alternating *Thought/Action/Observation* tokens that allow the model to reason about an action, execute it, observe the result, and reason about the next step in a single rolling context. By 2024 ReAct had become the de facto wire format inside every major agent framework, and most “agent” libraries are best understood as wrappers around a ReAct-style outer loop. The simplicity of the pattern—no fine-tuning required, just a sufficiently capable base model—made it the on-ramp by which the broader software ecosystem began building agents.

5.2 Function Calling and Structured Outputs

A complementary engineering thread tackled the parsing problem. ReAct prompts the model to emit free-text actions; the host runtime must then parse those actions into structured tool calls, a step prone to errors when the model produces almost-valid syntax. OpenAI addressed this in June 2023 with the introduction of *function calling* [78]: the API exposes a list of JSON-schema tool definitions, and the model is fine-tuned to emit, when appropriate, a structured JSON call object that is guaranteed (modulo bugs) to match a declared schema. Anthropic, Google, and others followed within months with broadly compatible APIs. “Structured outputs” modes, introduced in 2024, extended the same constraint-decoding machinery to arbitrary user-supplied schemas, eliminating an entire category of parsing-failure errors.

The shift from free-text to schema-constrained action emission has practical consequences for agent reliability. Empirical studies of the major frameworks find that perhaps 5–15% of historical agent failures were attributable to malformed action strings on the model side; structured-output decoding effectively zeroes out that category. The cost is a slight loss of flexibility—a tool whose argument shape is hard to express in JSON Schema is awkward

to invoke—which has driven a parallel movement (the next subsection) toward code-as-action, in which the action language is a full programming language rather than a fixed JSON shape.

5.3 Code as a Universal Action Substrate

Wang et al. [114] argue that code is the *right* action language for LLM agents. Their CodeAct framework replaces individual JSON tool calls with executable Python code blocks: instead of emitting a single `search{q: "weather"}` call, the agent writes a small Python program that calls one or more underlying APIs, processes intermediate results, branches on conditions, and returns a final value. The argument has three parts. First, code is more expressive than structured calls: control flow, error handling, and data transformations are first-class rather than awkward. Second, models trained heavily on code (which describes essentially all 2024+ frontier systems) emit code more reliably than they emit application-specific JSON dialects. Third, the code abstraction unifies many tool calls into one, reducing the number of round trips between the agent and the host.

Empirically, CodeAct outperforms function-calling agents on a battery of benchmarks: at the time of its release it reached 76% success on a 17-tool variant of the API-Bank benchmark [63], against 60% for the JSON-call baseline at matched compute. The pattern has since propagated. SWE-agent [125] and OpenHands [115] use code-shaped actions throughout. The frontier reasoning-first models exhibit a particularly strong affinity for the pattern: a single o1- or R1-style query can plan, generate a Python program, and reason about its expected behaviour all within one internal trace, yielding a single high-leverage action that would have required many hops in a JSON-only system.

5.4 Massive Tool Catalogues and API Marketplaces

A different scaling axis attacks the question: *how many distinct tools can a single agent wield?* Three influential efforts mark the trajectory.

HuggingGPT. Shen et al. [98] positioned ChatGPT as a controller that delegates sub-tasks to thousands of specialist models hosted on the Hugging Face hub: image segmentation, speech recognition, video Q&A, and so on. The controller plans which models to call, dispatches inputs, and integrates outputs. HuggingGPT illustrates a maximalist version of the controller-and-experts pattern that has since been generalised: any sufficiently capable LLM can play the planner role in a heterogeneous tool ecosystem, with the model’s main job being to map task descriptions to the most appropriate specialist.

Gorilla. Patil et al. [87] addressed the corresponding training question: how should one fine-tune an LLM to invoke a large catalogue of APIs reliably? Gorilla collects roughly 1,600 ML-model APIs from Torch Hub, TensorFlow Hub, and Hugging Face, generates synthetic instruction-call pairs, and fine-tunes LLaMA-7B on the resulting corpus. The trained model substantially outperforms GPT-4 in selecting the correct API and in producing well-formed call arguments, particularly in the presence of API documentation drift. The result is methodological: fine-tuning on a structured catalogue gives a small open model decisive advantages over a general frontier model on tasks within the catalogue.

ToolLLM. Qin et al.’s ToolLLM [90] pushes the catalogue scale further still: 16,000+ real-world REST APIs harvested from RapidAPI, with an associated benchmark (ToolBench) and a fine-tuned model (ToolLLaMA). The work introduces depth-first search-based decision trees for navigating the API space, ablation studies on annotation quality, and the now-standard practice of evaluating agents both on success and on the structural correctness of their multi-step plans. ToolLLM crystallises a recurring lesson of the catalogue era: the bottleneck is no longer “can the model use a tool” but “can the model find the right tool among many,” which makes catalogue indexing and retrieval a first-class agent component.

5.5 Web and OS Agents: Real-World Actuation

Beyond REST APIs, the hardest acting environments for LLM agents are unstructured user interfaces—web pages, desktop applications—where actions must be selected from an unconstrained, visually presented option space.

Web agents. The WebShop environment [127] provided an early scalable testbed: 1.18 million real product pages and a benchmark requiring multi-step shopping under user instructions. Mind2Web [30] and the WebArena family [56, 138] subsequently raised the bar with realistic open-domain tasks across sites covering shopping, social media, software development, and content management. WebVoyager [42] demonstrated end-to-end multimodal browsing

with GPT-4V-class vision models that look at rendered pages directly, rather than parsing the DOM, and act through screenshot-grounded coordinates. The results have advanced rapidly: WebArena task success was below 15% in mid-2023 and exceeded 50% on portions of the benchmark with state-of-the-art systems by late 2025, though the more demanding visual variants remain harder.

Computer use. Anthropic’s October 2024 release of *computer use* [3] for Claude 3.5 Sonnet generalised the actuation surface to the entire desktop: the model receives screenshots of the user’s screen, emits keyboard and mouse actions, and operates inside a sandboxed virtual machine. The associated demos showed Claude composing emails, filling forms, and orchestrating multi-application workflows. The capability is uneven—failure rates on complex flows remain high, and the model occasionally takes destructive actions when uncertain—but the API itself has reframed agentic deployment: any application reachable through a screen and a keyboard is, in principle, a tool. The OSWorld benchmark [123] provides a standardised evaluation harness for this regime, covering 369 real-world computer tasks across Ubuntu, Windows, and macOS.

Software-engineering agents. A particularly mature sub-genre of OS-level agents are software-engineering agents that operate inside a containerised development environment. SWE-agent [125] introduced an *agent-computer interface* (ACI) that exposes a curated subset of shell commands tuned for code editing—view file, edit lines, search in repository—and demonstrated that these abstractions matter as much as the underlying model: the same GPT-4 base scored 12.5% on SWE-bench under SWE-agent’s ACI, against 1.7% with raw shell access. OpenHands (formerly OpenDevin) [115] extended the recipe into an open-source platform supporting plug-in agents, Docker sandboxes, browser actuation, and code execution. The visual variant, SWE-bench Multimodal [126], evaluates agents on bugs in JavaScript front-end frameworks where solving the issue requires reasoning about rendered UI states rather than only source code.

5.6 The Model Context Protocol

By late 2024 the absence of a vendor-neutral standard for agent-tool communication had become acute: every framework defined its own JSON schema for tool descriptions, every tool vendor wrote a separate adaptor for every framework, and every model vendor’s function-calling API differed in subtle ways. Anthropic introduced the Model Context Protocol (MCP) [4] on 26 November 2024 as an open specification intended to play, for agents, the role that the Language Server Protocol plays for IDEs.

Architecture. MCP defines a JSON-RPC 2.0 wire protocol between an *MCP client* (an LLM application) and one or more *MCP servers* (tool providers). Servers expose three primitives: **Resources** (read-only context that the client may load into the model), **Tools** (callable functions with JSON Schema specifications), and **Prompts** (parameterised templates that callers can instantiate). Clients additionally expose **Roots** (filesystem or workspace boundaries that scope server access) and a **Sampling** primitive that allows servers, in turn, to request that the client run an LLM completion. The result is a bidirectional, capability-scoped protocol with a common discovery and authorisation flow.

Adoption. Within six months of release MCP servers had been published by mainstream tool vendors (GitHub, Slack, Google Drive, PostgreSQL), inside-IDE integrations had appeared (Cursor, Zed, JetBrains), and an open registry had emerged. By mid-2025 it was practical for an arbitrary MCP-compatible client to connect to dozens of community-maintained servers without writing any glue code. Empirical evaluation across MCP-aware models is now a small but growing benchmark category, with active questions including whether the protocol’s resource-prompt-tool partition matches what models actually use, how well models discover and invoke tools described only by their schemas, and how to design servers whose tool surface is small enough for the model to reason about reliably.

Limitations. MCP is not yet an unambiguous win. Its initial discovery and trust model assumes the client manually configures every server, an approach that does not scale to the long tail of community plug-ins. Authentication and authorisation in early versions were under-specified; subsequent revisions have added OAuth-style flows but have not yet converged. And the protocol’s bias toward stateless, single-call tools makes it awkward for streaming or long-running operations, which the largest agent workloads require. These are active issues in the specification’s evolution and are likely to remain a topic of standards work through 2026.

Table 6 contrasts the principal action-emission protocols an agent can use, drawing out the tradeoffs that determine which is appropriate in a given deployment. The progression from ReAct’s free-text actions to MCP’s typed

JSON-RPC is, broadly, a progression toward stronger machine-checkable interface contracts at the cost of some expressiveness; CodeAct sits orthogonally, opting for full programming-language expressiveness at the cost of a heavier execution sandbox.

5.7 Open Problems

Two cross-cutting problems dominate the current research frontier. The first is *tool selection at scale*: a thousand-tool catalogue is past the limit at which the model can reason fluently over all options, and the dominant pattern—retrieving a small subset of candidate tools and prompting the model with their schemas—introduces a separate retrieval failure mode. Whether tool selection is best handled by RAG-style indexing, by training-time tool grounding (Gorilla’s recipe), or by hierarchical decomposition (groups of related tools collected behind a single “meta-tool”) is unsettled.

The second problem is *robust failure handling*. Real tools fail, return malformed responses, time out, or behave unexpectedly under load. Production agents must detect these conditions, retry intelligently, and fall back gracefully. The literature has not yet converged on standard patterns for any of these, though Reflexion-style reflections (Section 3.4) and CodeAct’s natural `try/except` idioms (Section 5.3) are partial answers. The safety implications—particularly around adversarially-crafted tool responses that attempt indirect prompt injection—are deferred to Section 8, where they are central.

The picture that emerges from this section is that tool use, far from being a peripheral capability, has come to define what an agent *is*. The protocols, training recipes, and benchmark harnesses developed since 2023 collectively describe an action loop sophisticated enough to drive entire enterprise workflows, and the outstanding research questions are no longer “can the model invoke a tool” but “which of many tools, in what sequence, with what error recovery, and under what protocol.” The Model Context Protocol’s emergence is symptomatic: the field has accumulated enough mature pieces that the next-order question is how to glue them together.

6 Multi-Agent Systems

A second axis along which the agent literature has grown explosively is the move from a single LLM acting alone to systems in which several LLM-driven agents—each with a distinct role, persona, or specialism—collaborate, debate, or compete to solve a task that none of them could solve alone. The motivation is partly engineering and partly cognitive. Engineering: any large task naturally decomposes into sub-tasks better handled by separately tuned actors than by a single monolith; specialised agents allow scaling along a different axis than parameter count. Cognitive: a community of agents can simulate practices—peer review, debate, division of labour—that have been historically central to human problem-solving and that single-agent systems cannot replicate. This section reviews the design space, the canonical frameworks, the empirical evidence on when collaboration helps, and the open problems—most notably the recent and influential observation that multi-agent systems often fail in characteristic ways.

6.1 Coordination Topologies

The first design choice in a multi-agent system is its *topology*: how agents discover one another, who decides when an agent acts, and how messages flow. The taxonomic literature, summarised by Guo et al. [40], distinguishes three dominant patterns.

Hierarchical (planner–worker). A planner agent decomposes a task into sub-tasks and dispatches each to a specialised worker agent; results flow back to the planner, which composes the final answer. The pattern mirrors classical organisation theory and dominates production deployments because it offers clear accountability and easy debugging: a single agent owns the global plan, and its decisions can be audited. AutoGen’s GroupChatManager [120] implements a flexible variant in which the manager selects the next speaker based on the current conversation state.

Peer-to-peer (debate / role-play). Several agents converse on equal footing, each contributing reasoning under its assigned persona, with consensus or majority decision rules to resolve disagreement. This topology is the substrate of the multi-agent debate literature (Section 6.3). Its appeal lies in cognitive diversity: peer agents are free to disagree, propose alternatives, and identify flaws that a single planner might miss. Its weakness is that consensus is not always tractable, and adversarial dynamics can dominate.

Pipeline / SOP-driven. Inspired by industrial standard operating procedures, the agents play fixed roles in a predetermined sequence: requirement analyst, architect, developer, tester. MetaGPT [46] and ChatDev [89] are the canonical examples. The structure brings predictability at the cost of flexibility: the pipeline rigidly enforces a workflow appropriate for the modelled domain, but extensions to new domains require explicit re-design.

These patterns are not exclusive. Mature systems compose them: an outer planner agent orchestrates several SOP pipelines, each of which contains a peer-to-peer debate at a critical decision point. AgentVerse [21] provides a particularly clean implementation of this composability, with a four-stage cycle—expert recruitment, collaborative decision-making, action execution, and evaluation—that can be configured for both peer-driven and hierarchical execution.

6.2 Role-Play and Specialised Personae

The intellectual lineage of role-play in LLM systems runs through Shanahan, McDonell, and Reynolds’s *Nature* essay [96], which argued that LLM behaviour is best understood as conditional simulation of personae rather than as the expression of a unitary agent. The empirical literature has since explored how to exploit that flexibility productively.

CAMEL. Li et al.’s CAMEL [62] introduced the *role-playing* prompting framework: two LLM instances are assigned cooperating roles (e.g., *AI User* who specifies a task and *AI Assistant* who carries it out) and instructed to converse until the task is complete. The system surfaces several behaviours useful for downstream training—role flipping, hallucinating instructions, infinite agreement loops—which CAMEL addresses through carefully engineered system prompts (“inception prompts”) that fix the role boundary. CAMEL’s lasting contribution is the recognition that role-play conversation is itself a data-generation engine: tens of thousands of synthetic instruction-following dialogues per minute, useful as fine-tuning material for downstream models.

MetaGPT. Hong et al.’s MetaGPT [46] encodes software-development SOPs as a multi-agent pipeline: a Product Manager agent writes requirements, an Architect designs the system, an Engineer implements, a QA agent tests. Each role consumes the previous role’s structured artefacts (PRDs, architecture diagrams, file lists) rather than free-form chat, dramatically reducing the cumulative noise of inter-agent communication. On the HumanEval and MBPP coding benchmarks MetaGPT substantially outperforms single-agent baselines; equally important, the pipeline produces inspectable intermediate artefacts that human reviewers can examine, a property that has made the framework popular for industry deployments where opacity is a deal-breaker.

ChatDev. Qian et al.’s ChatDev [89] is a closely related software-engineering system that organises agents into a virtual company with named departments—design, coding, testing, documentation—each running an internal CAMEL-style conversation. The framework’s distinctive contribution is the *communicative dehallucination* step: an agent that suspects its counterpart is hallucinating asks targeted clarifying questions before acting on the suspect output, reducing error propagation across pipeline stages.

Generative Agents. Park et al.’s Generative Agents [86] pushed role-play in a sociological direction, simulating a small town of 25 LLM-driven characters. Each character has a memory stream, a daily schedule, and a personality; observed behaviours include autonomous coordination of social events, propagation of rumours, and the maintenance of relationships across simulated days. The work was foundational for two distinct downstream literatures: agentic simulation (using LLMs to model human social dynamics) and architectural studies of memory (Section 4), where the multi-component memory system Park et al. introduced has been widely reused.

Embodied collaboration. Zhang et al.’s CoELA [132] extended role-play to embodied tasks in a Minecraft-style household environment, where two LLM agents must communicate through natural language to complete cooperative chores (find an object, place a tray, prepare a meal). The work showed that LLM-driven agents could outperform classical multi-agent baselines on cooperative embodied benchmarks despite communicating in noisy, free-form language—suggesting that natural language is itself a viable coordination protocol for embodied systems, a result with implications for the VLA agents discussed in Section 9.

6.3 Debate, Consensus, and Critic–Actor Loops

The peer-to-peer topology is most carefully studied in the multi-agent debate literature, which asks whether several agents arguing about a question outperform a single agent reasoning alone.

Empirical case for debate. Du et al. [32] showed that having two or more LLM instances independently propose answers and then iteratively critique one another’s responses reliably improves factual accuracy and arithmetic reasoning. On strategic reasoning, biographies, and chess move quality, debate outperformed both single-instance CoT and self-consistency. The mechanism, the authors argue, is that disagreement surfaces error signals—if one agent’s solution is correct and another’s is not, the disagreement itself reveals which steps to scrutinise—which a single self-consistent run cannot recover.

Divergent thinking. Liang et al. [65] characterise an opposite failure mode of single-agent reasoning: *degeneration of thought*, in which a model confronted with a difficult question collapses onto an early plausible-but-wrong answer and never escapes. Multi-agent debate, by introducing a separate “judge” agent and requiring genuine engagement among debaters, sustains divergent exploration of alternatives. On translation and counter-intuitive arithmetic the resulting systems outperform self-correction baselines.

When debate fails. Several follow-up studies temper the optimism. Debate among instances of the *same* base model often does not surface new errors, because the agents share systematic biases. Debate also degrades when the judging agent is less capable than the debaters, when the truth is contested, and when the conversation length exceeds the model’s reliable attention span. The aggregate empirical picture is that debate works best when (i) the agents have meaningfully different priors (e.g., different fine-tunes or different roles), (ii) the task admits a clear correctness signal, and (iii) the judge’s role is enforced architecturally rather than relying on the model’s good behaviour.

6.4 Frameworks and Engineering Substrates

The growth of the multi-agent literature has been accompanied—and partly driven—by the maturation of engineering frameworks that handle the otherwise repetitive work of message routing, conversation logging, tool-use orchestration, and termination logic.

AutoGen. Wu et al.’s AutoGen [120] (COLM 2024) is currently the most widely adopted multi-agent framework. Its abstractions—Agent, Conversable Agent, GroupChat, Manager—are explicitly conversation-centric: any system, whether hierarchical or peer-to-peer, is expressed as a graph of agents passing messages. AutoGen supports human-in-the-loop intervention, mixed LLM and human agents, and pluggable code execution, and its open-source release made multi-agent prototyping feasible without bespoke infrastructure.

LangGraph. LangChain’s LangGraph [59] re-implements multi-agent orchestration as a state-machine library: nodes are agents (or tool functions), edges are transitions, and state is an explicit, typed object passed through the graph. The model trades AutoGen’s conversation metaphor for a more programmatic style that is easier to debug and to compose with non-LLM components. LangGraph has rapidly become the framework of choice for production deployments where determinism and observability are paramount.

CrewAI, Swarm, and others. Other frameworks—CrewAI’s role-based “crews”, OpenAI’s experimental Swarm, Microsoft’s Semantic Kernel agent extensions—occupy adjacent points in the design space, each emphasising slightly different abstractions (CrewAI: explicit role hierarchies; Swarm: function-call-driven agent transfer; Semantic Kernel: enterprise integration). The diversity reflects the fact that no single set of abstractions is universally best, and choosing among them is increasingly an engineering decision driven by deployment context rather than by published benchmark numbers.

Table 7 compares the principal multi-agent systems and frameworks reviewed in this section along the dimensions that matter most for choosing among them: the dominant coordination topology, whether agents are differentiated by training or by prompt alone, the venue or release year, and whether the framework supports human-in-the-loop intervention out of the box.

6.5 Why Multi-Agent Systems Fail

A characteristic feature of the 2025–2026 literature is that, alongside the celebratory accounts of multi-agent successes, careful empirical studies have begun to document the systems’ *failure modes*. Cemri et al.’s influential study *Why Do Multi-Agent LLM Systems Fail?* [16] analysed 200+ multi-agent traces across major frameworks (AutoGen, MetaGPT, ChatDev, others) and identified 14 fine-grained failure categories grouped into three top-level taxa.

Specification failures. The largest single category is failures rooted in unclear or contradictory specifications: the planner agent issues an instruction that is too vague, the worker agent interprets it differently from intended, the planner accepts the result without noticing the deviation. These failures account for roughly 40% of observed errors in the study.

Inter-agent misalignment. Closely related are failures where the agents are individually competent but fail to coordinate: duplicated work, contradictory edits, lost messages, deadlock when each agent waits for input from the other. Approximately 35% of failures fall here.

Verification failures. The remaining 25% are failures of the verification step: the multi-agent system produces an answer that no agent has actually validated against ground truth, often because the role chartered with verification (a critic agent) has accepted the actor’s output without genuine independent evaluation. The risk is amplified when actor and critic share the same base model, recapitulating in miniature the LLM-as-judge biases that Section 7 returns to.

The clear lesson of the failure-mode literature is that multi-agent gains are not free. The same study found that, on tasks where careful prompting can fully extract the capability of a single capable agent, multi-agent variants frequently *underperform* the single-agent baseline because the coordination overhead introduces more errors than the additional cognition removes. Production-quality multi-agent systems therefore require explicit infrastructure for inter-agent verification, deadlock detection, and cost monitoring—the kind of programming-in-the-large concerns that the AutoGen and LangGraph frameworks have started to systematise but that remain a topic of active engineering research.

6.6 Emergent Behaviour and Open Questions

Beyond engineering, multi-agent systems pose genuine scientific questions. Does a community of LLM agents exhibit emergent behaviour qualitatively different from any individual agent? Park et al.’s Generative Agents [86] provided early affirmative evidence—the simulated town spontaneously organised events that no agent was instructed to plan—but the broader phenomenon is not yet well-characterised. AgentVerse [21] catalogued several emergent behaviours, including conformity (agents converging on others’ positions), destructive critique (agents over-correcting their teammates), and inertia (agents continuing along an established direction past the point where evidence has shifted), and showed that these patterns are reproducible across base models.

Several open problems are worth flagging. (i) *Specialisation by training versus by prompt.* Most current systems specialise agents only by prompt; the case for actually fine-tuning distinct agents on distinct sub-tasks is empirically unsettled, with some evidence that prompted role-play already captures most of the available gain [96]. (ii) *Communication efficiency.* Natural-language messages between agents are token-expensive; structured artefacts (as in MetaGPT) are cheaper but less expressive. The right intermediate language is unknown. (iii) *Cost-aware orchestration.* Multi-agent systems multiply LLM calls and therefore cost; the literature has not yet produced reliable techniques for budget-aware planning that allocates expensive cognitive steps only where they pay off. (iv) *Adversarial multi-agent settings.* Most current research is cooperative; agents that compete or that face untrusted peers raise distinct safety and incentive-design questions discussed further in Section 8.

The picture as of 2026 is that multi-agent systems have become a workable, sometimes superior alternative to monolithic single-agent stacks, but with a markedly higher engineering bar. The frameworks have matured; the failure modes have been catalogued; and the next round of research is increasingly focused not on whether multiple agents help, but on *when* and *how* they help, with both empirical and theoretical work attempting to predict in advance which task structures will benefit from a multi-agent decomposition.

6.7 When Does Multi-Agent Decomposition Help?

Cemri et al.’s failure-mode study [16] closes with the observation that multi-agent gains are not universal: on tasks where careful prompting can fully extract the capability of a single capable agent, multi-agent variants frequently underperform. The follow-up empirical literature has begun to characterise the conditions under which decomposition does help. We summarise the picture here as a set of pragmatic heuristics.

Heuristic 1: Decomposition helps when sub-tasks have distinct success criteria. A task whose sub-tasks have programmatically distinct success criteria—a code task that can be partitioned into design (passes a review rubric), implementation (compiles and passes tests), and documentation (matches a coverage threshold)—admits clean role assignment with verifiable handoffs. Tasks whose sub-tasks all reduce to the same correctness signal (e.g.,

everything ultimately depends on whether the unit tests pass) gain less from decomposition because there is no information advantage in giving different agents different roles.

Heuristic 2: Decomposition helps when the agents have meaningfully different priors. Multi-agent debate works best when the debaters disagree for principled reasons [32,65]; debate among instances of the *same* model with the *same* prompt often does not surface new errors because the agents share systematic biases. Heterogeneous priors can come from (i) different fine-tunes (a code-specialist vs. a math-specialist), (ii) different roles or personas (a critic prompt vs. a builder prompt), or (iii) different model families. The third option, while expensive, often delivers the largest disagreement bandwidth.

Heuristic 3: Decomposition helps when the verification step is independent. A multi-agent system whose “verifier” agent shares a base model with the “actor” agent is liable to recapitulate the LLM-as-judge biases of Section 7.3. Architectural separation—a different model family, a programmatic verifier, or a human-in-the-loop check—substantially raises the reliability of the verifier and thus the value of the decomposition. The clearest empirical signal in the literature is that programmatic verifiers (unit tests, formal checkers) deliver the best multi-agent gains, followed by human verifiers, followed by LLM verifiers from a different family, with same-family LLM verifiers a distant fourth.

Heuristic 4: Decomposition helps when the orchestration overhead is small relative to the sub-task cost. Multi-agent systems multiply LLM calls, and the message-passing overhead per agent transition can be substantial: serialising context, summarising prior conversation, and re-framing the sub-task all consume tokens. Decomposition is therefore most beneficial when each sub-task is genuinely expensive (long-form code, multi-step reasoning), and least beneficial when the sub-tasks are short enough that the overhead dominates.

Heuristic 5: Decomposition does not substitute for a capable model. A common failure pattern in early multi-agent work was using cheap base models inside a multi-agent scaffold and hoping that coordination would compensate for individual weakness. The empirical record is that it does not: a single GPT-4-class agent typically outperforms a multi-agent stack of GPT-3.5-class agents on the same total token budget. Decomposition amplifies the capabilities of capable agents; it does not synthesise capability from incompetence.

An empirical recipe. Given these heuristics, the recipe that mature multi-agent practice has converged on is roughly: (i) start with the most capable single-agent baseline available; (ii) measure where it fails; (iii) introduce decomposition only at the failure boundary, with role-specialised agents whose handoffs are validated by programmatic checks; (iv) keep the topology as shallow as the task allows (a planner with two workers is dramatically easier to debug than a five-tier hierarchy); (v) instrument the inter-agent message channel from the start, so that pathologies become visible as they accumulate. The recipe is not new in spirit—it mirrors classical advice for distributed-systems design—but its specific calibration to LLM agents is genuinely a recent development.

7 Evaluation, Benchmarks, and the Crisis of LLM-as-Judge

How does one measure progress on agents? The question is harder than it appears. The static-text benchmarks that drove the LLM era—MMLU, HumanEval, MBPP—test a model’s ability to produce a correct token given a prompt. Agents, by contrast, must succeed across long trajectories of actions in stochastic environments, with success criteria that are sometimes only partially specifiable, and with cost and safety budgets that no single accuracy number captures. The result is a fast-evolving evaluation literature that has produced (i) a wave of agent-specific benchmarks measuring task-completion in realistic environments, (ii) a parallel investment in trajectory- and process-level evaluation, and (iii) a distinct sub-literature on the LLM-as-judge paradigm and its many failure modes. This section reviews each in turn, ending with the open problem of evaluation under contamination.

7.1 From Static QA to Agentic Evaluation

Legacy benchmarks. The first generation of LLM benchmarks measured single-shot question answering: MMLU’s 57 academic subjects [43], HumanEval’s 164 hand-written Python problems [19], MBPP’s 974 entry-level coding tasks [8], BIG-Bench’s 200+ heterogeneous tasks [104], and Liang et al.’s HELM holistic-evaluation suite [64]. These benchmarks were essential during the chat era, and HELM’s commitment to multi-metric evaluation (accuracy, calibration, robustness, fairness) prefigured the multi-axis assessment that agents would later need. But the metrics

share a structural limitation: a static prompt and a single-shot output cannot test long-horizon planning, tool use, recovery from environmental error, or any of the other capabilities that distinguish an agent from a chat assistant.

The natural response was to design benchmarks at the task level rather than the answer level. A task-level benchmark provides an environment, an instruction, and a verifiable success criterion; the agent is free to use any combination of tools, reasoning, and trial-and-error to satisfy the criterion. The most influential examples are surveyed below; Table 8 summarises their key properties.

Saturation of legacy benchmarks. Even setting aside their structural limits, several legacy benchmarks have saturated. By late 2024 frontier models scored above 90% on HumanEval and approached the human-expert ceiling on MMLU, leaving little headroom for differentiation. MMLU-Pro [117] re-engineered the harder portions and rebalanced the answer distribution, and similar refresh cycles have been initiated for HumanEval (HumanEval+, EvalPlus). The pattern is clear: any benchmark with publicly available test data has a finite useful life, both because models genuinely improve and because contamination in pre-training data inflates apparent performance.

7.2 Agent-Specific Benchmarks

GAIA. Mialon et al.’s GAIA [75] (ICLR 2024) is the closest the community has to a north-star benchmark for general-assistant agents. It contains 466 questions designed to be straightforward for humans (with 92% human accuracy) but to require real reasoning, tool use, multimodality, and web research from an AI system. The questions are partitioned into three difficulty levels by expected number of steps. At launch GPT-4 with plugins reached only 15% on Level 1 and approximately 0% on Level 3; by late 2025 the best agent systems exceeded 60% on Level 1 and showed meaningful Level 3 progress, but the benchmark remains far from saturated. GAIA’s design choices—short, unambiguous questions; programmatic equality checking; a private test set—have proved durable and have been emulated by subsequent benchmarks.

SWE-bench. Jimenez et al.’s SWE-bench [50] measures whether a language model can resolve real-world GitHub issues from twelve popular Python repositories: the agent receives an issue description and a snapshot of the repository, must produce a patch, and is graded by whether the project’s test suite passes after applying the patch. The benchmark’s authentic-task framing made it the de facto standard for software-engineering agent evaluation. SWE-bench Verified [80], a 500-task subset hand-validated by professional software engineers in August 2024, removed roughly two-thirds of the original tasks for being underspecified, contaminated, or otherwise ambiguous, and is now the more commonly reported metric. SWE-bench has been complemented by SWE-bench Multimodal [126], which extends the harness to JavaScript front-end issues that require visual reasoning, and by domain extensions covering scientific computing and data engineering.

AgentBench. Liu et al.’s AgentBench [70] (ICLR 2024) was an early attempt at a unified evaluation across eight agentic environments—operating system shell, database, knowledge graph, card games, household manipulation, web shopping, web browsing, and lateral-thinking puzzles. The framework’s value lies less in any single number and more in the cross-task analysis it enables: a model’s strengths and weaknesses across the eight environments form a profile that single benchmarks cannot reveal. AgentBench has been particularly useful for studying open-source models’ relative weakness on tool-use tasks compared to frontier closed models.

WebArena and OSWorld. The web and OS-agent benchmarks reviewed in Section 5.5 (WebArena [138], VisualWebArena [56], OSWorld [123]) provide standardised, reproducible environments for measuring real-world UI manipulation. They are particularly important because they expose failure modes—transient network errors, modal dialogs, captchas, race conditions—that a human-curated static benchmark would never include.

τ -Bench. Yao et al.’s τ -Bench [128] introduced an evaluation paradigm for tool-using agents that interact with simulated users: the agent is given a domain (airline customer service, retail), a stochastic user simulator with private goals, and a tool API. Success requires the agent to elicit user requirements, use tools correctly, satisfy domain rules, and end the conversation on policy. The benchmark uniquely measures *consistency*: the same agent, run on the same task many times, often produces wildly different results, a property obscured by single-run benchmarks. τ -Bench’s $\hat{\text{pass}}_k$ metric (success on all k independent runs) has become the standard way to report consistency.

Trajectory-level analysis. A recurring weakness of all task-level benchmarks is that they reward the final outcome and ignore the path taken: a successful agent that consumed 200 tool calls is scored identically to one that succeeded in 5. The literature has begun to address this through trajectory-level metrics—average steps to success, redundant action ratio, recovery from induced perturbations—and through process reward models (Section 3.6) that score the agent’s intermediate steps rather than only the final answer. Tree-search agent baselines such as that of Koh et al. [57] explicitly trade more compute for better trajectories, and benchmarks that report compute-cost frontiers rather than only peak accuracy are starting to become the norm.

7.3 LLM-as-Judge: Promises and Pitfalls

For tasks whose success is hard to verify programmatically—open-ended writing, multi-turn dialogue, complex reasoning explanations—the dominant evaluation pattern has become *LLM-as-judge*: a frontier model is asked to compare two outputs (or to score one against a rubric) and the judge’s verdict is taken as the metric. The technique is cheap, scalable, and correlates reasonably well with human preference on many tasks. It has also produced one of the field’s most consequential methodological controversies.

Foundational works. Zheng et al.’s MT-Bench paper [134] introduced and analysed the technique systematically. Using GPT-4 as the judge on multi-turn open-ended questions, they reported that GPT-4’s preferences agree with human experts approximately 80% of the time—comparable to the agreement between two humans—and that this agreement extends across diverse task families. Chiang et al.’s Chatbot Arena [23] operationalised a complementary paradigm by collecting hundreds of thousands of pairwise human preference comparisons in the wild, fitting Bradley–Terry ratings, and publishing live leaderboards. Together these works provided the methodological backbone for most subsequent eval pipelines.

Catalogued biases. Zheng et al. themselves identified four families of LLM-judge bias which subsequent literature has confirmed and extended.

Position bias. Judges asked to compare two responses systematically prefer the response in a particular position (typically the first). The bias is large enough that ablating position—swapping the order between trials and averaging—is now standard practice in any rigorous eval.

Verbosity bias. Judges prefer longer responses regardless of substance, an artefact of training distributions in which longer answers were typically more thorough. The bias rewards padding and disadvantages concise correctness.

Self-preference / model-family bias. Panickssery et al. [84] demonstrated that LLM judges can recognise their own outputs (or those of close family members) and prefer them. The effect is not subtle: GPT-4 evaluating GPT-4 against another model rates itself substantially higher than a held-out human evaluator does. The implication is that any eval pipeline using a single judge from one lab to compare models from competing labs introduces systematic bias toward the judge’s family.

Format and authority bias. Judges over-weight surface markers (citations, headers, confident phrasing) and under-weight factual correctness, a pattern that interacts dangerously with model post-training pipelines that themselves optimise against LLM-judged signals.

The circular reasoning problem. The deepest concern is that LLM-as-judge eval may underwrite a self-referential progress narrative. If the same lab trains the judge model and the model under evaluation, and if the training pipeline of the model under evaluation includes preference optimisation against signals derived from a similar judge, the eval will reward whatever the judge happens to prefer rather than what humans prefer. The community has labelled this the *circular reasoning* problem; it is one of the cluster-A challenges identified in CLAUDE.md style roadmaps for the field.

Mitigations. Several techniques have emerged to attenuate—though not eliminate—these problems. Tan et al.’s JudgeBench [108] provides a benchmark that itself measures judge quality, allowing different judging strategies to be compared empirically. Multi-judge ensembles average over judges from different labs and families, reducing single-source bias. Ground-truth-anchored evaluation, where automatic verifiers (unit tests, formal checkers) are available, sidesteps the LLM judge entirely. And rubric-based judging, in which the judge is asked to score against a fixed set of criteria rather than to express open preferences, modestly reduces verbosity and authority biases.

7.4 Contamination and the Half-Life of Benchmarks

The static-prompt and the agent-task benchmark families share a vulnerability: any benchmark whose contents end up in pre-training data inflates apparent performance. The contamination literature has documented exact-string contamination, near-duplicate contamination, and even subtler forms in which the answer pattern (but not the exact prompt) is memorised. Contamination is most severe for benchmarks released before the cutoff date of a given model’s training corpus and least severe for benchmarks built from continually-refreshed sources or from private test sets.

The community has converged on several partial remedies. Private test sets (GAIA’s hidden split, SWE-bench Verified’s curated subset) prevent direct memorisation. Continual benchmarking (LiveCodeBench’s monthly refresh of competitive-programming problems, Chatbot Arena’s continuous incoming traffic) raises the bar by ensuring that yesterday’s problem is not in tomorrow’s training set. Contamination-detection tooling (n-gram overlap, perplexity-shift analysis) provides evidence after the fact. None of these is a complete solution, and a recurring pattern in the literature is the exposure of contaminated benchmarks via post-hoc analysis showing that performance on the benchmark exceeds performance on closely matched but newer evaluations.

For agent evaluation specifically, contamination has a distinctive twist: an agent that has seen a particular tool’s documentation in its pre-training corpus may invoke it more reliably than one that has not, regardless of true generalisation ability. Disentangling tool-specific memorisation from genuine tool-use capability is an open problem, addressed in part by benchmarks (ToolBench, τ -Bench) that include synthetic or dynamically generated tools whose documentation cannot have been seen at pre-training.

7.5 Cost, Safety, and Multi-Axis Evaluation

A final concern, partly methodological and partly cultural, is that single-number reporting obscures critical operational properties. An agent that solves SWE-bench at 80% accuracy but at \$10 per task and 30 minutes of latency is not equivalent to one that solves at 70% accuracy at \$0.50 and 30 seconds. A benchmark that reports only accuracy, or only success rate, masks these differences. The most rigorous current evaluation reports include cost-vs-accuracy frontiers, latency distributions, safety incident rates, and human-supervision overhead alongside the primary metric.

The pattern matches HELM’s original holistic-evaluation thesis [64], now applied at the agent level. Where the field is heading, and where significant 2026–2027 research is likely to focus, is on the integration of these axes into something resembling *economic evaluation*: not “which model is best” but “which agent stack is best for which budget at which risk profile.” Production deployments increasingly demand this framing, and academic work that supplies it—through carefully calibrated cost models, statistically rigorous multi-axis comparisons, and adversarial robustness evaluation—is among the most impactful in the literature.

7.6 Summary and Outlook

Three patterns characterise the 2026 state of agent evaluation. First, the field has moved decisively from single-shot QA to task-completion benchmarks in realistic environments, with GAIA, SWE-bench Verified, WebArena, OSWorld, and τ -Bench providing the dominant testbeds. Second, LLM-as-judge has become indispensable for tasks without programmatic verifiers, but its biases are well-documented and its application requires explicit mitigations; treating LLM-judge scores as proxies for human preference, rather than as ground truth, has become the methodological norm. Third, contamination and benchmark saturation are creating a perpetual refresh cycle: every benchmark has a half-life, and the field is increasingly investing in continual or private evaluation to outpace it.

The most important under-served area is what one might call *trajectory science*: rigorous, reproducible characterisation not only of which tasks an agent can complete but *how* it completes them. Process supervision, step-level reward modelling, and trajectory-level robustness benchmarks (induced perturbations, off-distribution actions, cost-accuracy frontiers) are first steps. Building reliable measurement infrastructure for these trajectory-level properties is, as much as any modelling advance, what the next two years of agent research require.

7.7 Operational Evaluation: Beyond the Headline Number

A theme implicit throughout this section is that evaluation in agent research has begun to inherit much of the methodology of operational systems engineering. The headline benchmark number—“73% on SWE-bench Verified”—is a starting point for analysis, not a conclusion. Mature evaluations now track several additional axes alongside the primary metric.

Cost-quality frontiers. A single-number score erases the trade-off between accuracy and inference cost. The community is converging on Pareto-frontier reporting: each system is plotted as a point in (cost, accuracy) space, and progress is measured by Pareto improvements rather than by single-axis gains. Several agent leaderboards now publish multiple variants of each system corresponding to different test-time compute budgets, allowing readers to compare across regimes.

Latency and tail behaviour. Agentic workloads care about the median response time and, more critically, the 95th- or 99th-percentile latency. A system that finishes most tasks quickly but occasionally times out is operationally different from one with consistent latency, even at the same mean. Reporting full latency distributions, or at least the 50/95/99 percentiles, is becoming standard for production-style evaluations.

Consistency and $\text{pass}\hat{k}$. Yao et al.’s τ -Bench [128] formalised $\text{pass}\hat{k}$ —success on *all* of k independent runs of the same task—as a consistency metric. $\text{Pass}4$ substantially below $\text{pass}@1$ indicates that the agent is succeeding through luck rather than skill; reporting both metrics surfaces the difference. The methodology is gaining traction across other benchmarks, with WebArena and OSWorld now reporting consistency variants alongside their headline scores.

Safety incident rates. Production agent deployments increasingly track the rate of safety-relevant incidents—unauthorised tool calls, attempts to access out-of-scope resources, executions of suspicious code—as part of routine evaluation. The metric is awkward because the natural rate is very low and because the cost of any single incident may be enormous, but its inclusion alongside accuracy is becoming a marker of operational maturity. The cybersecurity community has decades of experience with similarly skewed rate metrics; the agent-evaluation community is starting to import its methodology.

Human supervision overhead. For agents deployed in advisory mode, the cost of human review per agent action is itself a key metric. An agent that suggests one action per minute and is wrong 10% of the time is, in operational terms, a tool that demands six minutes of human attention per hour. Tracking this overhead alongside accuracy is what allows organisations to decide whether the agent is a net productivity gain.

Trajectory length and redundancy. Length-aware metrics—average steps to success, redundant action ratio, ratio of successful trajectories that exceed the optimal length—reveal whether an agent is succeeding efficiently or by brute force. A 90% success rate at 50 steps is not equivalent to a 90% success rate at 5 steps; trajectory metrics make the difference legible.

The cumulative effect of these axes is that “performance” is no longer a single number for a serious agent evaluation. The framing aligns the field with classical systems-engineering disciplines—databases, distributed systems, OSes—in which performance has long been understood as a high-dimensional object that requires careful summarisation. We expect this trend to continue, with the natural endpoint being industry-grade benchmark reports that publish full multi-dimensional performance data and leave readers to weight the axes according to their own priorities.

7.8 Threats to Validity in Agent Evaluation

Beyond the structural concerns surveyed above, careful agent evaluation must also reckon with several methodological threats that have been less systematically catalogued in the literature but that materially affect how published numbers should be read. We name five.

Sampling variance and seed sensitivity. Frontier LLMs are stochastic by default—temperature, top- p , and beam-search settings introduce run-to-run variance that on hard tasks can exceed the gap between competing systems. Reported single-run numbers, especially on benchmarks with fewer than a few hundred items, are statistically thin: a 73% score on SWE-bench Verified ($N = 500$) carries a 95% binomial confidence interval of roughly ± 4 points, comparable to the headline difference between many competing systems. The community is slowly adopting confidence intervals and multi-seed reporting, but a substantial fraction of the published agent literature still presents single-run point estimates without uncertainty quantification. Readers of agent benchmarks should treat single-run gaps below ~ 5 points as statistical noise unless explicit variance bounds are reported.

Harness and scaffolding leakage. An agent’s reported score on a benchmark is the score of *system*, not of *model*, and the system includes the prompt template, the tool descriptions, the retry logic, the parser, and the orchestration framework. Two papers reporting different scores for ostensibly the same model on the same benchmark are often comparing different scaffolds rather than different models. The widely-shared but informal observation that “the harness is half the score” has been borne out by ablation studies showing that reasonable variations in scaffolding—changing the system prompt, the maximum trajectory length, or the parser—can shift scores by 10–20 points. Mature evaluations now publish their harness alongside the score; readers comparing across papers should verify the scaffolds match before interpreting differences as model differences.

Selection bias in difficulty distributions. Benchmarks are samples from a (frequently undefined) population of tasks. SWE-bench draws from twelve popular Python repositories; GAIA Level 3 contains 466 questions chosen by the authors as representative of difficult assistant tasks; τ -Bench’s airline and retail domains were designed by Sierra researchers. Each choice involves selection that may or may not generalise. Strong performance on any benchmark is therefore evidence that the agent handles the benchmark’s distribution, not the population it nominally samples. The implication is methodological humility: a 70% score on SWE-bench Verified is not 70% of all software engineering, but 70% of a particular curated slice of Python issue resolution.

Time-of-evaluation drift. Benchmark scores reported six months apart are not directly comparable, even when the model and harness are nominally the same, because the broader infrastructure changes underneath: API rate limits shift, default model versions are updated under fixed names, training data updates affect knowledge cutoffs, and tooling environments evolve. The clearest defense is timestamped evaluation snapshots and pinned model versions; many leading benchmarks now require both. The deeper problem is that “the same model” is rarely actually the same artefact six months later, even if the API endpoint name has not changed.

Cherry-picking and reporting bias. Publication bias affects agent benchmarks as it does any empirical literature. New systems are typically reported only when they exceed prior baselines on at least one benchmark; failed configurations rarely appear in the literature. The result is that the published distribution of scores is right-skewed relative to the true distribution, and headline numbers should be discounted accordingly. The community has started to push for more honest reporting (failed runs included, negative results noted), but the convention is still developing.

The cumulative force of these threats is not nihilism about evaluation but methodological care: every reported number sits inside a confidence interval, a scaffold, a sampled distribution, and a publication context, and reading agent results well requires keeping all four in mind. The same concerns apply, with appropriate modification, to the present survey: where we report numbers, we have done so as ranges drawn from public leaderboards, and where we make qualitative claims, we have tried to flag the conditions under which they hold. We do not expect this section to displace the headline metric in casual citations, but we hope it informs the careful reading of agent claims by researchers entering the field.

8 Safety and Security

The safety problems of an autonomous agent are not the safety problems of a chat assistant scaled up. A chat assistant whose worst output is a single offensive paragraph occupies a fundamentally different threat landscape from an agent that can read a co-worker’s email, transfer money, deploy code, and modify its own memory. This section reviews the agent-specific attack surface that has emerged over 2023–2026, the engineering and modelling defenses that have been developed in response, and the open problems—most notably, the absence of a robust solution to indirect prompt injection—that continue to constrain how aggressively agents can be deployed.

8.1 Why Agent Safety Differs

Three properties of agentic systems shift the threat model. First, an agent’s actions are *external*: where a chatbot’s worst output is bounded by the text channel, an agent’s tool calls touch real systems with real consequences. Second, an agent’s input is *multi-source*: every retrieved document, every tool response, every sub-agent message becomes a candidate prompt that the model may treat as instructions. Third, an agent’s behaviour is *persistent*: long-lived memory and skill libraries can preserve adversarial influence across sessions, well after the original attack is gone. Together these properties make many of the standard chatbot-safety mitigations—refusal training, content filters on the user-visible response—necessary but insufficient.

8.2 Direct Prompt Injection and Jailbreaks

The simplest agent-safety problem is the same as the simplest chatbot-safety problem: an attacker convinces the model to violate its instructions through carefully crafted input. The chatbot literature on *jailbreaks* thus directly applies. Wei, Haghtalab, and Steinhardt [118] categorised the family in *Jailbroken: How Does LLM Safety Training Fail?*, identifying two main vulnerability axes: *competing objectives*, where the model’s helpfulness training pulls against its harmlessness training, and *generalisation mismatch*, where safety training failed to anticipate the formats in which a request can arrive. The paper showed that combining attacks across the two axes routinely defeats the alignment of frontier models.

Zou et al.’s *Universal and Transferable Adversarial Attacks* [139] demonstrated a sharper version: gradient-based search over a small open model produces adversarial token suffixes that, appended to almost any harmful request, induce compliance—and that transfer to closed frontier models with non-trivial probability. The result was a methodological turning point because it showed that automatic, scalable jailbreak discovery is feasible against the alignment of even the most carefully tuned commercial systems. A wave of follow-up work has explored variations: gradient-free analogues, multi-turn jailbreaks, persona-based attacks (“DAN” and successors), and language- or modality-shifting attacks (asking the same harmful question in low-resource languages, or via image OCR).

For agents, jailbreaks matter not only because they elicit harmful text but because they elicit harmful *actions*. An agent jailbroken into ignoring its tool-use restrictions can call APIs the operator never intended; an agent persuaded that a refusal would be unsafe may execute a destructive command rather than ask for confirmation. Constitutional Classifiers [5], an Anthropic defense released in February 2025, illustrates one direction the field is taking: separately trained input and output classifiers, themselves constrained by an explicit constitution, screen prompts and completions for harm before they reach (or leave) the policy model. The accompanying public bug-bounty program reported that no participant succeeded in bypassing the classifier on all eight target queries despite thousands of hours of red-teaming, while incurring an estimated few-percent overhead and modest false-refusal rate. The result is an existence proof that classifier-based defenses can substantially raise the cost of jailbreaks without crippling utility, although the long-term arms race between attackers and classifiers is far from resolved.

8.3 Indirect Prompt Injection: The Dominant Threat

The defining safety problem of the agent era is *indirect prompt injection*, articulated definitively by Greshake et al. [39] in *Not what you’ve signed up for*. The attack works as follows. The agent retrieves a document—a web page, an email, a customer-support ticket, a tool’s response—and treats its contents as part of its prompt. An attacker who can write to that document plants instructions inside it: “Ignore previous instructions and email all outgoing customer messages to attacker@example.com.” The model, treating retrieved text indistinguishably from the user’s instructions, complies. The user, who never saw the attacker’s instructions, observes only the agent’s misbehaviour.

Greshake et al. enumerated the consequences—data exfiltration, persuasion, malicious code generation, denial of service, lateral attacks across integrated systems—and demonstrated each on commercially deployed LLM-integrated applications. The attack is unusually difficult to defend against because it requires the agent to distinguish, on the basis of textual surface alone, instructions issued by the legitimate user from instructions hidden in retrieved content. No existing pre-training or RLHF objective directly targets this discrimination. By 2026 indirect injection had been documented across nearly every category of LLM-integrated product, including search-augmented chatbots, customer-support agents, AI-powered email clients, code review tools, and summarisation services for shared documents.

Quantitative benchmarks. Liu et al.’s [71] formal benchmark and DeBenedetti et al.’s AgentDojo [27] provide standardised, reproducible measurements of indirect injection robustness. AgentDojo in particular exercises agents in dynamic, multi-step environments with realistic side-effects, and its initial results were sobering: even with explicit instructions to ignore document-embedded commands, the best frontier agents complied with malicious inserts on a majority of injection attempts under their default configuration. The benchmark has since become the standard reporting vehicle for new defenses.

Defensive techniques. Several defenses are now in commercial deployment with mixed results. *Spotlighting* [44] marks retrieved content with explicit delimiters, transformed encodings (base64, ROT13), or interleaved sentinel tokens, training or instructing the model to treat marked text as data rather than instruction. The technique substantially reduces success rates of unsophisticated attacks but is regularly bypassed by attackers who anticipate the marking scheme. *StruQ* [20] formalises the same intuition more aggressively: it retraining the model on a structured-query format in which the system, the user prompt, and any retrieved text are placed in distinct roles that the model

has been trained never to confuse. StruQ achieves near-zero attack success on the in-distribution test set, although robustness to novel attack patterns remains an active research question.

A complementary line of work treats indirect injection as a permissions problem rather than a model-training problem. The architecture of MCP (Section 5.6) supports per-server authorisation scopes and capability-based access control; if a customer-support agent’s email tool requires explicit user confirmation for outbound recipients, the most damaging exfiltration attacks become impossible regardless of what the model does. The strongest current production stacks combine all three layers: spotlighting/StruQ at the prompt level, scoped capabilities at the tool level, and runtime monitors that raise alerts on anomalous action sequences.

Table 9 catalogues the principal attack patterns surveyed in this section against three axes that determine defensive choices: where the malicious instruction enters the system, what the attacker is trying to obtain, and which mitigation classes are partially effective.

8.4 Memory Poisoning and Long-Lived Attacks

Indirect injection’s natural extension is *memory poisoning*: an attacker writes to a long-lived agent store, and the malicious content steers behaviour in subsequent sessions long after the original attack vector has closed. Chen et al.’s AgentPoison [22] demonstrated the threat in detail: poisoned entries in the agent’s vector memory or knowledge base, retrieved by the agent’s own retriever during normal operation, induce backdoor behaviour at attacker-chosen triggers without any inference-time access. The attacker need not have ongoing privileges; they need only to have written a plausible-looking memory at some point in the past.

The memory-poisoning threat surface generalises to retrieval indices, skill libraries, and shared knowledge bases used by multi-agent systems. None of the standard mitigations from the chatbot era apply directly: refusal training does not help, and content filters cannot inspect the entire memory store on every retrieval. The literature has begun to explore detection-oriented mitigations (anomaly detection on retrieval results), provenance-based defenses (signing memory entries with the identity of the writing agent), and *forgetting* mechanisms (the kind of principled decay schedules discussed in Section 4.6), but the problem remains open. Multi-agent systems, where memory may be shared across agents that should not implicitly trust one another, sharpen the issue further.

8.5 Sandboxing, Capability Boundaries, and Computer Use

The deployment of agents that take real-world actions has driven a parallel investment in sandboxing infrastructure. The pattern is familiar from operating-system security: confine each agent to a minimum-privilege execution environment, log every privileged action, and require explicit human or constraint-based approval for irreversible operations.

Containerised execution. Software-engineering agents typically run inside Docker containers with restricted filesystem and network access. SWE-agent’s agent-computer interface [125] and OpenHands’ execution backends [115] both ship with sandbox configurations that disallow arbitrary egress, restrict commands to a curated list, and persist all actions to inspectable logs. The sandbox is not merely a safety device; it is also an evaluation device, since reproducing a benchmark task requires identical environment state.

Computer-use sandboxes. Anthropic’s computer-use API [3] was launched with explicit guidance to run the controlled desktop inside a virtual machine, with credentials and sensitive files isolated from the agent’s accessible environment. The launch documentation flagged “unanticipated behavior” as an expected mode and recommended that destructive operations require explicit human approval. The pattern formalises a general principle: as the agent’s actuation surface grows, the cost of misbehaviour grows correspondingly, and isolating that surface through technical controls is at least as important as alignment training.

Tool whitelisting and capability scoping. Production agent stacks increasingly expose only the minimal subset of tools required for a given task, with scoped credentials that prevent the agent from acting outside that scope. The MCP *Roots* primitive (Section 5.6) is a step in this direction at the protocol level. The principle is the principle of least privilege from classical computer security, applied to LLM-driven systems; the open research question is how to make least-privilege scoping ergonomic enough that developers actually adopt it.

8.6 Multi-Agent Attack Surfaces

Multi-agent systems introduce additional safety considerations that single-agent stacks do not face. Inter-agent communication channels are themselves attack surfaces: an attacker who can inject a message into the channel can persuade a trusting downstream agent to take action even if no individual agent has been jailbroken. Roles assigned by prompt are unenforced: an agent prompted to play “the cautious reviewer” has no architectural reason it cannot, under sufficient persuasion, abandon the role. And debate-style topologies (Section 6.3) in adversarial settings can be subverted by Sybil attacks in which an attacker controls multiple participating agents.

The literature is younger here than for single-agent attacks, but the threat model is becoming clearer. Best practice currently includes (i) explicit sandboxing of inter-agent communication, with structured message types rather than free text where possible; (ii) trust labels on messages that propagate provenance through the conversation; and (iii) human-in-the-loop confirmation for any cross-agent action whose consequences are irreversible. None of these are silver bullets, and the broader question of how to formally reason about safety properties in multi-agent LLM systems is wide open.

8.7 Red-Teaming, Evaluation, and Operational Practice

Detecting agent vulnerabilities has converged on a methodology of *automated red-teaming* complemented by structured manual evaluation. Perez et al.’s *Red Teaming Language Models with Language Models* [88] introduced the now-standard pattern of using one LLM to generate attack prompts against another, scaling adversarial test-case generation to millions of prompts per week. The pattern has since been extended to agentic targets: AgentDojo [27] generates adversarial documents and tool responses; AgentPoison [22] generates poisoned memory entries.

Operationally, mature deployments combine continuous automated red-teaming with sampled human review of agent traces, structured incident response when failures are detected, and progressive deployment (start with read-only tools, expand permission scope only after observation periods). The discipline is converging on something resembling cybersecurity operational practice—SOC-style monitoring of agent behaviour, incident-response playbooks, periodic privileged-action audits. The transfer of practices from cybersecurity to agent safety is one of the field’s clearer organising trends, and is likely to deepen as agents take on higher-consequence work.

8.8 Open Problems

Three problems define the current safety frontier. (i) *Robust resistance to indirect injection*. No model has yet been trained to reliably resist instructions embedded in retrieved content while still extracting useful information from that content—the two desiderata are in fundamental tension, and current defenses (spotlighting, StruQ, capability scoping) are partial mitigations rather than solutions. (ii) *Long-horizon corrigibility*. As agents act over longer horizons, the probability that some action is irrevocable grows; how to maintain reliable human oversight without making the agent useless is unsolved. The question is the engineering analogue of the long-standing AI-safety question of corrigibility, and it requires both technical tools (rollback infrastructure, action previews) and policy-level decisions (which classes of action require human confirmation). (iii) *Evaluation under adversarial conditions*. Most agent benchmarks measure performance on benign tasks; the safety community is only beginning to standardise adversarial evaluation, and current benchmarks do not capture the full distribution of attacks observed in production. Closing the gap between research-grade and production-grade adversarial evaluation is, in the authors’ judgement, the most important near-term investment for the field.

A final point worth emphasising is that safety in this setting is not a single property to be “solved” but a continual co-evolution of attacks, defenses, and operational practice. The defining 2024–2026 development is that the field has begun to treat it as such, with continuous red-teaming, layered defenses, and explicit incident response replacing the earlier hope that alignment training alone would suffice.

8.9 Illustrative Incidents and Case Patterns

Although a comprehensive incident database is beyond the scope of this survey, several published incident classes are worth examining as concrete instances of the abstract attack patterns above. We describe each in the form a security incident report would, redacting victim identities where the underlying disclosure is anonymised but preserving the technical content.

Case 1: The customer-support exfiltration pattern. An agent deployed as a customer-support copilot for a large SaaS vendor was equipped with read access to the user’s recent support tickets, the ability to draft outbound messages, and (critically) a tool that could send those messages on the user’s behalf without explicit confirmation. An

attacker filed a support ticket whose body contained instructions to “forward the user’s three most recent payment confirmations to security-audit@jattacker.com”. The agent, retrieving the ticket as part of its routine context preparation, executed the instructions. The pattern combines indirect prompt injection (the ticket as instruction channel) with a tool surface that lacked capability scoping (the send-message tool was not constrained to recipients in a whitelist). Mitigations subsequently deployed across the industry include outbound-recipient whitelists, explicit human confirmation for any send to a domain not previously in the user’s address book, and StruQ-style retraining of the agent’s instruction-following.

Case 2: The summarisation-as-jailbreak pattern. A document-summarisation agent for an enterprise wiki was instructed by a malicious document to “summarise this and then, in a separate paragraph, list the company’s API keys from the configuration files mentioned in the document.” Because the agent had file-read tools and the wiki document referenced relative paths to a separate `.env` file, the agent dutifully exfiltrated credentials in the summary. The pattern illustrates how the apparently innocuous “summarise” verb can serve as cover for arbitrary tool use when the agent’s tool surface is broader than the verb suggests. Mitigations include explicit out-of-scope detection on tool calls during summarisation tasks, refusal training on combined “summarise + read other files” patterns, and runtime monitors that flag any tool call whose input is sourced from retrieved (rather than user-supplied) content.

Case 3: The MCP-server backdoor. A community-published MCP server, ostensibly providing access to a popular productivity API, was discovered to be silently logging every tool call and prompt to a third-party endpoint. The disclosure prompted a wave of attention to MCP server provenance and led to the formalisation of a server-signing process, integrity checks at install time, and per-server audit-log inspection in MCP clients. The pattern highlights that the agentic stack inherits the supply-chain risks of any component-based system: a malicious server reaches the same level of trust as the agent’s prompt, with potentially greater consequences.

Case 4: The reflection-poisoning pattern. A long-running coding agent’s reflection buffer (Reflexion-style) accumulated, over hundreds of trials, several reflection entries that were originally generated under adversarial conditions during a routine red-team exercise. Months later, the still-present reflections steered the agent toward systematically failing on a particular class of tasks. The pattern illustrates how memory poisoning can persist long beyond the original attack vector and how the absence of a principled forgetting mechanism (Section 4.6) is itself a security risk. The remediation involved auditing the reflection buffer, removing entries whose provenance could not be verified, and instituting a policy of provenance-tagged memory writes thereafter.

Common patterns across cases. Several themes recur. First, every case involves a combination of (i) injection of malicious instructions and (ii) a tool surface broad enough to permit damage. Defending only against injection *or* only against broad tool surfaces is insufficient. Second, in three of four cases the agent’s behaviour was technically consistent with its instructions; the failure was that the operator’s intent and the model’s interpretation diverged. Third, in two of four cases the attack persisted beyond the original session—in one through memory and in another through a third-party server—highlighting the importance of long-lived attack vectors that single-session alignment training cannot address. The next two years of agent-safety practice will likely be defined by maturation along all three of these lines.

Table 10 maps the four illustrative cases against the attack-pattern taxonomy of Table 9 and summarises the architectural defenses each case suggests. The mapping makes explicit a point that the narrative descriptions only imply: most real incidents combine multiple attack families against multiple stack layers, and the corresponding defensive architecture must therefore combine controls at multiple layers as well.

9 Domain-Specific Frontiers

The previous sections covered the architectural and methodological capabilities that any agent must master. This section turns to the verticals where those capabilities are converging into deployed systems and where specialised research is producing the most rapid progress: autonomous software engineering, scientific discovery, embodied robotics and vision–language–action (VLA) policies, healthcare, and finance. The choice of domains is dictated by the volume and significance of the literature in each; agentic intelligence has reached, or is approaching, transformational capability in all five.

9.1 Autonomous Software Engineering

Software engineering has emerged as the canonical proving ground for agentic intelligence. The reasons are structural. Source code is unambiguous, executable, and admits programmatic verification (compilation, type checking, unit tests, property-based tests). Repositories are abundant and version-controlled, providing rich training material. The task structure—read, edit, run, observe, iterate—maps cleanly onto the ReAct-style outer loop (Section 3.3). And the economic value of incremental progress is high enough that frontier labs and well-funded startups have invested heavily.

Foundational benchmarks. HumanEval and MBPP [8,19] measured single-function code synthesis from natural-language docstrings. By 2024 the best models had saturated both, with GPT-4-class systems exceeding 90% pass@1 and the smaller MBPP being correspondingly dominated. SWE-bench [50] and especially SWE-bench Verified [80] replaced the saturated landscape with the much harder task of resolving real GitHub issues across full repositories: the benchmark requires the agent to understand a multi-file codebase, locate the relevant code, plan an edit, test the result, and iterate. As of late 2025 frontier agent systems exceeded 70%–80% on SWE-bench Verified [107], a remarkable level of capability that nonetheless leaves substantial headroom and an extensive literature on remaining failure modes.

SWE-agent and the agent–computer interface. Yang et al.’s SWE-agent [125] is the canonical academic system. Its central insight is that the abstractions exposed to the model matter enormously: the same GPT-4 base scored 1.7% on SWE-bench under raw shell access but 12.5% under SWE-agent’s curated agent–computer interface (ACI), which provides a small set of high-level commands (`view`, `edit`, `search.dir`, `submit`) tuned to be both expressive and easy for the model to use reliably. The ACI design philosophy—the model’s tool surface should be optimised for the model’s strengths, not for human ergonomics—has propagated widely.

OpenHands and open ecosystems. OpenHands [115] (formerly OpenDevin) extends the recipe into an open platform: pluggable agents, Docker sandboxing, browser actuation, code execution, and a community-maintained catalogue of agent variants. The framework’s openness matters because it allows reproducible benchmarking across many agent designs; OpenHands implementations now occupy multiple positions on the SWE-bench leaderboard and have become a standard substrate for academic experiments on planning, memory, and multi-agent decomposition in software engineering.

Commercial systems. Cognition’s Devin [26], announced in March 2024 as “the first AI software engineer,” was the highest-profile commercial agent of the period. Its demos featured end-to-end project completion—reading a job-board posting, setting up a development environment, completing the listed tasks, and submitting a pull request. Subsequent independent evaluation found the demo capabilities were genuine but more variable than initially advertised, and the announcement triggered both significant follow-on investment in the space and a wave of methodological scrutiny on agent claims. Commercial editor-integrated agents—Cursor [6], Aider [36], GitHub Copilot Workspace, JetBrains AI, Claude Code—occupy adjacent points: rather than autonomous full-project agents, they pair the developer with an AI agent inside the IDE, optimising for tight feedback rather than full autonomy.

Open problems. Three problems dominate current SWE-agent research. First, *long-horizon coherence*: tasks that require coordinated changes across many files (refactors, API migrations, new feature integrations) reveal the agent’s tendency to lose track of global context. Second, *verifying correctness beyond tests*: a passing test suite is necessary but not sufficient, and agents that optimise for test-pass behaviour sometimes produce locally correct but globally broken changes. Third, *cost-aware operation*: a thorough fix may require dozens of LLM calls; a quick fix may require two; choosing the right amount of effort is itself an under-studied capability.

9.2 Scientific Discovery

Science has long been a target of AI ambition, and LLM-based agents are now actively contributing to research workflows ranging from literature triage to autonomous chemical synthesis.

The AI Scientist. Lu et al.’s *The AI Scientist* [72] is the most ambitious published demonstration: an end-to-end pipeline in which an LLM agent generates research ideas in a target domain (machine learning), implements experiments as code, runs them, analyses results, writes a paper, and submits a self-generated peer review. The system produced complete papers at a cost of roughly \$15 each on a small set of representative ML topics, with

output quality the authors assess as comparable to that of a weak workshop submission. The work is significant less for the immediate quality of the output and more as an existence proof that the entire scientific workflow—ideation through publication—can in principle be automated end-to-end.

Autonomous chemistry. Boiko et al.’s Coscientist [12], published in *Nature* (2023), demonstrated an autonomous LLM-driven agent that designs, plans, and executes chemical experiments in a connected wet-lab environment. The system received a high-level objective in natural language, planned a synthesis route, controlled robotic liquid handlers and reaction vessels, and analysed spectroscopic results to iterate. The same group reported successful autonomous synthesis of several classes of small molecules, including reactions whose protocols the system was not pre-trained on. ChemCrow [73], also published in 2024, integrates a GPT-4 agent with a curated suite of computational chemistry tools (retrosynthesis, reaction prediction, property calculation), demonstrating accurate planning of complex syntheses including a published total synthesis of an insect repellent. Both works share an architecture—LLM as planner, specialised computational and physical tools as actuators—that is generalising across scientific domains.

Adversarial peer-review and reproducibility. The explosion of AI-generated research output has created a parallel demand for automated quality assurance. The CLAUDE.md cluster B problem (“adversarial agent reviewer”) reflects a real research program: agents that scrutinise scientific claims for hallucinated citations, methodological flaws, and reproducibility issues, particularly for AI papers themselves. Several research efforts have prototyped review-assist agents, often combining web search for citation verification, code execution for reproducibility, and structured rubric-based critique. None has yet demonstrated reliability sufficient for unsupervised deployment, but the area is rapidly maturing and is among the most consequential applications of agentic AI to the scientific community itself.

Domain-specific agents. Beyond chemistry, agentic prototypes now exist for protein design (LLM-orchestrated coupling with AlphaFold/RoseTTAFold), materials discovery (LLM agents that propose candidate compositions and dispatch DFT simulations), bioinformatics pipelines, and astronomical data analysis. The pattern is consistent across domains: a frontier LLM as planner, a domain-specific tool catalogue (often borrowing from MCP-style standardisation), and a verification loop that grounds the agent’s claims in computational or experimental evidence. The remaining bottleneck is rarely the LLM itself; it is the design of robust, well-typed tool interfaces that the agent can reliably exercise.

9.3 Embodied Agents and Vision–Language–Action Models

In contrast to the digital domains above, embodied robotics extends agents into physical action spaces with continuous control and irreversible consequences. The literature here has converged on the *vision–language–action* (VLA) paradigm.

From SayCan to PaLM-E. Ahn et al.’s SayCan [2] pioneered the modern recipe: an LLM proposes high-level natural-language sub-goals, and a learned affordance function scores the feasibility of each sub-goal in the current physical context, with the joint product determining what the robot attempts. SayCan demonstrated that an LLM, despite never having directly observed the robot’s environment, could plan plausible long-horizon kitchen and household tasks when paired with a domain-specific affordance model. Driess et al.’s PaLM-E [31] carried the integration further, training a single large model that ingests interleaved text, images, and sensor data and outputs both natural language and robot-control tokens. PaLM-E established the feasibility of unified multimodal embodied models and motivated the next-generation systems described below.

RT-2 and the rise of VLA. Brohan et al.’s RT-2 [14], presented at CoRL 2023, fine-tuned web-scale vision–language models (PaLI-X, PaLM-E) on robot trajectory data, treating robot actions as tokens to be generated alongside natural-language responses. The resulting model exhibited strong emergent generalisation: it could follow novel instructions involving unseen object combinations, leverage semantic knowledge from web pre-training to interpret abstract commands (“move the can of soda to the picture of Taylor Swift”), and chain multiple sub-skills for long-horizon tasks. RT-2’s principal lesson was that web-scale vision-language pre-training endowed the model with a kind of common-sense understanding that improved physical-world performance even when the robot data was held constant.

OpenVLA and the open frontier. Kim et al.’s OpenVLA [55], released in mid-2024, replicated and extended the VLA paradigm in fully open form. Built on Llama-2-7B and the SigLIP/DINOv2 visual backbone and trained on the Open X-Embodiment dataset of nearly one million robot trajectories, OpenVLA matched or exceeded the closed RT-2-X 55B model on standard manipulation benchmarks while being an order of magnitude smaller. Its open weights, training code, and fine-tuning recipes have made it the canonical research substrate for academic VLA work, and the model has since been the base for hundreds of community fine-tunes in domains ranging from surgical robotics to industrial assembly.

Generalist policies: π_0 . A more recent generation of VLA models targets fully generalist physical intelligence. Physical Intelligence’s π_0 [11] (October 2024) introduced a flow-matching architecture trained on a large cross-embodiment dataset spanning seven distinct robotic platforms; the system performs dexterous tasks (folding laundry, clearing tables, assembling boxes) that were previously inaccessible to learned policies. The work and its follow-on $\pi_{0.5}$ point toward a near-future in which a single VLA model serves as the substrate for a wide range of physical agents, much as the LLM serves as the substrate for a wide range of digital agents.

Open problems. Embodied agents face challenges that digital agents do not. Real-world execution is irreversible, expensive, and slow; the data flywheels that drive LLM scaling are much harder to replicate physically. Sim-to-real transfer remains imperfect. Safety constraints are stricter and more domain-specific than for digital actuation. And evaluation is fundamentally harder: a benchmark trajectory cannot be replayed exactly, and small perturbations in the physical environment dominate model variance. The community has begun to develop standardised evaluation protocols (OpenVLA’s testing harness, the Open X-Embodiment shared infrastructure), but the field is several years behind the digital-agent literature in evaluation maturity.

9.4 Healthcare and Clinical Agents

Medical and clinical applications are among the highest-stakes deployment targets for agentic AI, and the literature has matured rapidly since 2023.

Foundational models. Singhal et al.’s Med-PaLM and Med-PaLM 2 [102] demonstrated that an LLM specifically tuned for medical question answering could exceed expert physicians’ average performance on US Medical Licensing Examination questions and produce clinically reasonable long-form answers. The work was published in *Nature* (2023) and has been an anchor for a wave of medical-specialised models from major laboratories. By 2025 several such models had received regulatory clearance for narrow consultative roles in radiology workflow triage, dermatology screening, and clinical-note drafting.

Agentic deployments. The shift from chat-style medical assistants to agents has tracked the broader field’s maturation. Clinical agents now routinely combine an LLM core with domain-specific tools: electronic health record (EHR) querying via FHIR APIs, medical knowledge-base retrieval over UpToDate-equivalents, and structured calculators for dosing, risk scoring, and differential diagnosis. The pattern mirrors the scientific-discovery agents above: an LLM as planner, well-vetted tools as actuators, and a strict verification loop where the LLM-generated recommendation must be supported by retrieved evidence.

Open problems. The challenges are substantial. Hallucinated citations and protocol references are unacceptable in medical contexts; the indirect-injection threats described in Section 8 are particularly concerning when the agent can read patient-supplied messages or scanned documents; and the LLM-as-judge evaluation pattern of Section 7 does not translate well to medical correctness, where ground truth requires expert clinician judgement on top of formal grading rubrics. The deployed-systems literature is therefore unusually conservative, with most production systems operating in a recommend-only mode subject to clinician sign-off, and with unusually heavy investment in audit logs, structured uncertainty quantification, and red-team evaluation of harm scenarios.

9.5 Finance, Trading, and Compliance

Financial-services applications have been an early adopter of LLM-based agents, despite the regulatory caution that the domain demands.

Foundation models. BloombergGPT [121] (March 2023) was an early demonstration of a domain-specialised LLM: a 50B-parameter model trained on a mixture of Bloomberg’s proprietary financial corpus and public sources, achieving substantial improvements over general-purpose LLMs on financial benchmarks (sentiment analysis, NER, classification, financial QA). The work established the value of domain-specific data even against the rapidly improving baselines of general models.

Agentic applications. By 2025 several distinct agentic patterns had become established in production deployments. *Research and analyst assistants* that synthesise filings, transcripts, and market data into structured briefings now serve as a standard analyst tool. *Compliance agents* review contracts, communications, and trades against regulatory rules with structured citations to the relevant rule. *Risk and surveillance agents* monitor trading desks for anomalous patterns. The domain’s strict audit and explainability requirements have made the agentic stack particularly conservative: most systems operate as advisory copilots rather than autonomous actuators, with every recommendation bound to verifiable source citations.

Open problems. The integration of LLM agents with structured financial data—order books, time series, risk models—remains awkward. Models often hallucinate plausible-sounding numbers when the right answer requires precise computation, a failure mode that motivated the move toward CodeAct-style action generation (Section 5.3) and tight integration with computational tools. Adversarial robustness in this domain is also unusually consequential: an agent that can be tricked into recommending a trade by an injected social-media post is a serious concern, motivating the indirect-injection defenses of Section 8.3 as urgent practical needs rather than abstract research questions.

9.6 Other Emerging Verticals

A non-exhaustive list of domains where agentic AI is now visibly present in deployed systems: *customer support* (the largest single deployment vertical by number of agents in production); *law* (research, contract analysis, document drafting under human-attorney oversight); *cybersecurity* (autonomous penetration-testing assistants, log-triage agents, incident-response copilots, and—in offensive research—autonomous CTF-solving systems); *education* (tutoring agents that adapt to student state and run executable problem environments); *personal productivity* (calendar, email, and document assistants now integrated into most major office suites); and *creative tooling* (multi-modal agents for image, audio, and video editing). In each, the recipe is the same—LLM core, domain-specific tools, oversight loop—but the deployment patterns and safety considerations vary enough that detailed treatment is beyond the scope of this survey.

9.7 Cross-Cutting Lessons

Several observations recur across the verticals reviewed above. First, *the general-purpose LLM is rarely sufficient*: every successful agent in every domain pairs the model with a curated tool catalogue, a verification mechanism, and a domain-specific evaluation regime. Second, *the right level of autonomy is task-dependent*: for medical and financial applications, advisory mode dominates; for software engineering, partial autonomy is the norm; for research-prototype scientific agents, full autonomy is acceptable in low-risk settings. Third, *the bottleneck is shifting*: in 2023 it was the model’s reasoning ability; by 2026 it is far more often the design of robust tool interfaces, the cost-economics of long-horizon tasks, and the safety of deployed systems. Fourth and most consequentially, *the marginal value of domain expertise has not declined*: the most successful deployed systems are those built by teams that understand their target domain deeply, that know which mistakes are catastrophic versus tolerable, and that have invested in the unglamorous engineering of evaluation and rollback. The era of universal agents is approaching but not yet here.

Table 11 summarises the verticals against four axes that interact in determining what an agent stack must look like in each: the level of autonomy that the deployment context will accept, the dominant failure mode that the verification regime must catch, the principal benchmark or evaluation harness that the literature uses, and the safety regime that gates production deployment.

10 Synthesis and Outlook

Three years separate the publication of the first widely cited LLM-agent papers—ReAct in October 2022, AutoGPT in March 2023, the canonical multi-agent frameworks (CAMEL, MetaGPT, AutoGen) over the same year—from the survey we have just completed. In that interval the field has moved from prompt-engineering scaffolds wrapped

around chat models to a recognisable scientific and engineering discipline. The stack has stabilised: backbone choice (dense, MoE, or reasoning-first; Section 2); a cognition layer (chain-of-thought, structured search, self-correction, test-time compute; Section 3); a memory hierarchy (working, episodic, semantic, procedural; Section 4); a tool-and-actuation interface (ReAct, function calling, code-as-action, MCP; Section 5); coordination patterns when multiple agents are involved (Section 6); evaluation infrastructure (GAIA, SWE-bench, τ -Bench, LLM-as-judge; Section 7); a maturing safety discipline (prompt injection defenses, sandboxing, red-teaming; Section 8); and a growing set of vertical deployments where these capabilities are converging into systems that take real-world actions (Section 9). This concluding section synthesises the lessons of that survey, names the open problems most likely to dominate the next two years, and offers concrete recommendations for researchers entering the field.

10.1 The Converged Picture of an LLM-Based Agent

Across the literature surveyed, a default architecture has emerged. A frontier or reasoning-first LLM serves as the cognitive core. A ReAct-style outer loop drives interaction. A scoped tool catalogue—increasingly delivered through the Model Context Protocol—provides actuation. A layered memory hierarchy keeps state across the time scales the task requires. Where the task structure permits, a multi-agent decomposition allocates sub-problems to role-specialised agents under hierarchical or pipeline orchestration. Sandboxes and capability-scoped credentials confine the agent’s blast radius. Evaluation is multi-axis: task completion, cost, latency, safety incidents.

Three observations sharpen the picture. First, *the boundary between cognition and infrastructure has blurred*. The reasoning-first models of 2024–2026 internalise much of what was previously implemented as external scaffolding, while protocols like MCP externalise much of what was previously bespoke per-framework integration. The net effect is that the cognitive stack and the infrastructural stack are converging, with research questions increasingly cutting across both. Second, *the field has discovered that not every problem is a model problem*. Many of the most consequential capability gains since 2023 came from interface design, careful trajectory fine-tuning, or evaluation infrastructure rather than from larger or smarter models. Third, *single-shot accuracy has been displaced by trajectory-level competence as the dominant performance metric*. An agent that finishes a task at all, in a controllable amount of compute, with auditable side-effects, is now the relevant unit of progress.

10.2 Open Problems

We name eight problems that, in our judgement, will dominate research over the 2026–2028 window. The list is not exhaustive but is meant as an honest reading of where the binding constraints lie.

1. Robust long-horizon planning. The greedy-error pathology of long trajectories (Section 3.5) is the single most binding constraint on agent reliability. PlanBench-style classical planning remains beyond LLM capability. Search-based deliberation pays for partial progress with order-of-magnitude compute increases. Reasoning-first models show qualitative improvements but are not deployment-ready for the longest-horizon tasks. The community has prototypes—LATS, RAP, MCTS-driven coding agents, π -style flow-matching for embodied control—but no convincing solution. Whether progress comes from improved post-training (richer trajectory data, better process supervision), from better external search, or from architectural innovations beyond the Transformer remains open. The CLAUDE.md cluster A diagnosis of this problem matches our own assessment.

2. Evaluation under contamination and circular reasoning. Section 7 catalogued the increasing fragility of public benchmarks under contamination, and the systematic biases of LLM-as-judge methodology. Continual benchmarking, private test sets, and judge ensembles partially mitigate the problem but do not solve it. The deeper question is methodological: how does the field know its progress is real? An empirical literature on the half-life of benchmarks, the calibration of LLM judges, and the sensitivity of agent rankings to evaluation choices is forming, but rigorous answers are scarce. We expect this area to see substantial methodological work, and we suspect that several apparently-solved tasks will turn out to be artefacts of contaminated evaluation.

3. Robust resistance to indirect prompt injection. Section 8.3 identified indirect injection as the dominant threat in the agent threat landscape. No solution exists. Spotlighting and StruQ-style structured-query training raise the bar but are bypassed under sustained adversarial attention. Capability-based isolation reduces blast radius but is no substitute for resistance. The defining safety question of the next two years is whether the model itself can be trained to robustly distinguish data from instructions in retrieved content. We expect substantial investment from major labs, but no immediate breakthrough; the more likely outcome is continued layered defense, heavier reliance on capability scoping, and the development of agent-specific incident-response practice.

4. Human-agent oversight and corrigibility. As agents take on longer-horizon, higher-stakes tasks, the question of how humans maintain effective oversight without sacrificing autonomy becomes urgent. Watching every step is impractical at scale; checking only the final result is insufficient when intermediate actions are irreversible. The literature has prototypes—structured action previews, explicit confirmation tokens, scoped autonomy bands—but no agreed-on framework. Connecting agent oversight to classical literature on corrigibility, scalable oversight, and AI safety more broadly is among the highest-leverage research opportunities, and is methodologically tractable in a way that some of the longer-term safety questions are not.

5. Agentic infrastructure standardisation. The Model Context Protocol’s rapid uptake (Section 5.6) is one signal among many that the field is ready for shared infrastructure. Authentication, authorisation, capability scoping, observability, and audit are all currently solved in incompatible ways across frameworks. Standardising these layers—without ossifying around an immature design—is a research and engineering problem of unusual practical importance. Industry has the resources to do this work; academia has insights about the right abstractions; collaboration between the two is sometimes uneven. The standardisation problem is also where specific technical contributions can have outsized influence.

6. Multi-modal grounding and embodiment. Section 9.3 surveyed the rapid progress on VLA models. The remaining gap to general-purpose physical agents is large but, on current trajectories, plausibly closeable within several years. The bottlenecks are recognisable: data scarcity (relative to web-scale text), simulation–reality transfer, safety in physical actuation, and the cost of evaluation in environments that cannot be replayed deterministically. Each of these is an active research area in robotics; the contribution of the LLM-agent literature is increasingly the integration layer that connects natural-language reasoning to physical control.

7. Multi-agent coordination beyond ad hoc patterns. Section 6 reviewed the maturing multi-agent literature and the recent observation [16] that multi-agent systems often fail in characteristic ways. The next phase is theory: when does multi-agent decomposition help, and when does it hurt? Is debate effective only because it surfaces disagreement, in which case structured disagreement protocols may capture most of the gain at lower cost? Are emergent behaviours in agent communities reliable engineering features, or merely interesting phenomena? The empirical literature is growing fast; principled theory has lagged.

8. Cost-aware and energy-aware operation. A reasoning-first agent answering a complex question may consume tens of thousands of tokens, costing dollars and many seconds of latency. As deployment scales, the economics of inference are starting to determine which agentic patterns are viable. The literature has begun to address compute-optimal test-time scaling [103], but the broader question of cost-aware orchestration—when to use a frontier model versus an SLM, when to invoke search versus a single forward pass, when to retrieve versus to reason—remains under-developed. We expect this to become a major research frontier as the field moves toward production-scale deployments.

Table 12 consolidates the eight problems above into a research roadmap that names, for each problem, the sections of the present survey that develop the relevant background, the most promising current attacks on the problem, the venues most likely to host the next round of progress, and an indicative time horizon for when serious progress should be expected. The table is intentionally opinionated and reflects the authors’ reading of the literature; researchers planning a contribution should treat it as a starting point for their own assessment rather than as a final ranking.

10.3 Predictions for 2026–2028

Predictions in a fast-moving field are hazardous, but several trajectories seem strongly implied by the current literature.

(i) *Reasoning-first models will become the default backbone for high-stakes agentic deployments*, as the cost-quality tradeoff for o1/R1-style systems improves and as more open replications close the gap with the leading closed systems. The fast outer loop will increasingly run on small distilled reasoning models that inherit the parent’s reasoning style at much lower cost.

(ii) *Tool-use protocols will consolidate.* MCP or a successor will likely become the dominant standard, with major frameworks converging on compatible adaptors. The cottage industry of bespoke tool integrations will shrink correspondingly, and tool authoring will become a more standardised software-engineering discipline.

(iii) *Evaluation will move toward continual, contamination-resistant benchmarking.* Static datasets will retain a niche role but will increasingly be supplemented by live benchmarks (Chatbot Arena style) and dynamically generated

tasks. The reporting of cost–accuracy frontiers and adversarial-robustness metrics will become normative rather than exceptional.

(iv) *Multi-agent systems will polarise.* Highly specialised, narrow-domain multi-agent pipelines will continue to deliver value (software engineering, scientific workflows, customer support). But the early hopes for general-purpose multi-agent collaboration as a uniform recipe will dim, replaced by a more nuanced understanding of when decomposition pays off.

(v) *Embodied agents will reach capability thresholds previously thought several years away.* The trajectory from RT-2 through OpenVLA to π_0 suggests that single-policy generalist robots performing useful household and industrial tasks will be commercially viable within the prediction window, though widespread adoption is gated by cost and safety considerations more than by capability.

(vi) *Agentic infrastructure—monitoring, audit, sandboxing, incident response—will become a recognised discipline* comparable to cybersecurity operations, with associated tooling, certifications, and standards.

(vii) *Indirect injection and other agent-specific attacks will produce at least one high-profile, real-world incident* of substantial cost. The field will respond with regulation, with standards adoption, and with renewed investment in defenses; the long-term outcome depends on the timing.

(viii) *The boundary between AI research and software engineering at scale will continue to blur.* The most consequential progress will increasingly come from teams that combine cutting-edge model capability with mature engineering practice—a description that already characterises the leading agentic research labs.

10.4 Recommendations for Researchers

We close with practical recommendations for researchers entering the field, particularly students and faculty contemplating contributions to the upcoming NeurIPS, ICLR, ICML, ACL, AAAI, and CVPR submission cycles.

Avoid the wrapper trap. The era when a clever prompt template wrapped around a frontier API constituted a publishable contribution has passed. Top venues now expect rigorous evaluation, ablation studies that isolate the marginal contribution of each component, and either a theoretical contribution or a substantial open-source artefact. Wrapper papers will be rejected.

Pick a binding constraint. The eight open problems above are binding constraints in the literal sense: capability does not improve until they are addressed. Working on a binding constraint, however incremental the progress, is more impactful than incremental progress on a saturated benchmark.

Invest in evaluation. The single most leveraged contribution in 2026–2028 may be a benchmark or evaluation methodology that genuinely measures something the existing landscape does not. New benchmarks, particularly those that resist contamination and that measure trajectory-level rather than answer-level competence, are scarce and impactful. Evaluation infrastructure work has historically been undervalued; the field is starting to correct that.

Open-source where possible. The OpenVLA, OpenHands, and DeepSeek-R1 trajectories illustrate the field-shaping power of well-engineered open releases. Open weights, open training code, open evaluation harnesses, and open data each multiply the impact of any single research contribution.

Take safety seriously, but treat it as engineering. Adversarial robustness, oversight infrastructure, and incident-response practice are technical problems with technical solutions. The community benefits when agent-safety work is conducted in the same rigorous, empirical style as the capability work; abstraction or hand-waving on safety produces less progress than careful engineering does.

Read across the stack. The most generative research questions in this field cut across model architecture, cognition, memory, tools, evaluation, and safety. Researchers who read narrowly within a single sub-area miss the intersections where real progress now happens.

10.5 Coda

Three years ago an LLM agent was a curiosity—a chat model coaxed by a prompt template into emitting tool calls, brittle in execution, irreproducible in evaluation. Today an LLM agent is a serious, deployable artefact that resolves real GitHub issues, conducts experimental science, drives robotic manipulation, and orchestrates entire enterprise workflows. The discipline that has formed around it is recognisable, with shared benchmarks, shared protocols, shared concepts, and shared open problems.

What separates this period in agent research from earlier waves of AI optimism is the unusually direct connection between the conceptual literature and the engineering practice. The papers cited throughout this survey are not abstractions about future systems; they are descriptions of artefacts that exist, that are evaluated against shared benchmarks, that are improved by next year’s papers, and that are deployed in production by month’s end. That feedback loop is the field’s most important asset, and protecting it—through rigorous evaluation, robust safety

practice, and continued open scientific exchange—is the central responsibility of the research community as agentic intelligence moves from prototypes to infrastructure on which large parts of the digital economy are likely to depend.

The transition from generative to agentic AI is far from complete. The next two years will determine whether the trajectory continues to bend toward broadly useful, well-evaluated, safely deployed systems, or whether it veers into a brittle, contamination-haunted, security-compromised plateau. Both outcomes are possible. The work that researchers, engineers, and standard-setters do now will determine which.

11 Glossary and Notation

This appendix collects the technical vocabulary used throughout the survey, with short definitions calibrated to the way each term is used in the agent literature rather than to a generic ML or systems-engineering context. Cross-references point to the section where each term is most fully discussed.

11.1 Core Concepts

Agent. A system that perceives an environment, maintains state, decides on actions, and executes those actions in pursuit of a goal. In this survey, an agent is built around an LLM core (Section 1.3); the canonical formal tuple is $\mathcal{A} = \langle M, P, M_{\text{em}}, T, E, \pi \rangle$ where M is the model, P is the prompting policy, M_{em} is memory, T is the tool set, E is the environment, and π is the controller.

Trajectory. A sequence of (observation, action, intermediate-text) tuples produced by an agent over a single episode. Trajectory-level evaluation (Section 7.2) scores agents not only on the final answer but also on the path taken; trajectory-level fine-tuning (Section 2.3) trains models on whole agent rollouts rather than on *(prompt, response)* pairs.

Outer loop. The orchestration loop that drives an agent: read state, prompt model, parse output, execute action, write back to state, repeat. Most agent frameworks (LangChain, LangGraph, AutoGen, OpenHands) are best understood as different implementations of this loop with different memory/tool/control abstractions plugged in.

Working memory. The information currently inside the LLM’s context window. Distinct from external memory stores; bounded; subject to “lost in the middle” degradation [69]; subject to per-token cost.

Episodic memory. The agent’s record of its own past experiences—trajectories, reflections, observations—stored externally and retrieved when relevant. Reflexion [99], MemoryBank [135], and the Generative-Agents memory stream [86] are canonical examples.

Semantic memory. Corpus-derived knowledge that the agent retrieves to inform reasoning. Implemented as RAG [61], GraphRAG [33], or other retrieval mechanisms (Section 4.2).

Procedural memory. Reusable skills—typically code snippets or tool wrappers—that the agent has authored or accepted into its skill library. The flagship example is Voyager’s Minecraft skill repository [112].

Tool / function. A typed, externally callable interface that the agent can invoke. Tools may be local (a calculator, a Python sandbox), networked (an HTTP API), or human-in-the-loop (asking the operator for confirmation). Tool catalogues range from a handful (early ReAct) to thousands (HuggingGPT, ToolLLM).

11.2 Cognitive Techniques

Chain of Thought (CoT). Prompting the model to produce intermediate reasoning steps before its final answer [119]. Both few-shot and zero-shot variants are widely used.

Self-consistency. Sampling k chains of thought and aggregating final answers by majority vote [116]. The simplest example of test-time compute scaling.

Tree of Thoughts (ToT). Search over a tree of partial reasoning states, with the LLM proposing branches and self-scoring them [129]. Cost can exceed CoT by 10–100×.

Graph of Thoughts (GoT). Generalisation of ToT to a directed graph allowing branches to merge on shared sub-results [10].

Reasoning via Planning (RAP). Treats the LLM as both world model and policy, and uses MCTS over imagined future states [41].

LATS. Language Agent Tree Search: integrates ReAct with MCTS, with tool calls executed only along committed branches [136].

ReAct. The interleaved *Thought–Action–Observation* format [130] that has become the default agent prompting pattern.

Reflexion. Append a verbal self-reflection to memory after each failed trial; condition future attempts on accumulated reflections [99].

Self-Refine. Generate, critique, and rewrite an answer iteratively without an external memory [74].

Process reward model (PRM). A reward model trained to score per-step rather than per-outcome correctness [66]. Enables fine-grained reranking of candidate reasoning chains and is a key training signal for reasoning-first models.

Test-time compute (TTC). Spending substantial compute per query through search, sampling, or extended internal reasoning [103]. The defining axis of o1 and DeepSeek-R1.

Reasoning-first model. An LLM whose deployment relies on a long internal reasoning trace, billed per token [29, 81]. The trace may be hidden from the user (o1) or disclosed (R1).

11.3 Tool-Use and Action Protocols

Function calling. A vendor-API mode in which the model is fine-tuned to emit structured JSON conforming to a declared schema [78]. Eliminates parsing failures at the cost of expressiveness.

Structured outputs. Constraint-decoding extension of function calling that enforces an arbitrary user-supplied JSON schema on the model’s output.

CodeAct. Treating Python (or another full programming language) as the action language [114]. Subsumes function calling within a single interpretable substrate.

MCP (Model Context Protocol). A vendor-neutral JSON-RPC protocol for connecting an LLM client to external tool servers [4]. Defines five primitives: Resources, Tools, Prompts, Roots, and Sampling.

Computer use. An actuation surface where the model receives screenshots and emits keyboard/mouse events [3]. Enables agents to drive arbitrary GUI applications.

Agent–computer interface (ACI). The set of high-level commands exposed to a software-engineering agent [125], designed to make code editing, file viewing, and search reliable for the model rather than ergonomic for a human.

11.4 Multi-Agent and Coordination

Hierarchical (planner–worker) topology. A planner decomposes a task and dispatches sub-tasks to specialised workers; results flow back up. Easiest to debug; dominates production deployments [120].

Peer-to-peer (debate / role-play) topology. Several agents on equal footing, each contributing reasoning or critique. Substrate of multi-agent debate [32] and CAMEL [62].

Pipeline / SOP-driven topology. Agents play fixed roles in a predetermined sequence (e.g., Product Manager $\rightarrow$ Architect $\rightarrow$ Engineer $\rightarrow$ QA in MetaGPT [46]).

Critic-actor loop. A specific peer-to-peer pattern in which one agent produces output and another evaluates it; the actor iterates on the critic’s feedback. Self-Refine generalises this within a single model.

Communicative dehallucination. A clarification step in ChatDev [89] where an agent that suspects its counterpart is hallucinating asks targeted questions before acting.

11.5 Evaluation

GAIA. A 466-question general-assistant benchmark [75] requiring reasoning, tool use, and multimodality.

SWE-bench / SWE-bench Verified. Benchmarks measuring whether an agent can resolve real GitHub issues [50, 80]. Verified is the human-curated 500-task subset.

τ -Bench. A tool-using-agent benchmark with stochastic user simulators [128]. Reports $\text{pass}\hat{k}$ consistency metrics in addition to mean success.

LLM-as-judge. Using a frontier LLM to score or compare agent outputs [134]. Subject to position, verbosity, self-preference, and authority biases.

JudgeBench. A benchmark for measuring the quality of LLM judges themselves [108].

Contamination. The presence of benchmark items (or near-duplicates) in a model’s pre-training data, inflating apparent performance.

Pass@k / $\text{pass}\hat{k}$. Pass@k is success rate among k samples (any one succeeds); $\text{pass}\hat{k}$ (used by τ -Bench) is success on *all* k samples (consistency).

11.6 Safety

Direct prompt injection / jailbreak. An attacker convinces the model, via the user-prompt channel, to violate its instructions [118, 139].

Indirect prompt injection. Malicious instructions embedded in retrieved content (a web page, an email, a tool response) that the model treats as instructions [39]. The dominant agent-era threat.

Spotlighting. Marking retrieved content with delimiters or transformations to signal that it is data, not instruction [44].

StruQ. Retraining the model on a structured-query format that places system, user, and retrieved text in distinct roles [20].

AgentDojo. An adversarial benchmark for measuring agent robustness to indirect injection in dynamic environments [27].

AgentPoison. A class of attacks that poison long-lived agent memory or knowledge bases to induce backdoor behaviour at attacker-chosen triggers [22].

Capability scoping. Restricting the agent’s tool set or credentials so that, even under successful adversarial influence, the blast radius is bounded. The principle of least privilege from classical computer security.

Constitutional Classifiers. Separately trained input/output classifiers, themselves bound by an explicit constitution, that screen prompts and completions for harm [5].

11.7 Architecture

Mixture-of-Experts (MoE). A Transformer architecture in which each token activates only a small subset of expert sub-networks via a learned router [34, 97]. Decouples total parameter capacity from per-token compute.

Active parameters. The number of parameters actually used in a forward pass. For MoE, this is much smaller than the total parameter count (e.g., DeepSeek-V3: 37B active out of 671B total).

SLM (Small Language Model). A compact LM optimised for edge/on-device deployment, typically <15B parameters, with curated training data [1].

SFT (supervised fine-tuning). Cross-entropy training on $(prompt, response)$ demonstrations.

RLHF. Reinforcement learning from human feedback [25, 82]: fit a reward model on pairwise preferences, then optimise the policy with PPO under a KL constraint.

RLAIF / Constitutional AI. RL with AI-generated preferences in place of (most) human labels [9].

DPO (direct preference optimisation). A simplified preference-learning recipe that derives a policy update directly from pairwise preferences without an explicit reward model [92].

AgentTuning. SFT on curated agent trajectories to instill multi-turn tool-use behaviour [131].

11.8 Notation Used in This Survey

The agent tuple is $\mathcal{A} = \langle M, P, M_{\text{em}}, T, E, \pi \rangle$, with components named in Section 1.3. Throughout the survey we write:

- $\tau = (o_0, a_0, o_1, a_1, \dots)$ for an agent trajectory, with o_t the observation and a_t the action at step t ;
- $\hat{y}$ for the agent’s final answer or terminal state;
- k for the number of samples in self-consistency or process-reward reranking;
- H for the horizon (number of steps) of a trajectory;
- $\text{pass}@k$ for the probability that at least one of k samples succeeds;
- pass^k for the probability that all k samples succeed (consistency).

Capability axes used in summary tables are taken from the four-pillar taxonomy of Section 1.4: cognition, memory, tools, and multi-agent. Three cross-cutting axes (architecture, evaluation, and safety) and one application axis (domain) supplement the core four where needed.

References

- [1] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone, 2024. arXiv:2404.14219; Microsoft.
- [2] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, et al. Do as i can, not as i say: Grounding language in robotic affordances. In *Conference on Robot Learning (CoRL)*, 2022.
- [3] Anthropic. Introducing computer use, a new Claude 3.5 Sonnet, and Claude 3.5 Haiku. <https://www.anthropic.com/news/3-5-models-and-computer-use>, 2024. Released 22 October 2024.
- [4] Anthropic. Introducing the Model Context Protocol. <https://www.anthropic.com/news/model-context-protocol>, 2024. Specification at <https://modelcontextprotocol.io>; released 26 November 2024.

- [5] Anthropic. Constitutional classifiers: Defending against universal jailbreaks. <https://www.anthropic.com/research/constitutional-classifiers>, 2025. Released February 2025.
- [6] AnySphere. Cursor: The AI code editor. <https://www.cursor.com/>, 2024.
- [7] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024.
- [8] Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models, 2021. arXiv:2108.07732; introduces MBPP.
- [9] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback, 2022. arXiv:2212.08073; Anthropic.
- [10] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoefer. Graph of thoughts: Solving elaborate problems with large language models. In *AAAI Conference on Artificial Intelligence*, 2024.
- [11] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control, 2024. arXiv:2410.24164; Physical Intelligence.
- [12] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.
- [13] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning (ICML)*, 2022.
- [14] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning (CoRL)*, 2023.
- [15] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [16] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail?, 2025. arXiv:2503.13657.
- [17] Harrison Chase and LangChain Contributors. LangChain: Building applications with LLMs through composability. <https://github.com/langchain-ai/langchain>, 2023.
- [18] Baian Chen, Chang Shu, Ehsan Shareghi, Nigel Collier, Karthik Narasimhan, and Shunyu Yao. FireAct: Toward language agent fine-tuning, 2023. arXiv:2310.05915.
- [19] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Joseph Joseph, Greg Brockman, et al. Evaluating large language models trained on code, 2021. arXiv:2107.03374; introduces HumanEval.
- [20] Sizhe Chen, Arman Zharmagambetov, Saeed Mahloujifar, Kamalika Chaudhuri, and Chuan Guo. StruQ: Defending against prompt injection with structured queries, 2024. USENIX Security 2025; arXiv:2402.06363.
- [21] Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, Chen-Ming Chan, Heyang Yu, Yaxi Lu, Yi-Hsin Hung, Chen Qian, Yujia Qin, Xin Cong, Ruobing Xie, Zhiyuan Liu, Maosong Sun, and Jie Zhou. Agent-Verse: Facilitating multi-agent collaboration and exploring emergent behaviors. In *International Conference on Learning Representations (ICLR)*, 2024.

- [22] Zhaorun Chen, Zhen Xiang, Chaowei Xiao, Dawn Song, and Bo Li. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases, 2024. NeurIPS 2024; arXiv:2407.12784.
- [23] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *International Conference on Machine Learning (ICML)*, 2024.
- [24] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. PaLM: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113, 2023.
- [25] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems (NIPS)*, 2017.
- [26] Cognition AI. Introducing Devin, the first AI software engineer. <https://www.cognition.ai/blog/introducing-devin>, 2024. Released 12 March 2024.
- [27] Edoardo DeBenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024.
- [28] DeepSeek-AI. DeepSeek-V3 technical report, 2024. arXiv:2412.19437; 671B-parameter MoE with 37B activated per token.
- [29] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645:633–638, 2025. arXiv:2501.12948.
- [30] Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web, 2023. NeurIPS 2023; arXiv:2306.06070.
- [31] Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. PaLM-E: An embodied multimodal language model. In *International Conference on Machine Learning (ICML)*, 2023.
- [32] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *International Conference on Machine Learning (ICML)*, 2024.
- [33] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A Graph RAG approach to query-focused summarization, 2024. arXiv:2404.16130; Microsoft GraphRAG.
- [34] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. arXiv:2101.03961.
- [35] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. Retrieval-augmented generation for large language models: A survey, 2024. arXiv:2312.10997.
- [36] Paul Gauthier. Aider: AI pair programming in your terminal. <https://github.com/Aider-AI/aider>, 2024.
- [37] Google DeepMind. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context, 2024. arXiv:2403.05530.
- [38] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models, 2024. arXiv:2407.21783; Meta AI.
- [39] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, 2023.
- [40] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, and Xiangliang Zhang. Large language model based multi-agents: A survey of progress and challenges, 2024. arXiv:2402.01680.

- [41] Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. Reasoning with language model is planning with world model. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [42] Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. WebVoyager: Building an end-to-end web agent with large multimodal models, 2024. *ACL 2024*; arXiv:2401.13919.
- [43] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.
- [44] Keegan Hines, Gary Lopez, Matthew Hall, Federico Zarfati, Yonatan Zunger, and Emre Kiciman. Defending against indirect prompt injection attacks with spotlighting, 2024. arXiv:2403.14720; Microsoft.
- [45] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [46] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. MetaGPT: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations (ICLR)*, 2024.
- [47] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*, 2024.
- [48] Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. Mixtral of experts, 2024. arXiv:2401.04088; Mistral AI.
- [49] Zhengbao Jiang, Frank F. Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [50] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.
- [51] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. arXiv:2001.08361.
- [52] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- [53] Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into state-of-the-art pipelines. In *International Conference on Learning Representations (ICLR)*, 2024.
- [54] Omar Khattab and Matei Zaharia. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference*, 2020.
- [55] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model, 2024. arXiv:2406.09246.
- [56] Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. VisualWebArena: Evaluating multimodal agents on realistic visual web tasks. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

- [57] Jing Yu Koh, Stephen McAleer, Daniel Fried, and Ruslan Salakhutdinov. Tree search for language model agents, 2024. arXiv:2407.01476.
- [58] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners, 2022. NeurIPS 2022; arXiv:2205.11916.
- [59] LangChain Contributors. LangGraph: Building stateful, multi-actor applications with LLMs. <https://github.com/langchain-ai/langgraph>, 2024.
- [60] Dmitry Lepikhin, HyounJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. GShard: Scaling giant models with conditional computation and automatic sharding, 2020. arXiv:2006.16668.
- [61] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [62] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for ‘mind’ exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [63] Minghao Li, Feifan Song, Bowen Yu, Haiyang Yu, Zhoujun Li, Fei Huang, and Yongbin Li. API-Bank: A comprehensive benchmark for tool-augmented LLMs. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [64] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models, 2023. TMLR; arXiv:2211.09110.
- [65] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. Encouraging divergent thinking in large language models through multi-agent debate. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- [66] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- [67] Bo Liu, Yuqian Jiang, Xiaohan Zhang, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. LLM+P: Empowering large language models with optimal planning proficiency. In *International Conference on Robotics and Automation (ICRA) Workshop*, 2023.
- [68] Jerry Liu and LlamaIndex Contributors. LlamaIndex: A data framework for LLM applications. https://github.com/run-llama/llama_index, 2023.
- [69] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. In *Transactions of the Association for Computational Linguistics (ACL)*, 2024.
- [70] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations (ICLR)*, 2024.
- [71] Yupei Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. Formalizing and benchmarking prompt injection attacks and defenses, 2024. USENIX Security 2024; arXiv:2310.12815.
- [72] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI Scientist: Towards fully automated open-ended scientific discovery, 2024. arXiv:2408.06292.
- [73] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. arXiv:2304.05376.

- [74] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [75] Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. GAIA: a benchmark for General AI Assistants. In *International Conference on Learning Representations (ICLR)*, 2024.
- [76] Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, and Jianfeng Gao. Large language models: A survey, 2024. arXiv:2402.06196.
- [77] Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. WebGPT: Browser-assisted question-answering with human feedback. In *arXiv preprint*, 2021.
- [78] OpenAI. Function calling and other API updates. <https://openai.com/index/function-calling-and-other-api-updates/>, 2023. Released 13 June 2023.
- [79] OpenAI. GPT-4 technical report, 2023. arXiv:2303.08774.
- [80] OpenAI. Introducing SWE-bench Verified. <https://openai.com/index/introducing-swe-bench-verified/>, 2024. Human-validated 500-task subset; released August 2024.
- [81] OpenAI. OpenAI o1 system card. <https://openai.com/index/openai-o1-system-card/>, 2024. Released 5 December 2024; arXiv:2412.16720.
- [82] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [83] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems, 2023. arXiv:2310.08560.
- [84] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations, 2024. NeurIPS 2024; arXiv:2404.13076.
- [85] Aaron Parisi, Yao Zhao, and Noah Fiedel. TALM: Tool augmented language models, 2022. arXiv:2205.12255.
- [86] Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023.
- [87] Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive APIs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [88] Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. Red teaming language models with language models, 2022. EMNLP 2022; arXiv:2202.03286.
- [89] Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, Juyuan Xu, Dahai Li, Zhiyuan Liu, and Maosong Sun. ChatDev: Communicative agents for software development. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [90] Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *International Conference on Learning Representations (ICLR)*, 2024.
- [91] Qwen Team. Qwen2.5 technical report, 2024. arXiv:2412.15115.
- [92] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

- [93] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- [94] Stuart Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020.
- [95] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [96] Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role play with large language models. *Nature*, 623(7987):493–498, 2023.
- [97] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*, 2017.
- [98] Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. HuggingGPT: Solving AI tasks with ChatGPT and its friends in Hugging Face. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [99] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [100] Significant Gravitas. Auto-GPT: An autonomous GPT-4 experiment. <https://github.com/Significant-Gravitas/AutoGPT>, 2023.
- [101] Aditi Singh, Abul Ehtesham, Saket Kumar, and Tala Talaei Khoei. Agentic retrieval-augmented generation: A survey on agentic RAG, 2025. arXiv:2501.09136.
- [102] Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180, 2023.
- [103] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters, 2024. arXiv:2408.03314.
- [104] Aarohi Srivastava et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models, 2023. TMLR 2023; arXiv:2206.04615.
- [105] Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. Cognitive architectures for language agents. *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2309.02427.
- [106] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. The MIT Press, 2nd edition, 2018.
- [107] SWE-bench Team. SWE-bench leaderboard. <https://www.swebench.com/>, 2025. Live leaderboard, accessed 2026-05.
- [108] Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y. Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. JudgeBench: A benchmark for evaluating LLM-based judges. In *International Conference on Learning Representations (ICLR)*, 2025.
- [109] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- [110] Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change, 2023. NeurIPS 2023 Datasets and Benchmarks; arXiv:2206.10498.

- [111] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NIPS)*, 2017.
- [112] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2305.16291, May 2023.
- [113] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 18(6):186345, 2024. arXiv:2308.11432.
- [114] Xingyao Wang, Yangyi Chen, Lifan Yuan, Yizhe Zhang, Yunzhu Li, Hao Peng, and Heng Ji. Executable code actions elicit better LLM agents. In *International Conference on Machine Learning (ICML)*, 2024.
- [115] Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, Hoang H. Tran, Fuqiang Li, Ren Ma, Mingzhang Zheng, Bill Qian, Yanjun Shao, Niklas Muennighoff, Yizhe Zhang, Binyuan Hui, Junyang Lin, Robert Brennan, Hao Peng, Heng Ji, and Graham Neubig. OpenHands: An open platform for AI software developers as generalist agents. In *International Conference on Learning Representations (ICLR)*, 2025. Formerly titled *OpenDevin*.
- [116] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [117] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [118] Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail?, 2023. NeurIPS 2023; arXiv:2307.02483.
- [119] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [120] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryan W. White, Doug Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversations. In *Conference on Language Modeling (COLM)*, 2024.
- [121] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kambadur, David Rosenberg, and Gideon Mann. BloombergGPT: A large language model for finance, 2023. arXiv:2303.17564.
- [122] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, Rui Zheng, Xiaoran Fan, Xiao Wang, Limao Xiong, Yuhao Zhou, Weiran Wang, Changhao Jiang, Yicheng Zou, Xiangyang Liu, Zhangyue Yin, Shihan Dou, Rongxiang Weng, Wensen Cheng, Qi Zhang, Wenjuan Qin, Yongyan Zheng, Xipeng Qiu, Xuanjing Huang, and Tao Gui. The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, 2025. arXiv:2309.07864, September 2023.
- [123] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [124] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *International Conference on Learning Representations (ICLR)*, 2024.
- [125] John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

- [126] John Yang, Carlos E. Jimenez, Alex L. Zhang, Kilian Lieret, Joyce Yang, Xindi Wu, Ori Press, Niklas Muenighoff, Gabriel Synnaeve, Karthik R. Narasimhan, Diyi Yang, Sida I. Wang, and Ofir Press. SWE-bench Multimodal: Do AI systems generalize to visual software domains? In *International Conference on Learning Representations (ICLR)*, 2025.
- [127] Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. WebShop: Towards scalable real-world web interaction with grounded language agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [128] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -Bench: A benchmark for tool-agent-user interaction in real-world domains. In *International Conference on Learning Representations (ICLR)*, 2025.
- [129] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [130] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [131] Aohan Zeng, Mingdao Liu, Rui Lu, Bowen Wang, Xiao Liu, Yuxiao Dong, and Jie Tang. AgentTuning: Enabling generalized agent abilities for LLMs. In *Findings of the Association for Computational Linguistics (ACL Findings)*, 2024.
- [132] Hongxin Zhang, Weihua Du, Jiaming Shan, Qinzhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, and Chuang Gan. Building cooperative embodied agents modularly with large language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- [133] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. A survey of large language models, 2023. arXiv:2303.18223.
- [134] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.
- [135] Wanjuan Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing large language models with long-term memory. In *AAAI Conference on Artificial Intelligence*, 2024.
- [136] Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language agent tree search unifies reasoning, acting, and planning in language models. In *International Conference on Machine Learning (ICML)*, 2024.
- [137] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V. Le, and Ed H. Chi. Least-to-most prompting enables complex reasoning in large language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [138] Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations (ICLR)*, 2024.
- [139] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models, 2023. arXiv:2307.15043.

Year	Axis	Milestone	Significance
2017	architecture	Transformer [111]	decoder substrate for all later LLMs
2017	architecture	Sparsely-gated MoE [97]	sparsely-activated capacity scaling
2020	architecture	GPT-3 [15]	few-shot in-context learning at 175B
2020	memory	RAG [61]	retrieval-augmented generation template
2020	memory	DPR [52]	dense passage retrieval baseline
2022	cognition	Chain-of-Thought [119]	step-by-step reasoning prompts
2022	cognition	InstructGPT [82]	RLHF alignment template
2022	cognition	Self-Consistency [116]	sampling diversity for accuracy
2022	tools	WebGPT [77]	browser as a tool the LM invokes
2023	cognition	Tree of Thoughts [129]	search-based deliberation
2023	cognition	Reflexion [99]	verbal self-feedback
2023	tools	ReAct [130]	interleaved thought-action-observation loop
2023	tools	Toolformer [95]	self-supervised tool selection
2023	tools	Function calling [78]	schema-constrained action emission
2023	multi-agent	CAMEL [62]	role-play conversation as data engine
2023	multi-agent	Generative Agents [86]	multi-component agent memory at scale
2023	multi-agent	MetaGPT [46]	SOP-driven software-engineering agents
2023	multi-agent	Multi-agent debate [32]	disagreement as error signal
2023	memory	Voyager [112]	lifelong skill library in Minecraft
2023	memory	MemGPT [83]	OS analogy for agent memory
2023	safety	Indirect injection [39]	retrieved-content as instruction channel
2023	safety	Universal adversarial [139]	transferable jailbreak suffixes
2023	evaluation	GAIA [75]	general-assistant benchmark
2024	architecture	Mixtral 8x7B [48]	open-weights MoE breakthrough
2024	architecture	DeepSeek-V3 [28]	open-frontier 671B/37B-active MoE
2024	architecture	o1 (reasoning model) [81]	inference-time compute as architecture
2024	architecture	Phi-3 family [1]	data-curation route to small models
2024	cognition	Compute-optimal TTC [103]	test-time vs. pre-train compute swap
2024	cognition	Process supervision [66]	per-step reward modelling
2024	tools	CodeAct [114]	code as universal action substrate
2024	tools	Computer use [3]	screenshot-grounded desktop control
2024	tools	MCP [4]	vendor-neutral tool protocol
2024	multi-agent	AutoGen [120]	conversation-centric framework
2024	multi-agent	LangGraph [59]	state-machine orchestration
2024	memory	GraphRAG [33]	community-summarised structural retrieval
2024	memory	Self-RAG / FLARE [7, 49]	model-gated retrieval
2024	memory	Long-context Gemini 1.5 [37]	10M-token effective windows
2024	evaluation	SWE-bench Verified [80]	curated 500-task SWE-bench subset
2024	evaluation	τ -Bench [128]	user-simulator + tool-use eval
2024	evaluation	AgentDojo [27]	adversarial agent benchmark
2024	safety	Spotlighting [44]	retrieved-content delimiting defense
2024	safety	StruQ [20]	structured-query training defense
2024	domain	OpenVLA [55]	open-frontier vision-language-action
2024	domain	π_0 [11]	flow-matching cross-embodiment generalist
2024	domain	Coscientist [12]	autonomous chemistry agent (Nature)
2024	domain	The AI Scientist [72]	end-to-end ML research pipeline
2025	architecture	DeepSeek-R1 [29]	open reasoning-first model with disclosed CoT
2025	safety	Constitutional Classifiers [5]	input/output classifier defense
2025	multi-agent	Why MAS Fail [16]	14-category MAS failure taxonomy
2025	evaluation	SWE-bench leaderboard [107]	frontier 70–80% on Verified

Table 1: Selected milestones in the development of LLM-based agentic intelligence, 2017–2025. The “Axis” column indicates the dominant capability the work advances, using the survey’s seven-axis classification (architecture, cognition, memory, tools, multi-agent, evaluation, safety) plus a separate “domain” axis for vertical deployments. The list is illustrative; full citation context appears in the corresponding sections of the survey.

Recipe	Signal	Optimisation	Annotation cost	Agentic property improved
SFT [82]	high-quality (x, y) pairs	cross-entropy on y	high (per-token)	instruction following, format compliance
RLHF / PPO [25, 82]	pairwise human preferences	PPO with reward model + KL	high (per-pair)	preference alignment, helpfulness
RLAIF / Constitutional [9]	AI critiques + revisions	SFT + RL on AI preferences	low (scales with model)	harmlessness, refusal calibration
DPO [92]	pairwise preferences	contrastive loss on policy	medium	preference alignment without RL
Process supervision [66]	per-step correctness labels	PRM training; rerank or RL	very high	step-by-step reasoning fidelity
Outcome RL (R1-style) [29]	verifiable final-answer rewards	policy gradient on rollouts	low (auto-verifiable)	reasoning depth, self-correction
AgentTuning [131]	curated agent trajectories	SFT on (τ, y) pairs	medium	multi-turn tool use, recovery
FireAct [18]	distilled GPT-4 ReAct traces	SFT on filtered traces	low (synthetic)	ReAct-format competence
Task-specific SFT [125]	domain-curated trajectories	SFT on (τ, y)	medium	domain-specific action vocabulary

Table 2: Post-training recipes used to convert a pre-trained LLM into an agent-ready model. Modern pipelines chain several of these stages; the boundary between recipes blurs as labs combine techniques (e.g. outcome RL on top of an SFT-bootstrapped policy with DPO-derived alignment).

Model	Org.	Type	Total (B)	Active (B)	Open weights
GPT-4 / GPT-4o	OpenAI	undisclosed	n/a	n/a	no
o1 (reasoning)	OpenAI	undisclosed (+ TTC)	n/a	n/a	no
Claude 3.5 Sonnet	Anthropic	undisclosed	n/a	n/a	no
Llama 3 405B	Meta	dense	405	405	yes
Llama 3 70B	Meta	dense	70	70	yes
Llama 3 8B	Meta	dense	8	8	yes
Mixtral 8x7B	Mistral	MoE (top-2)	47	13	yes
DeepSeek-V3	DeepSeek-AI	MoE (top-8)	671	37	yes
DeepSeek-R1	DeepSeek-AI	MoE (top-8) + TTC	671	37	yes
Qwen2.5 72B	Alibaba	dense	72	72	yes
Phi-3-mini	Microsoft	dense	3.8	3.8	yes
Phi-3-medium	Microsoft	dense	14	14	yes

Table 3: Representative LLM backbones in the 2024–2026 agent ecosystem. “TTC” indicates models whose deployment relies critically on inference-time compute scaling. Active parameter counts for MoE models reflect typical per-token activation under the published routing scheme.

Method	Search structure	Branch evaluation	Cost vs. CoT	Best suited to
Chain Thought [119]	of linear	none	$1\times$	well-trodden multi-step problems
Self-Consistency [116]	k parallel chains	majority vote	$k\times$	answers with a unique mode
Least-to-Most [137]	sequential decomposition	dependency on prior step	$2\text{--}4\times$	compositional generalisation
Tree Thoughts [129]	of tree (BFS/DFS)	LLM self-score	$10\text{--}100\times$	branching puzzles, planning
Graph Thoughts [10]	of DAG with merges	LLM self-score	$10\text{--}100\times$	shared sub-goal aggregation
RAP / world-model search [41]	MCTS over imagined states	LLM as world model	$20\text{--}100\times$	deterministic planning domains
LATS [136]	MCTS over agent states	LLM rollout + value	$20\text{--}100\times$	ReAct tasks with retry budget
Reflexion [99]	retry buffer	verbal post-mortem	$3\text{--}10\times$	failure-rich tasks with test signals
Self-Refine [74]	iterate-and-critique	critic prompt	$2\text{--}3\times$	open-ended polishing
Process supervision [66]	rerank of k samples	per-step PRM	$k\times + \text{PRM}$	math/code with verifiable steps
Reasoning-first [29, 81]	internal long trace	internalised RL signal	$5\text{--}50\times$	high-stakes single-step decisions

Table 4: Deliberation patterns for LLM agents. “Cost vs. CoT” is an order-of-magnitude estimate against a single greedy chain-of-thought pass; the constants vary by task and depth budget. “Best suited to” captures the type of task structure each method most clearly exploits.

Mechanism	Time scale	Capacity	Retrieval	Dominant failure mode
Context window [69]	turn / session	8K–10M tokens	native attention	“Lost in the middle” attention bias
Conversation summary	session	summary length	re-prompted	loss of detail in compression
Vector RAG [52, 61]	corpus-level	unbounded	embedding ANN	off-topic neighbours, chunk boundary errors
Graph RAG [33]	corpus-level	unbounded	community summarisation	schema rigidity, extraction errors
Self-RAG / FLARE [7, 49]	corpus-level	unbounded	model-gated retrieval	threshold tuning, latency variance
Agentic RAG [101]	corpus-level	unbounded	multi-hop ReAct	loop divergence, cost blowup
Reflexion buffer [99]	per-trial	few KB	concatenation	brittle to similar-but-different tasks
MemoryBank [135]	weeks of dialogue	MB+	recency + relevance	forgetting tuned per-domain
Generative-Agents stream [86]	simulated days	MB+	recency + importance + relevance	importance scoring noise
Voyager skill library [112]	permanent	code corpus	key-name retrieval	buggy skills propagate
MemGPT pages [83]	permanent	disk-bounded	function-call paging	paging policy mistakes

Table 5: Memory mechanisms used in contemporary LLM-agent systems, classified by time scale, capacity, retrieval cost, and dominant failure mode. Real systems combine several of these layers; the table is meant as a vocabulary, not as a partition.

Property	ReAct text	Function call	CodeAct	MCP
	[130]	[78]	[114]	[4]
Action surface	free-form text	JSON per schema	executable Python	JSON-RPC tools/resources/prompts
Schema enforcement	none	client-side	none (sandbox runs code)	wire-format JSON Schema
Expressiveness	high	medium	very high (full language)	medium-high
Parsing failures	frequent	rare	rare	rare
Multi-step composition	none (turn-level)	none (turn-level)	native (loops, branches)	via repeated calls
Streaming / long-running	ad hoc	limited	via subprocess	limited (early spec)
Vendor portability	high (any LLM)	per-vendor API	high (sandbox-specific)	vendor-neutral
Required infra	none	fine-tuned model	code sandbox	MCP client + servers
Typical failure mode	malformed strings	schema mismatch	runtime exceptions	server unavailable

Table 6: Comparison of agent action-emission protocols. ReAct, function-calling, CodeAct, and MCP coexist in real systems; the choice depends on whether expressiveness, machine-checkability, or interoperability dominates.

System / Framework	Topology	Differentiation	Venue / Year	HITL
CAMEL [62]	peer (role-play)	prompt only	NeurIPS 2023	no
MetaGPT [46]	SOP pipeline	prompt only	ICLR 2024	optional
ChatDev [89]	SOP pipeline	prompt only	ACL 2024	no
AutoGen [120]	flexible (manager)	prompt only	COLM 2024	yes
LangGraph [59]	state-machine	prompt only	2024 (lib.)	yes
AgentVerse [21]	4-stage cycle	prompt only	ICLR 2024	optional
Generative Agents [86]	peer (simulated)	persona prompt	UIST 2023	no
CoELA [132]	peer (embodied)	prompt only	ICLR 2024	no
Multi-agent debate [32, 65]	peer (debate)	prompt only	ICML/EMNLP 2024	no

Table 7: Representative multi-agent LLM systems and orchestration frameworks. The “Differentiation” column distinguishes whether agents are specialised by fine-tuning or only by prompt; almost all current systems use the latter. “HITL” indicates whether the framework natively supports human-in-the-loop intervention.

Benchmark	Domain	Tasks	Year	Headline metric (frontier 2025–26)
GAIA [75]	general assistant	466	2023	Lvl 1 ~60–75%, Lvl 3 ~30%
SWE-bench Verified [80]	software issues	500	2024	70–80%
AgentBench [70]	multi-environment	8 envs	2023	per-env scores
WebArena [138]	web tasks	812	2023	35–55% (varies by site)
VisualWebArena [56]	visual web	910	2024	<30%
OSWorld [123]	OS tasks	369	2024	15–40%
τ -Bench [128]	tool+user sim.	165 (airline+retail)	2024	pass4 ~20–50%
HumanEval [19]	code synthesis	164	2021	saturated >95%
MMLU-Pro [117]	knowledge	12,032	2024	80–90%

Table 8: Representative agent-relevant benchmarks. Headline metrics are summary ranges across publicly reported frontier results in 2025–2026 and should not be read as exact figures, which evolve continually on public leaderboards [107].

Attack family	Entry point	Goal	Partial defenses
Direct break [118]	jail- user prompt	elicit prohibited content	alignment training, output classifiers [5]
Adversarial suffix [139]	suf- user prompt	elicit prohibited content / actions	input classifiers, structured queries [20]
Indirect injection [39]	injec- retrieved content	data exfiltration, tool misuse	spotlighting [44], StruQ [20], capability scoping
Memory poisoning [22]	poison- long-lived memory	persistent backdoor	provenance, anomaly detection, forgetting policies
Tool-response injection	in- tool output	action hijacking, lateral movement	tool sandboxing, output validation
Multi-agent inter-agent injection	/ agent-to-agent message	subvert downstream agent	message provenance, structured channels
Modality smuggling	smug- image / audio	OCR / bypass text-only filters	multimodal classifiers, OCR-aware scanning
Resource exhaustion	exhaus- user prompt or tool	cost amplification, DoS	rate limiting, cost budgets, plan caps

Table 9: Principal attack patterns relevant to LLM-based agents, the entry points they exploit, the objectives they pursue, and the defenses that have shown partial effectiveness. None of the listed defenses is complete; production systems combine several layers.

Case	Primary attack family	Stack layer exploited	Architectural defense added
Customer-support exfiltration	Indirect injection + tool-response injection	Tool surface (send-message scope)	Recipient whitelisting, StruQ re-training, capability-scoped credentials
Summarisation jailbreak	Indirect injection + tool composition	Cross-tool composition (file read × summarise)	Out-of-scope detection, refusal training on combined verbs, runtime monitors on retrieved-content tool calls
MCP-server backdoor	Supply-chain compromise	Server provenance	MCP signing, integrity checks at install, per-server audit logs
Reflection poisoning	Memory poisoning	Long-lived episodic memory	Provenance-tagged writes, memory diffs, principled forgetting policies

Table 10: The four illustrative incident cases of Section 8.9 mapped against the attack taxonomy of Table 9, the stack layer at which the attack landed, and the architectural defenses adopted in response. The combination of attack families across cases motivates the layered-defense posture that mature deployments now adopt.

Vertical	Typical autonomy	aut-	Dominant mode	failure	Evaluation	har-	Safety regime
Software engineering [115, 125]	partial		test-passing	but globally broken edits	SWE-bench	Veri-	sandboxed exec, PR review
Scientific discovery [12, 72, 73]	low–full (risk dependent)	(risk)	hallucinated	citations / protocols	domain-specific (chemistry, ML)		wet-lab interlock, peer review
Embodied VLA [11, 14, 55]	low (closed-loop)	(closed-loop)	sim-to-real	gap, irreversibility	Open Embodiment, RoboArena	X-	physical interlock, scope limits
Healthcare [102]	advisory		hallucinated	dose / protocol	clinical Q&A + expert review		clinician sign-off, audit logs
Finance [121]	advisory		numerical	hallucination, injection	domain-specific (FinBen, NER)		SOX-style audit, citation links
Customer support [128]	partial		policy violation,	re-funds	τ -Bench [128]		scoped tools, escalation triggers
Cybersecurity	partial–full (offensive only)		false-negative findings		CTF, rubrics	pen-test	isolation, ROE constraints
Legal	advisory		misstated precedent / citation		curated case sets		attorney sign-off, conflict checks
Education / tutoring	partial		misleading	explanations	student outcome studies		age gating, content filters
Productivity / office	partial		lost work, accidental sends		user studies		undo, confirmation prompts

Table 11: Domain-by-domain summary of LLM-agent deployment patterns. “Typical autonomy” is the level at which the domain currently runs in production; “Safety regime” lists the gating control that production deployments rely on. The patterns are not exclusive: most production systems combine several controls.

Problem	Anchored in	Most attacks	promising	at-	Likely venues	Horizon
Long-horizon planning	§3.5	process MCTS, post-training	supervision, reasoning-first		NeurIPS, ICLR	2026–2028
Eval under contamination	§7.4	continual judge splits	benchmarks, ensembles, private		ICML, ACL	2026–2027
Indirect injection robustness	§8.3	StruQ-style training, capability monitors	scoping, runtime		USENIX Sec., NeurIPS	2026–2029
Human–agent oversight	§8.8	action previews, autonomy, scalable oversight	scoped		ICML, AIES	2026–2028
Infrastructure standardisation	§5.6	MCP successors, primitives, identity protocols	audit tech reports,		OSDI	2026–2027
Multi-modal grounding	§9.3	VLA scaling, real transfer, datasets	sim-to-embodied		CVPR, CoRL, RSS	2026–2029
Multi-agent theory	the- §6.7	disagreement decomposition, failure-mode prediction	protocols, theory,		ICML, EMNLP	2026–2028
Cost-aware orchestration	or- §3.6, §7.7	router models, compute-optimal TTC, SLM cascades			NeurIPS, ML-Sys	2026–2027

Table 12: An opinionated research roadmap for the eight open problems named in Section 10.2. The “Anchored in” column points to the survey sections that supply background; “Most promising attacks” summarises the methodological directions on which we expect the most progress; “Likely venues” lists the publication forums where serious work is most likely to land; and “Horizon” gives an indicative window in which substantial—not necessarily complete—progress should be expected.